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Upper Grand Coulee: New views of a channeled scabland megafloods enigma

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ABSTRACT

New findings about old puzzles occasion rethinking of the Grand Coulee, greatest of the scabland channels. Those puzzles begin with antecedents of current upper Grand Coulee. By a recent interpretation, the upper coulee exploited the former high-level valley of a preflood trunk stream that had drained to the southwest beside and across Coulee anticline or monocline. In any case, a constriction and sharp bend in nearby Columbia valley steered Missoula floods this direction. Completion of upper Grand Coulee by megaflood erosion captured flood drainage that would otherwise have continued to enlarge Moses Coulee.

Upstream in the Sanpoil valley, deposits and shorelines of last-glacial Lake Columbia varied with the lake's Grand Coulee outlet while also recording scores of Missoula floods. The Sanpoil evidence implies that upper Grand Coulee had approached its

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present intake depth early the last glaciation at latest, or more simply during a prior glaciation. An upper part of the Sanpoil section provides varve counts between the last tens of Missoula floods in a stratigraphic sequence that may now be linked to flood rhythmites of southern Washington by a set-S tephra from Mount St. Helens.

On the floor of upper Grand Coulee itself, recently found striated rock and lodgement till confirm the long-held view, which Bretz and Flint had shared, that cutting of upper Grand Coulee preceded its last-glacial occupation by the Okanogan ice lobe. A dozen or more late Missoula floods registered as sand and silt in the lee of Steamboat Rock.

Some of this field evidence about upper Grand Coulee may conflict with results of recent two-dimensional simulations for a maximum Lake Missoula. In these simulations only a barrier high above the present coulee intake enables floods to approach high-water marks near Wenatchee that predate stable blockage of Columbia valley by the Okanogan lobe. Above the walls of upper Grand Coulee, scabland limits provide high-water targets for two-dimensional simulations of watery floods. The recent models sharpen focus on water sources, prior coulee incision, and coulee's occupation by the Okanogan ice lobe.

Field reappraisal continues downstream from Grand Coulee on Ephrata fan. There, some of the floods exiting lower Grand Coulee had bulked up with fine sediment from glacial Lake Columbia, upper coulee till, and a lower coulee lake that the fan itself impounded. Floods thus of debris-flow consistency carried outsize boulders previously thought transported by watery floods.

Below Ephrata fan, a backflooded reach of Columbia valley received Grand Coulee outflow of small, late Missoula floods. These late floods can—by varve counts in post-S-ash deposits of Sanpoil valley—be clocked now as a decade or less apart. Still farther downstream, Columbia River gorge choked the largest Missoula floods, passing peak discharge only one-third to one-half that released by the breached Lake Missoula ice dam.

GENERAL PROLOGUE
(*Waitt*)

We went north through the coulee, its perpendicular walls forming vistas like some grand old ruined, roofless hall, down which we traveled hour after hour.

—T.W. Symons (1882)

Grand Coulee is the great central element of Washington's Channeled Scabland. It was a focus of J Harlen Bretz's evidence in the 1920s for his hypothesis of an enormous Pleistocene "Spokane" flood. A famous controversy ensued for three decades. But by the late 1950s, Bretz's idea that Grand Coulee formed as tandem recessional-cataract gorges excavated under giant flood(s) had gained general acceptance.

This GSA field trip will focus on capacious upper Grand Coulee (Fig. 1). Even now it remains in ways enigmatic. Why did the coulee form in just this place and not some other? How did preglacial drainage influence flood-carved topography? Just how large are floodflows needed to cut so great a gash by cataract retreat? How many of them? When did the great cataract eat back into Columbia valley to open the coulee's north end?

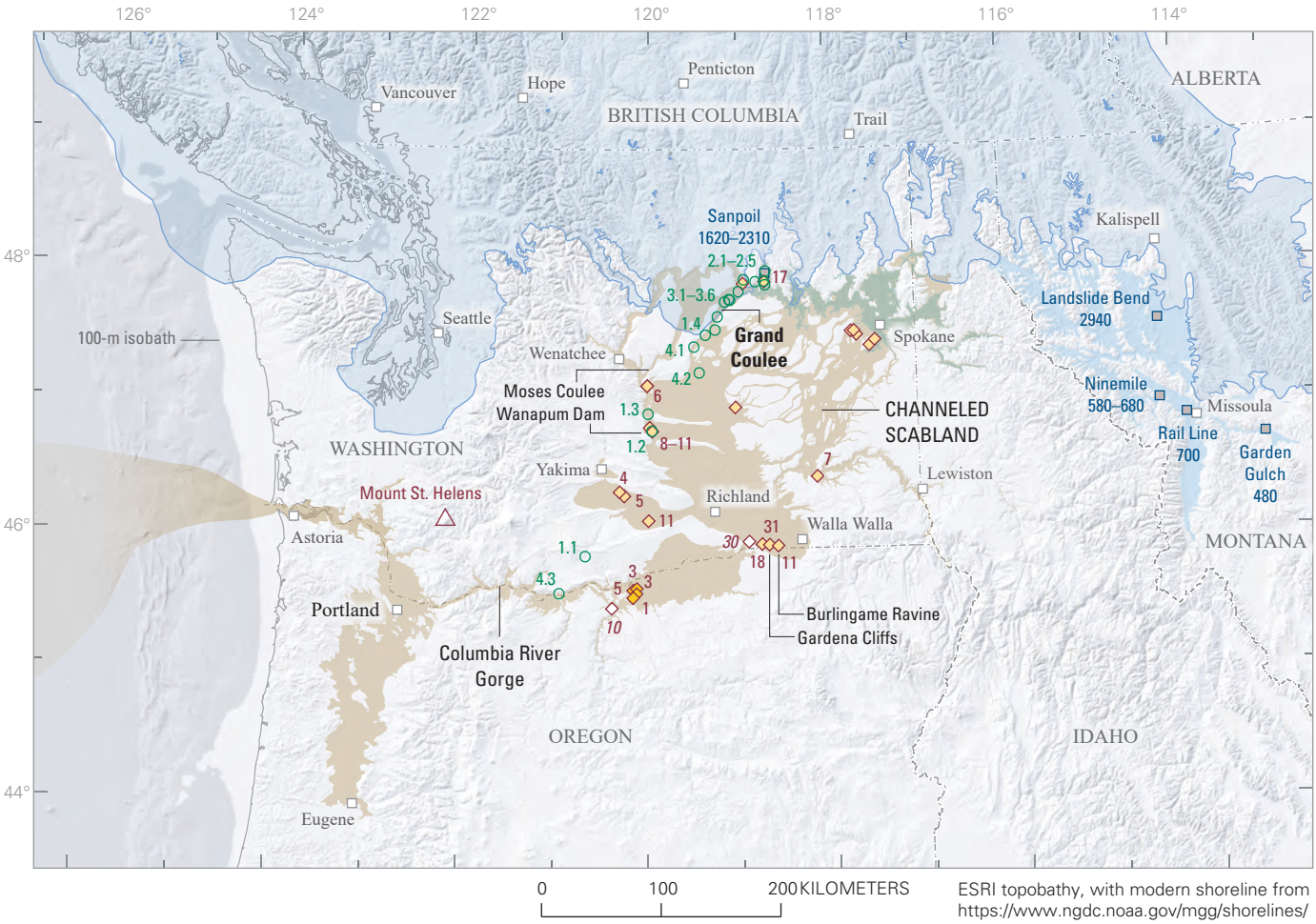
The trip's co-leaders have worked mostly separately or in pairs. They hold some different ideas about hydrologic processes and geologic history of upper Grand Coulee. Indeed, the field trip will explore field and modeling evidence on the coulee's development that may seem to conflict. Though better understood now than in Bretz's time, upper Grand Coulee retains no end of puzzles about Earth's largest freshwater floods and many of its lesser ones.

Trip Outline, Guidebook Contents, and Land Acknowledgment

Clues considered on this trip lie not just within upper Grand Coulee but farther upstream and downstream. Topics surveyed range from bedrock geology and preglacial landscapes, through glacial limits and flood routings, to computer simulations of largest Missoula floods.

The first day's geological stops begin far downstream of Grand Coulee with stratigraphic evidence for relatively small, late Missoula floods amidst scabland that had been carved by larger floods (Stop 1.2). Nearby cataracts and gravel bars attest to some of the largest floods (Stop 1.3). And no Grand Coulee trip can neglect the great cataracts of iconic Dry Falls (Stop 1.4).

The upstream stops provide panoramas of two parts of Columbia valley (Stops 2.1, 2.2) and afford hands-on inspection of Missoula-flood beds that alternate with varved deposits



EXPLANATION

Glacier, lake, or flood—At maximum extent during last glaciation, after approximately 20,000 cal yr BP. Adapted from O'Connor et al. (2020, Fig. 1)

- Cordilleran ice sheet
- Glacial Lake Missoula—1295 m
- Glacial Lake Columbia—715 m
- Area reached by ice-age floods—Submarine continuation, by hyperpycnal flows, evidenced offshore Astoria by plunge pools and giant ripples

1.2 ○ **Field trip stop**—Numbered by day and that day's sequence of stops. Example at left: second stop of first day

Ninemile 580–680 **Flood-era lake beds**—Number or range gives counts of fine-grained deposits that commonly contain rhythmic bedding. These deposits were interpreted as varved in reports on sections labeled Sanpoil (Atwater, 1986, pl. 3), Landslide Bend (Levish, 1997, p. 106, 111, 120–121), Ninemile (Chambers and Curry, 1989; Hanson et al., 2012), and Rail Line (Hanson et al., 2012, p. 72), but not at Garden Gulch (Smith, 2017, p. 120–121). In the Sanpoil section, includes only those varves intercalated with beds ascribed to Missoula floods by Atwater (1984, 1986), Hanson et al. (2015), and Hanson and Clague (2016). Sanpoil counts were reported as ranges. The two Ninemile totals are from independent observations 35 years apart. All counts have been rounded here to the nearest ten. The various totals provide clues to the lifespan of last-glacial Lake Missoula

◇ **Volcanic ash among ice-age flood deposits**—Assigned to Mount St. Helens set S, which has been dated at least in part to approximately 16,000 cal yr BP (Clynne, 2008, p. 619). Provides basis for stratigraphic correlation among late-glacial deposits north of, within, and south of the Channeled Scabland. Published sources: Waitt (1980; 1985, p. 1284), Bjornstad (1980), Baker and Bunker (1985, p. 30–31), Moody (1987, sites 1, 2, 9, 10, 13, 18, 20, 23, 25, 35, 36), Smith (1993, p. 82, 84), Kiver and Stradling (1995, p. 143), O'Connor and Waitt (1995, p. 103), Benito and O'Connor (2003, p. 631, 636), Clague et al. (2003), Hanson (2013, p. 206). Numbers plotted give counts of flood rhythmites observed or inferred to post-date tephra of set S in their vicinity:

11 ◇ Number of flood rhythmites observed above S tephra

30 ◇ Number of flood rhythmites inferred to post-date S tephra—Low-altitude exposure in which tephra layers were not found. Beneath terrace likely inset into deposits old enough to contain S tephra

Figure 1. Regional setting of Grand Coulee field trip stops (*Atwater*).

of glacial Lake Columbia (Stops 2.3–2.5) into which Missoula floods ran. The GSA group visits these five stops under a one-time permit from the Confederated Tribes of the Colville Reservation.

Consideration of the upper coulee itself begins at a viewpoint above its intake (Stop 3.1). A granitic hill near Steamboat Rock and a nearby gravel bar serve as stages for field geology and for discussing computer simulations of coulee-cutting floods (Stops 3.2, 3.3). An outcrop in the lee of Steamboat Rock displays evidence for relatively small, late floods (Stop 3.4). Day 3 concludes at an enigmatic large gravel bar above Dry Falls (Stop 3.5).

The return trip includes a brief roadside view of lower Grand Coulee and deposits of a postflood lake there (Stop 4.1), a longer stop among boulder fields of a dissected giant floodlaid fan 10 km downstream from the lower coulee (Stop 4.2), and an overview of Columbia Gorge east of Portland (Stop 4.3).

In the Grand Coulee area this trip ranges across historical lands of Nespelem, Sanpoil, and Sinkiuse (Moses-Columbia) peoples (Ruby and Brown, 1989; Ruby et al., 2010). Relocations brought additional groups to the Colville Indian Reservation established in 1872 (Johnson, undated).

Permission for any of the Day 2 stops should be obtained from the Confederated Tribes of the Colville Reservation.

Tectonic Setting

The oceanic Juan de Fuca Plate dives beneath the continental North American Plate along the Cascadia subduction zone.

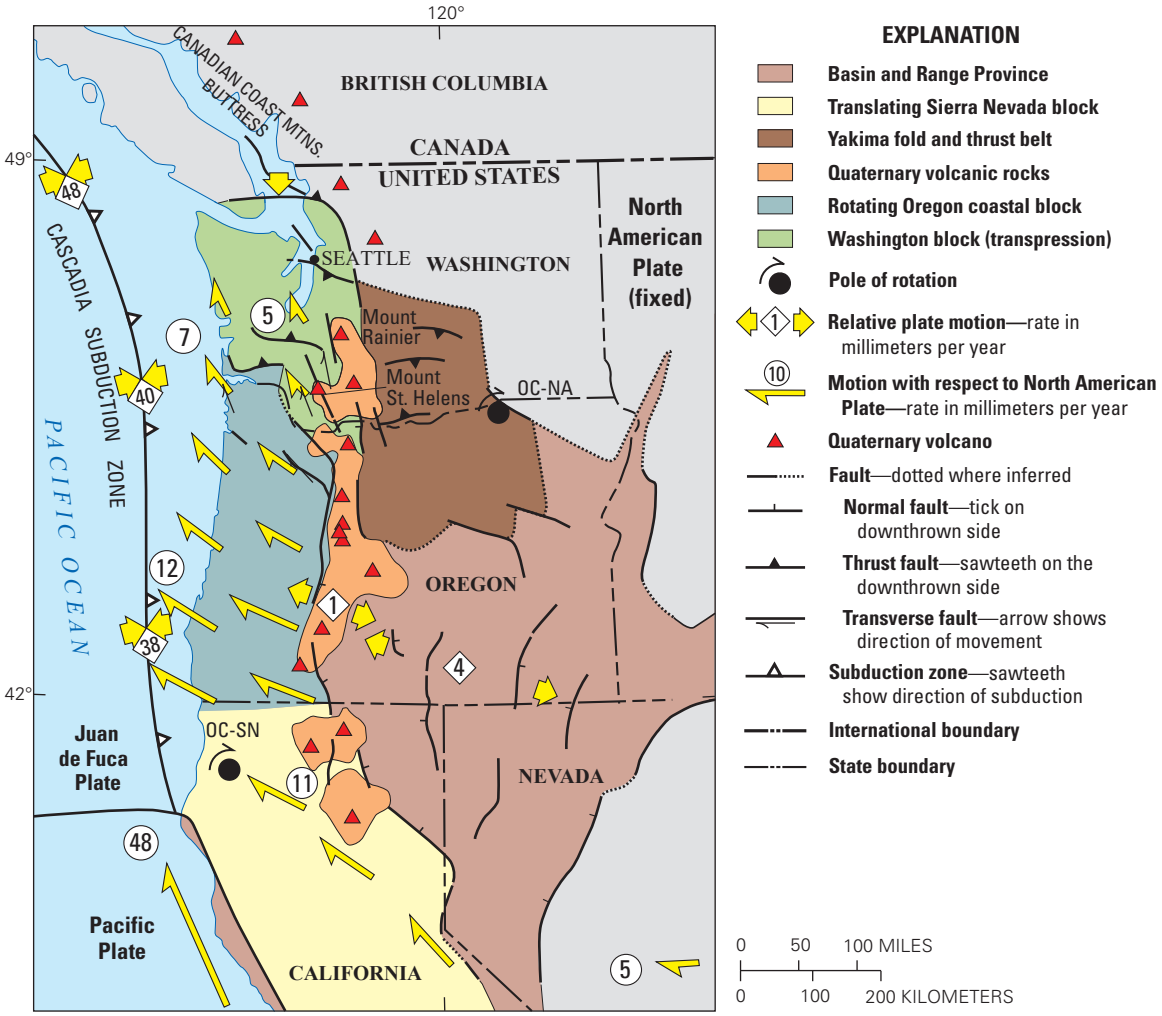


Figure 2. Bedrock setting. (A) Tectonic setting of Cascadia. The oceanic Juan de Fuca Plate dives beneath the continental North American Plate along the Cascadia subduction zone (white toothed). The migrating Cascadia forearc terrane is parceled into Sierra Nevada (SN), Oregon coastal (OC), and Washington blocks. Velocity of the tectonic blocks (yellow arrows) is calculated from a pole of rotation at point OC–NA (North America). The north end of the Oregon block squeezes the forearc area of Washington (green) against a buttress of crystalline rocks of the Canadian Coast Mountains. This north-south compression causes uplift and thrust faulting, spectacularly in the Yakima foldbelt. Orange areas are Quaternary volcanic rocks. Modified from Wells et al. (2002, fig. 1). (*Continued on facing page.*)

Southern parts of the migrating Cascadia forearc terrane squeezes the forearc area of Washington against a buttress of crystalline rocks of the Canadian Coast Mountains (Fig. 2A). This north-south compression causes uplift and thrust faulting—spectacularly so in the Yakima foldbelt that we cross on Days 1 and 4.

Grand Coulee and Environs

The northwest part of the Great Columbia Plain is underlain by the Miocene Columbia River basalt flows. The basalt overlies a high-relief surface that had been eroded upon Mesozoic to early Cenozoic granodiorite, gneiss, quartzite, meta-argillite, and other crystalline rocks (Stoffel et al., 1991). Columbia River formed in a natural gutter along the edge of the basalt field. Since the Miocene it has incised into these rocks as much as 720 m.

Columbia River Basalt

There would be no Channeled Scabland without the Columbia River basalt (Bretz, 1923, 1924, 1928a, 1959; Baker, 1978). The Miocene basalt field comprises more than 350 lava flows packaged by formal stratigraphic nomenclature into group, subgroups, formations and members with a bewildering array of names comprehensible by charts and tables (Reidel et al., 2013a, 2013b). This field trip about Missoula flooding needs only informal "Columbia River basalt" and a few names from higher parts of the basalt section of the areas we mostly visit (Fig. 2B).

Enormous water floods can quarry into the wide-jointed columnar parts of basaltic lava flows ("colonnade"). The more coherent close-jointed parts ("entablature") then fall into a flood and become its largest transported boulders. In such stripping along erosionally weak zones within basalt flows, floods erode a raw tiered landscape characteristic of the Channeled Scabland.

Megafloods and Sequence

J Harlen Bretz (1923, 1928a, 1928b) mapped a vast network of water-gouged steep-walled coulees cut into the basalt that diverge and rejoin. Within them lie plucked rock basins, giant cataracts, and great gravel bars. He inferred that a stupendous Pleistocene flood—source unknown—had shaped the Channeled Scabland (Fig. 1). Pardee (1942) then showed in high eddy bars and giant current ripples that enormous Pleistocene glacial Lake Missoula had discharged suddenly. Giant current dunes identified about the Channeled Scabland further proved immense flooding (Bretz et al., 1956; Bretz, 1959). Baker (1973) showed the Scabland landforms, however large, to be quantitatively consistent with open-channel hydraulics. The field trip explores the deepest scabland coulees, giant former cataracts, and a few high-energy gravel bars.

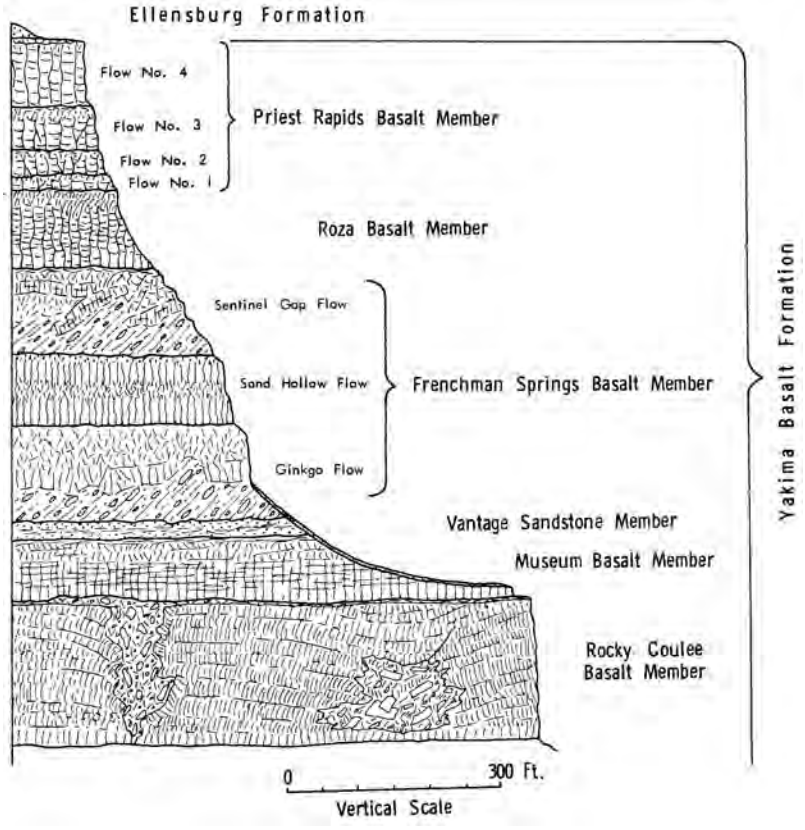


Figure 2 (*continued*). (B) Partial Columbia River basalt column portraying common internal structures (300 ft is 91.5 m). From Mackin, (1961, fig. 2). In more recent terminology (e.g., Reidel et al., 2013a), units below the Vantage Sandstone are members of the Grande Ronde Basalt, those above members of the Wanapum Basalt.

The term "coulee" comes from the Canadian French coulée, from the French "couler" meaning "to flow." In Washington, a coulee is a large, steep-walled trough, commonly an element of large flood channels cut into basalt of the Columbia plain or plateau.

Moraines register the late Wisconsin ice margin west of Grand Coulee. But with no clear moraine east of the coulee, Bretz (1923, 1928a, 1959) and Flint (1936, 1937) inferred bouldery debris in uplands south of Spokane to be drift (Fig. 3). These limits were later revised in stages to 25–70 km farther north (Weis and Richmond, 1965; Waitt and Thorson, 1983; Carrara et al., 1996; Waitt et al., 2016; O'Connor et al., 2020). Most Missoula-flood deposits proved to be late Wisconsin age (chronology below).

Within tall stacks of floodlaid graded beds in southern Washington, intercalated loess and volcanic ash imply subaerial episodes between late Wisconsin flood beds. Each of at least a few dozen graded beds seemed laid by a separate flood (Waitt, 1980). The field trip visits one such site.

Glacial lakes in northern valleys had accumulated hundreds of annual varves. Coarse graded beds laid by huge floods repeatedly interrupt such varve beds of glacial Priest Lake and glacial Lake Columbia. The varves confirmed that *separate Missoula floods each registered in distal settings as a single graded bed* (Waitt, 1984; Atwater, 1984). Field trip Day 2 explores such stratigraphy. Many stratigraphic sites about the Scabland reveal scores of last-glacial Missoula megafloods (Waitt, 1985, 1994; Atwater, 1986, 1987; Benito and O'Connor, 2003; Hanson et al., 2015; O'Connor et al., 2020).

Sanpoil valley's flaring mouth received every Missoula flood running down the Columbia. Most of the time an arm of glacial Lake Columbia accumulated varved bottom sediment there. Counting these varves between the coarse beds counts, with some uncertainty, the years between Missoula floods. Early Missoula floods came several decades apart, middle ones a few decades apart, late ones only years apart (Atwater, 1986, 1987). At least one giant Missoula flood came before the Okanogan lobe dammed glacial Lake Columbia (Waitt, 1994, 2016; Waitt et al., 2019). By this reckoning the late Wisconsin Missoula floods sum to 90 or more. The field trip explores parts of this rich stratigraphic record.

Modern ice-dammed lakes in Iceland, Alaska, and British Columbia have slowly filled and abruptly self-dumped—repeated floods called jökulhlaups. Here's how giant Pleistocene glacial Lake Missoula must have behaved: slowly filling, then abruptly spilling through glacier-bed tunnels—again and again and again (Waitt, 1980, 1985; Clarke et al., 1984).

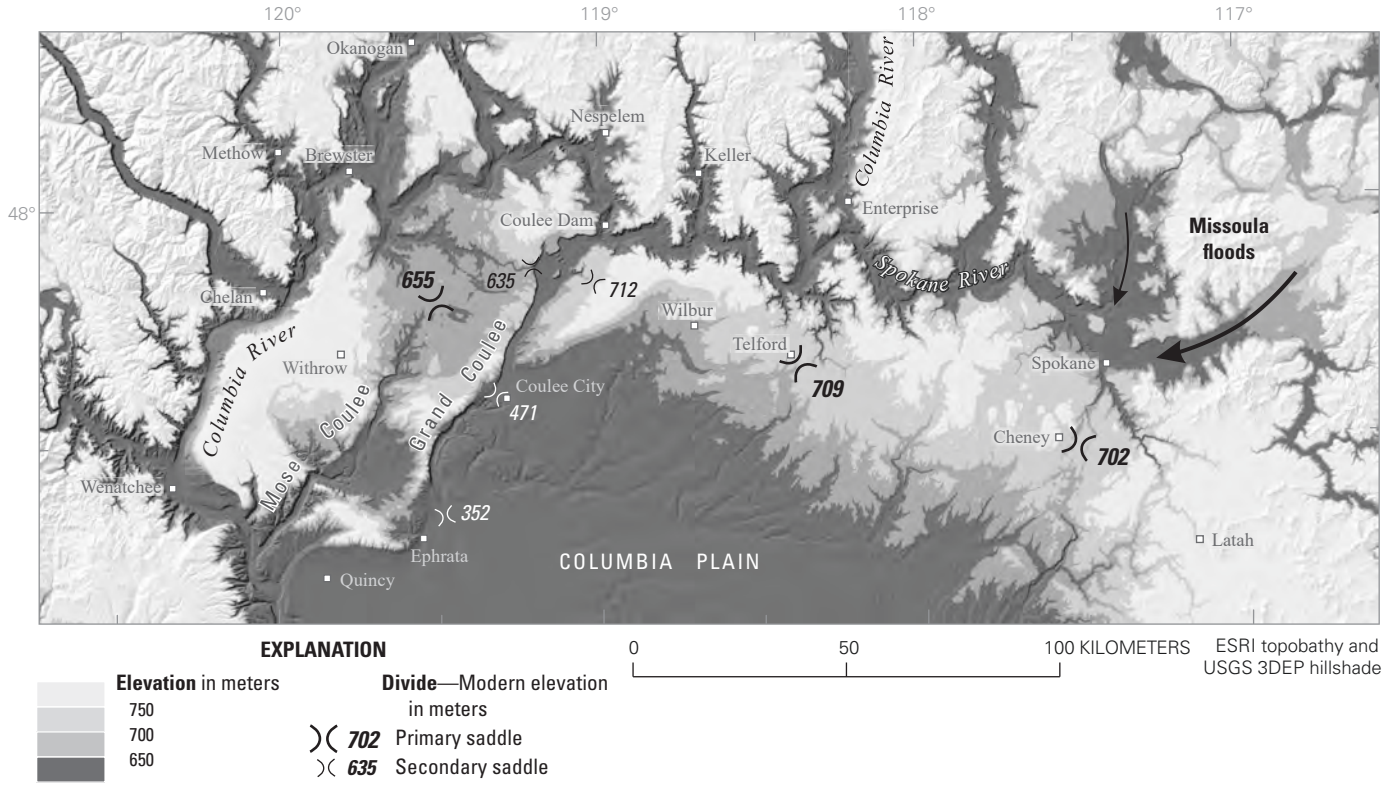


Figure 3. Drainage divide between Spokane–Columbia River drainage and the Columbia Plain (Waitt, 2021, fig. 2C) on a base map with selected altitude ranges shaded. When the Okanogan lobe of the Cordilleran icesheet dams Columbia valley, glacial Lake Columbia and diverted Columbia River spills south across one of the northern saddle divides. Missoula megafloods from the east channeled down Spokane-Columbia valley overflow south at one or several divide saddles, depending on topography developed by then and on extent of the Okanogan lobe then (*Atwater and Waitt*).

Divide South of Spokane and Columbia Valleys

South of Spokane and Columbia valleys a 230-km-long drainage divide lies 200–500 m above river grade (Fig. 3). When Okanogan ice dams a glacial Lake Columbia, its outlet—diverted Columbia River—spills across the lowest saddle on this divide. When a great flood engulfs glacial Lake Columbia, water pours over the divide at more than one saddle. One high saddle eventually eroded down into the deepest spillway across the long divide: upper Grand Coulee (Bretz, 1932). Farther-east floodways lie near Cheney and near Telford. This overall geography is vital to what follows and important when on the field trip Day 3 we consider advance and retreat of late Wisconsin Cordilleran ice in upper Grand Coulee.

Upper Grand Coulee and Glacial Lake Columbia

Preglacial and Early Glacial Landscape

Upper Grand Coulee lies between the Coulee monocline and anticline on the east and a gentle ridge (a low anticline?) mostly above 720 m on the west (Waitt, 2021). A structural trough lies between these two ridges. East-side draws—Klobuschar, Devils, Rusho, Paynes, others—descend regularly west till plunging 200–250 m over the walls of the later-cut coulee (Fig. 4). These drainages likely fed a high-level preflood trunk stream draining southwest where upper Grand Coulee later formed. (The field trip will also entertain a different preflood hypothetical paleogeography.)

Just below Grand Coulee's intake, Columbia valley turns sharply north, its lower 500 m of valley tightly constricted (the Grand Coulee damsite). Floodwater hydraulically ponds upstream of a valley narrows (Baker, 1973; O'Connor, 1993; Benito and O'Connor, 2003). Flume experiments and hydraulic modeling show that a large flood also ponds above a sharp valley bend (e.g., Frazão and Zech, 2002; Denlinger and O'Connell, 2010). The narrows and sharp bend in Columbia valley *together* pond megaflood water upvalley. This hydraulic ponding aids flood overflow into the high "Coulee creek" vale just upvalley (Fig. 4). Floodwater down this vale feeds the two incipient giant cataracts—near Hartline basin and near Soap Lake. Such flooding through the high-level vale could also spill west through a high-level saddle into Foster and Horse Lake Coulees—toward Moses Coulee.

Upper Grand Coulee apparently began on the limb of the Coulee monocline near Hartline basin as a flood cataract. It then receded northeast to form an immense deep gash (Bretz, 1932; Bretz et al., 1956). When the cataract finally intersected Columbia valley, it opened the coulee's north end.

Glacial Lake Columbia at Low Level

Late Wisconsin Cordilleran ice flowing down Okanogan valley blocked the northwest part of Columbia valley, damming glacial Lake Columbia and diverting the river down Grand Coulee. Glacial features of the Waterville plateau have been mapped in some detail, aided by vertical aerial photos (Fig. 5) (Hanson, 1970, fig. 19-2). As ice on the Waterville Plateau receded north, outwash streams flowing southeast along channels like Foster Coulee built small deltas into a late-stage glacial Lake Columbia in upper Grand Coulee.

Varved silt topped by only feeble soil in upper Grand Coulee and adjacent Columbia valley record a sustained late phase of glacial Lake Columbia that drained down Grand Coulee over the 471 m rock threshold at Coulee City (Bretz, 1932; Flint, 1935, 1936; Flint and Irwin, 1939; Atwater, 1987). This route having been also Columbia River, water flowed many meters deep over the river bed. From a water *surface* thus at or above 475 m at Coulee City, the lake plane as approximated by delta tops below Foster Coulee and Barker Canyon, and minimally by lacustrine silt, rises upcoulee to 485 m near Paynes Gulch to 506 m near Steamboat Rock to 509 m at the coulee head and to 512–530 m in Columbia valley. During the field trip we have a stunning view of this "Nespelem" terrace (Stop 2.1).

The gentle northeast uptilt of a former water surface must be by rebound from late Wisconsin time when the weight of Cordilleran ice and glacial Lake Columbia had isostatically depressed the upcoulee reaches. Such a half-meter or so per kilometer northeastward postglacial uptilt is two-thirds what Thorson (1989) documents for postglacial isostatic uptilt in Puget Sound.

Glacial Lake Columbia at High Level

Recent field evidence by Jim O'Connor in Columbia valley—fresh erratics and till topped by only thin brown soil—shows the late Wisconsin Columbia River ice lobe reached downvalley to well below Spokane valley (O'Connor et al., 2020, fig. 4; Waitt, 2021).

Measured sections and paleocurrent data in Sanpoil valley record beds of silty varves of glacial Lake Columbia repeatedly interrupted by coarse beds laid by Missoula floods (Atwater, 1986). Further stratigraphy and paleocurrent data there and in Columbia valley (Hanson et al., 2015; Hanson and Clague, 2016) sustain and elaborate that megafloods flowed from glacial Lake Missoula westward down Spokane-Columbia valley, they numbered in the dozens, and early big ones decades apart gave way to more frequent smaller ones. Day 2 of the field trip explores the field evidence.

Atwater (1986, 1987) infers a brief high stand of glacial Lake Columbia at ~715 m draining through an eastern spillway. Faint shorelines and patches of washed beach gravel at ~715 m altitude on Mount Tolman record such a high-level glacial Lake Columbia (Atwater, 1986, p. 6–7). A lake standing many centuries or millennia should leave voluminous and widespread deposits. But such deposits around of the lake are meager, none of them bulky. This sparseness shows that the lake stood at the high level only briefly, and it complements stratigraphic evidence that the lake stood at the high level for a few centuries at most (Stop 2.3).

Diverted Columbia River

Tracing the 715 m contour reveals glacial Lake Columbia's possible outlets—routes of diverted Columbia River over

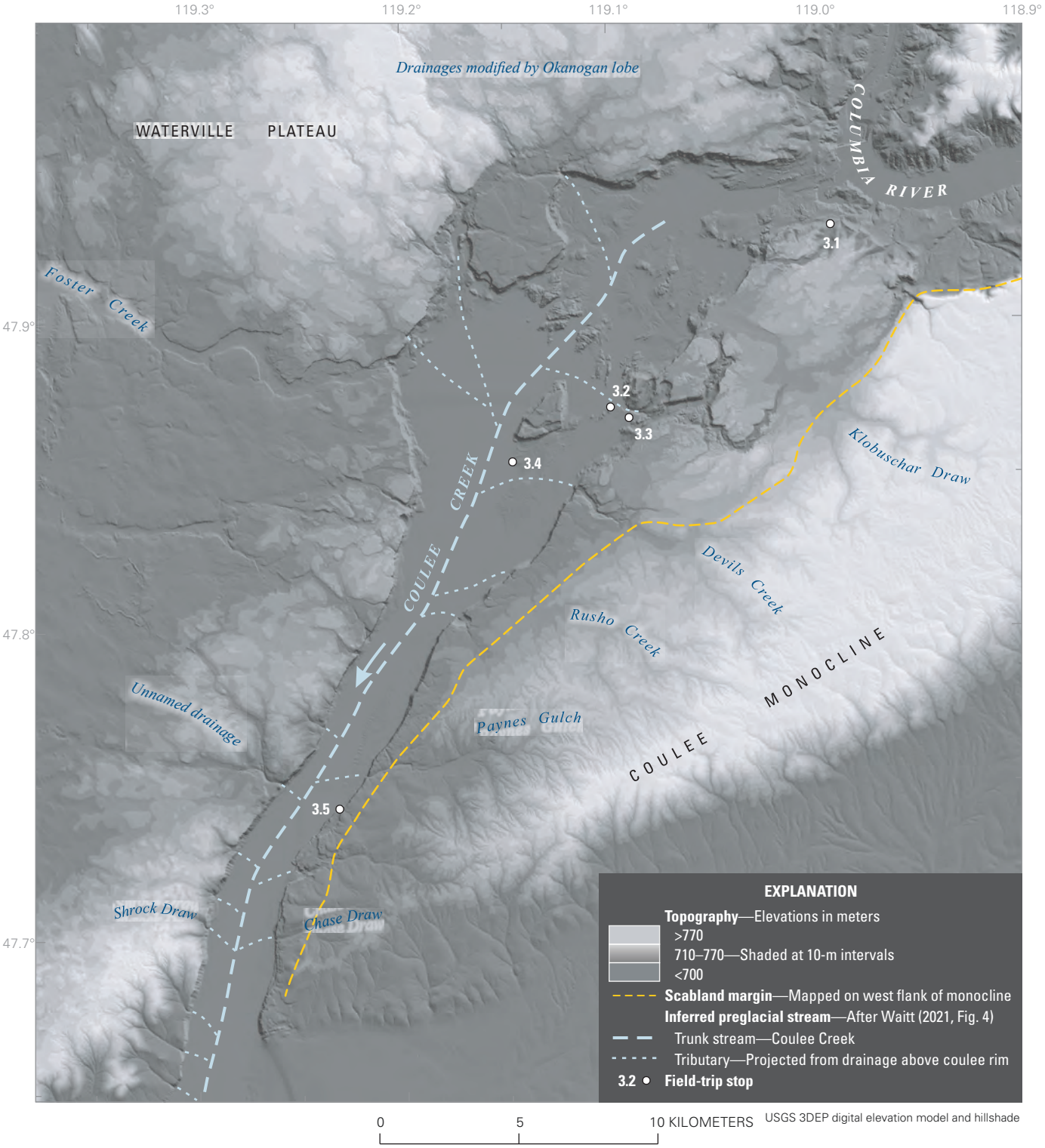


Figure 4. Preglacial drainage through inferred preglacial ("Coulee creek," dashed line) along later axis of upper Grand Coulee (Waitt, 2021, fig. 4). Dashed yellow line depicts approximate limit of upper east scabland tract. The scabland saddle on the west shows how, before upper Grand Coulee was excavated, Missoula floodwater utilizing the high-level "Coulee creek" vale overflowed west into Foster and Horse Lake Coulees, which from there overflowed south into Moses Coulee (off this map) (*Atwater and Waitt*).

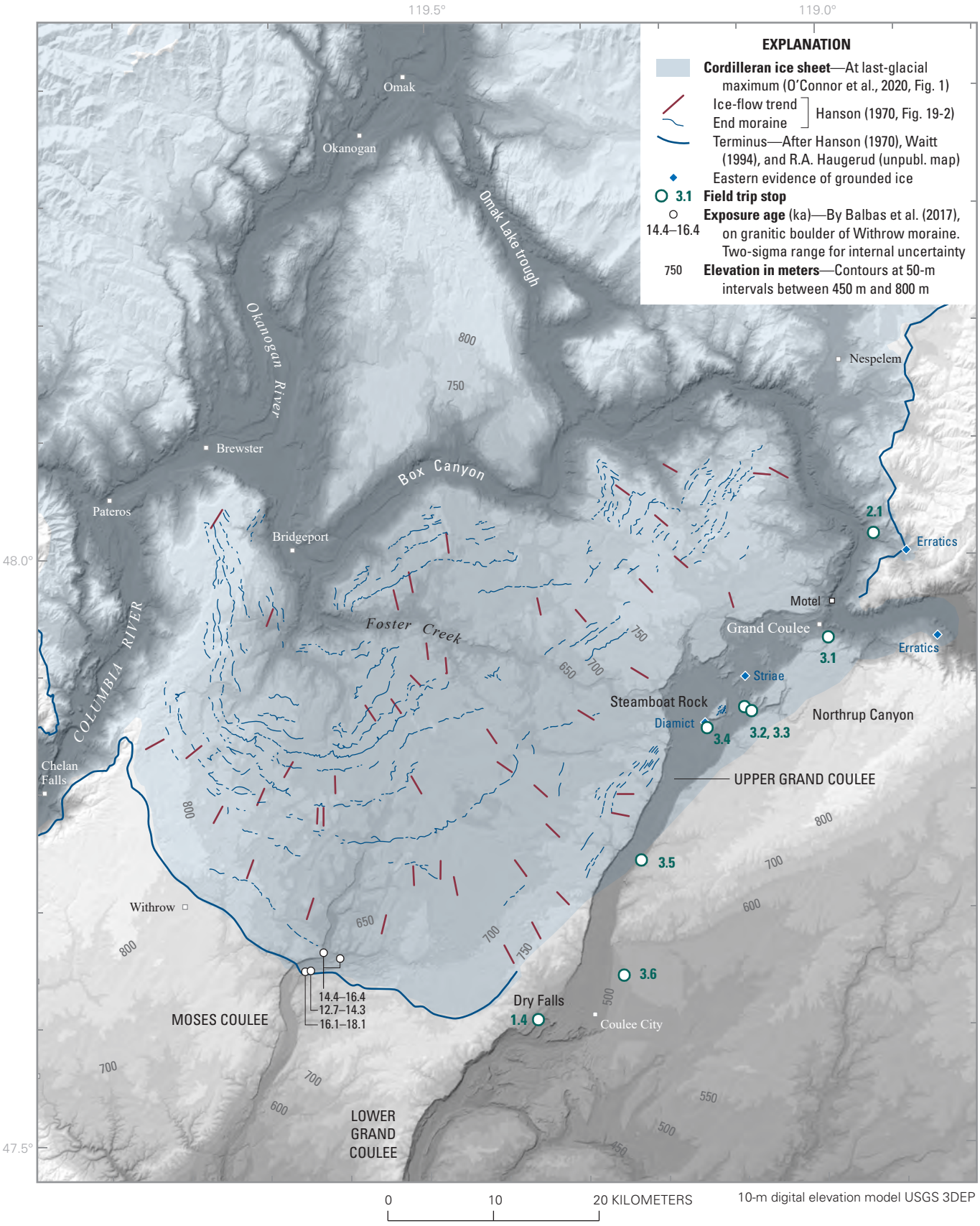


Figure 5. Glacial features on Waterville Plateau including a summary of Hanson's (1970, fig. 19-2) geomorphic map, ice margin to the west as by Waitt (1994) and to the northeast as by Ralph Haugerud (unpublished) (*Atwater*).

the south divide: a broad saddle east of Cheney, a narrow one at the head of Hawk Creek near Telford, and a saddle along the east scabland of upper Grand Coulee (Fig. 3). Digital terrain analysis utilizing U.S. Geological Survey (USGS) topography approximates the divide saddles as: Cheney at 702 m, Telford at 709 m, and complicated Grand Coulee upper east scabland at ~712 m or perhaps higher. But restoring 30 m of postglacial isostatic rebound in upper Grand Coulee to late Wisconsin levels lowers that outlet to or below the eastern ones. All three saddles are scablands 5 m or so in relief. The 702 m level near Cheney is only a low thread winding through scabland—a detail of the *bed* of Columbia River that flowed several meters deep. Atwater's (1986) 715 m records a lake *surface* held dynamically by water flowing deeply over one of these ragged beds.

Late Wisconsin Deposits and Landforms

The Cordilleran-icesheet limit descends Okanogan and Methow valleys and down Columbia valley to Chelan Falls (Figs. 1, 5), where moraines and a head of outwash mark the maximum late Wisconsin stand (Waters, 1933; Flint, 1935). Deposits of glacial drift and megafloods have been mapped far down the Columbia (Waitt, 2016, 2021; Waitt et al., 2019).

Far to the east, the Purcell Trench lobe of Cordilleran ice held in glacial Lake Missoula to levels as high as 1295 m (Pardee, 1942; and highest shoreline as at 46.8708°N, 113.9651°W). At that level, lake volume is as much as 2500 km^3. The hydraulically unstable Purcell Trench ice dam repeatedly released immense jökulhlaups (Waitt, 1985)—the largest at peak discharge above 15 million m^3/s (O'Connor and Baker, 1992; Denlinger and O'Connell, 2010; O'Connor et al., 2020).

Radiocarbon ages broadly limit the late Wisconsin floods to ~19,000–15,500 calibrated years ago (19.0–15.5 ka) (Waitt, 2016). Cosmogenic ^{10}Be and ^{26}Al exposure ages may peg the range slightly younger (Balbas et al., 2017; Gombiner et al., 2017; O'Connor et al., 2020). Different late Wisconsin floods followed different routes steered by changing Okanogan-ice margins and thus levels of glacial Lake Columbia.

Columbia Megaflood

Ice-rafted granitic erratics on the sides of Columbia valley near Wenatchee, some ^{10}Be dated to 18.2 ± 1.5 ka (Balbas et al., 2017), register a late Wisconsin flood as high as 320 m above Columbia River in a valley reach little if at all incised since the late Wisconsin (Waitt, 1982). Giant current dunes on the surface of huge Pangborn bar near Wenatchee and downvalley-dipping foreset beds show the bar grew under immense flood(s). Such deposits disappear upvalley beneath glacial moraine and outwash near Chelan. So, the largest late Wisconsin flood(s) down Columbia's "big bend" came before Okanogan ice blocked it (Fig. 6B) (Waitt, 1994, 2016, 2017; Waitt et al., 2019). Though the Columbia mainstem is the low route to discharge flooding from the east, only one monstrous late Wisconsin flood is proven down this reach.

Once advancing Okanogan ice blocked Columbia valley south of Brewster, Missoula floodwaters from the east diverted into not only the Cheney-Palouse and Grand Coulee but at times across the divide from Foster Creek into Moses Coulee (Fig. 6C). At Moses Coulee's mouth they built a giant gravel bar into Columbia valley. Four floodlaid basalt-gravel beds each overlain by slackwater silt (Waitt, 1982, 1985, 2016; Waitt et al., 2019) record at least four separate late Wisconsin giant floods down Moses Coulee that dammed Columbia River. This field trip cannot explore these western routes, but some of this history informs discussions.

Further advance of Okanogan ice blocked the intakes to Moses Coulee, and the westmost floodway became Grand Coulee (Fig. 6D). This seems the dominant geographic setting during the late Wisconsin. With Columbia valley blocked below Grand Coulee, scores of megafloods poured down the Cheney-Palouse tract and Grand Coulee, many also down the higher less-floodable Telford tract. This setting is explored at several field trip stops.

Large Missoula floods via Grand Coulee entered Quincy basin from the north (Fig. 6D). High gravel bars in northern Quincy basin descend and fine south and west to sand. Floods carved Quincy basin's natural south outlet to 16-km-wide Drumheller Channel's wild scabland. Though spilling more than 100 m deep through this broad gap, floodwater hydraulically ponded, backflooding Quincy basin. It eventually spilled from the basin west across Babcock-Evergreen ridge at three saddles into Columbia valley, cutting huge ragged cataracts (Bretz, 1959; Bretz et al., 1956). The field trip visits one of these (Stop 1.3).

During deglaciation, the thinning Purcell Trench lobe dammed ever-smaller glacial Lakes Missoula, and floods down the Cheney-Palouse and Grand Coulee became smaller and smaller (Waitt, 1985; Atwater, 1986, 1987). Uninterrupted varves above the topmost flood bed in Sanpoil valley and upper Grand Coulee show that after the last Missoula flood, glacial Lake Columbia persisted two centuries or more (Atwater, 1987). (Stops 2.5 and 3.4.) Eventually, in Waitt's (1994, 2016) view, this glacial Lake Columbia too released a deep flood down the Columbia.

Glacier and Floods in Upper Grand Coulee

Fresh moraines lie in uplands northeast and southwest of upper Grand Coulee, till overlies glacial-lake deposits at Grand Coulee Dam (Flint and Irwin, 1939), and fresh glacial striae and drift including erratics and moraines lies atop midcoulee Steamboat Rock (Bretz, 1932; Crosby and Carson, 1999). At its maximum late Wisconsin stand, the Okanogan lobe seems to have filled the coulee. Yet Bretz (1932) found only vague glacial striae on the "crumbly rock" of the granitic knolls in north parts of the coulee. Bruce Bjornstad, kayaking on Banks Lake in 2011, recorded fresh glacial striae 2.7 km northeast of Steamboat Rock (Fig. 7A). Much of the coulee floor had been mantled by late-glacial Lake Columbia bedded silt covering underlying rock from weathering. With Banks Lake's filling in 1951, silt has washed from parts of its shores, exposing once-covered rock

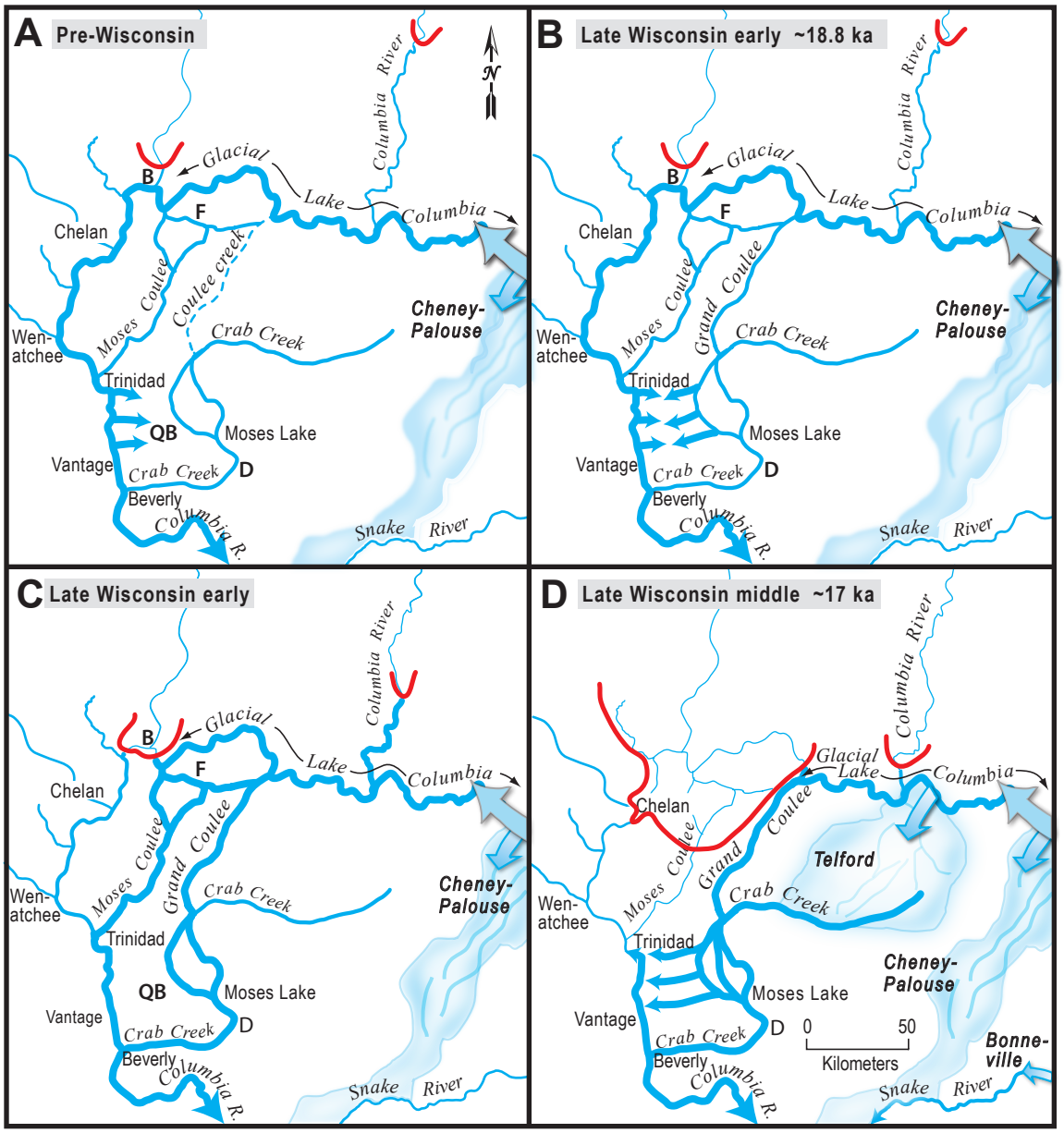


Figure 6. Map panels portraying schematic Missoula-floods routings at four times in four different geographic patterns. Red, hypothetical Cordilleran-ice limits at different times. D—Drumheller Channels; B—Brewster; F—Foster Creek; QB—Quincy basin. (A) Dashed line for future Grand Coulee, which at this stage is still largely blocked by the cataract in its head. From Waitt (2021, fig. 8).

and revealing lodgement till as well (Atwater, 1987) (Stop 3.4). The striated rock surface is clear evidence that late Wisconsin ice occupied this northeast reach of coulee to its east side.

Most late Wisconsin Missoula floods funneled down upper Grand Coulee. Some of them were gigantic. Many floods must have passed while the glacier receded west across the coulee. Some of them piled high-energy gravel bars into the mouth of Northrup canyon (Waitt, 1994). Such floods deposited coarse gravel into northern Quincy basin (Bretz et al., 1956; Baker, 1973), including granodiorite boulders derived from upper Grand Coulee ^{10}Be date ca. 15.5 ka (Balbas et al., 2017; Gombiner et al., 2017).

Atwater's (1986, fig. 17) varve-and-floodbed stratigraphy in Sanpoil valley suggest that Missoula floods arrived every five decades while glacial Lake Columbia stood near 715 m. The coulee's upper-east scabland as wide as 5.6 km (Fig. 4) must owe to such broad-spreading floods—for the natural width of Columbia River itself is only 300 m. Though Cordilleran ice filled northeast parts of the coulee it may most of the time have left open the coulee's east side below Steamboat Rock. Were lower Northrup

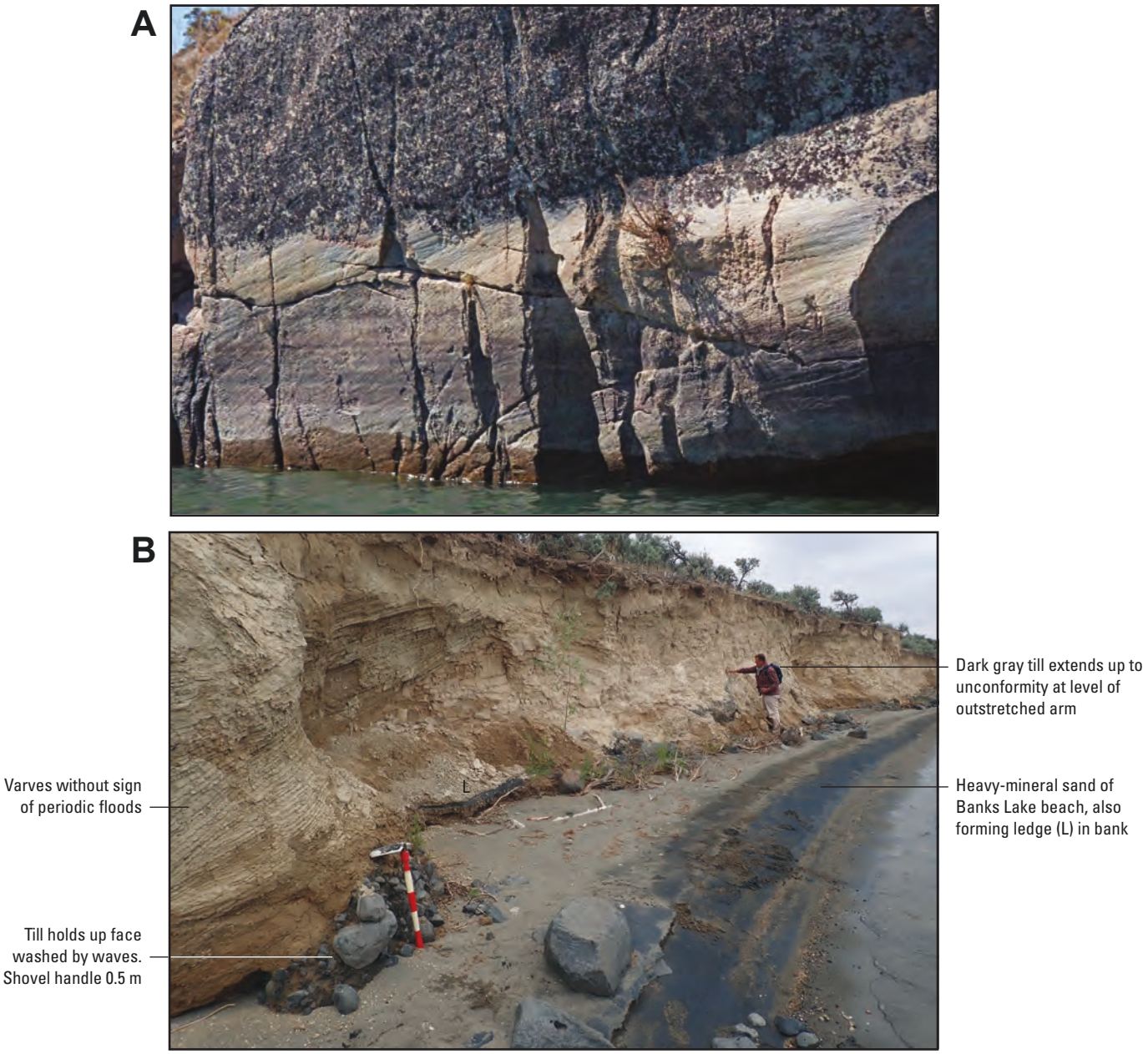


Figure 7. Late Wisconsin glacial features in upper Grand Coulee. (A) Striated rock surface along Banks Lake in upper Grand Coulee, August 2011. Above the pristine part of surface, striae and grooves also mark the lichen-covered area above normal waterline, to top of exposure. On the left, the visible part of this surface is ~4 m high (*Bjornstad*). (B) At part of compound section at Stop 3.4 southwest of Steamboat rock, till overlain by varved bottom sediments of a late phase of glacial lake Columbia (*Atwater*).

Canyon thus unblocked (Waitt, 2021, fig. 1A), water down the east scabland would drop 220 m through Northrup. It thus would function as a cataract.

From flood-plucked granodiorite surfaces in Northrup canyon, ^{10}Be exposure ages average 15.6 ± 1.3 ka (Balbas et al., 2017), which these investigators contend dates the main excavation of upper Grand Coulee. Yet when late Wisconsin ice filled the upper coulee, many large Missoula floods down the east scabland must have enlarged Northrup cataract. The ^{10}Be ages may date latest Missoula flooding in Northrup canyon but not the probably earlier excavation of upper Grand Coulee.

Hydraulic Modeling

Hydrologic modeling of Missoula floods helps quantify what decades of accumulated field evidence has revealed.

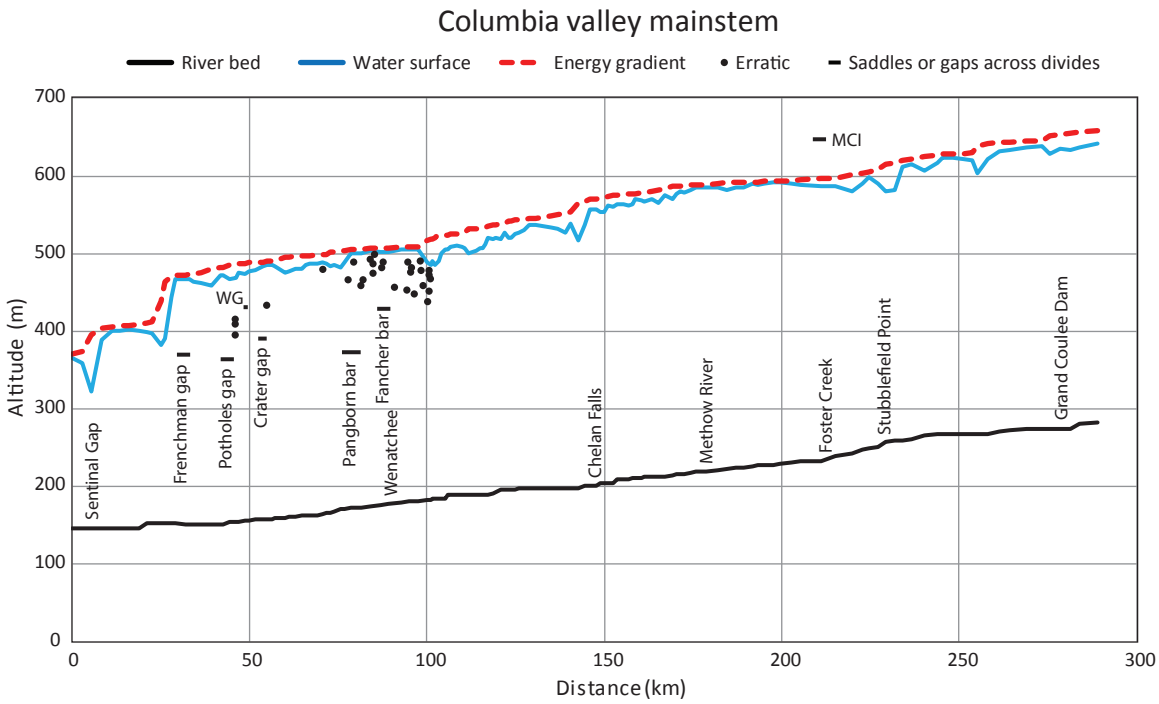


Figure 8. Plot of 1D HEC-RAS model of Columbia valley from above Grand Coulee down to edge of Pasco basin through 153 cross sections for hypothetical discharge 13 million m³/s. MCI—lowest saddle of Moses Coulee intakes; WG—Weber gulch. Cluster of high erratics near Wenatchee are from Waitt et al. (2019). Hypothetical discharge figure is 25% or so too high because roughness value was inadvertently much too low. Profile is close to Chris Harpel's 1996 model run with better roughness value. From Waitt (2021, fig. 15).

1-D Step-Backwater

Peak Missoula-flood profiles down individual channels may be estimated by the U.S. Army Corps of Engineers HEC-RAS (earlier, HEC-2) program, a one-dimensional routine run through measured channel cross sections to find calculated water levels that match field evidence of high water such as highest ice-rafted erratics. O'Connor (1993) summarizes theory, method, and rationale.

Through Rathdrum valley—the main conduit of floods from glacial Lake Missoula—peak flow at levels of highest field evidence passes ~17 million m³/s (O'Connor and Baker, 1992). Peak flow through Wallula Gap and narrows in Columbia Gorge model to ~10 million m³/s (O'Connor and Baker, 1992; Benito and O'Connor, 2003). Mid-1990s Modeling by C.J. Harpel (scarcely published) has helped understand peak floodflow profiles down a few other large Scabland channels (Waitt et al., 2009).

In 2017, a 291 km reach of Columbia valley from above Grand Coulee to below Quincy basin through Harpel's cross sections was run to generate a surface level to fit late Wisconsin field evidence such as ice-rafted erratics near Wenatchee up to 495 m altitude (Fig. 8). Upvalley the profile passes Foster Creek at ~560 m altitude. The lowest spot of the divide saddle just south toward Moses Coulee lies nearly 100 m higher. In this geographic setting (Columbia valley not ice-blocked), even a maximal Missoula flood cannot overflow into Moses Coulee.

2-D Dambreak

2-D dambreak modeling of the whole Channeled Scabland (Denlinger and O'Connell, 2010) is superseding 1-D modeling of individual channels. Different patterns of ice margins and ponding make different flood-routing patterns (Waitt, 1994; Waitt et al., 2009). New 2-D hydraulic models test six such geographic settings (O'Connor et al., 2020; Denlinger et al., 2021). The head of Grand Coulee may be modeled open as now, or largely shut when the former rock cataract still stood. Different positions of the Okanogan ice lobe either dam or leave open upper Grand Coulee, Moses Coulee, and the mainstem Columbia (Fig. 9). Such changing geographic conditions greatly influence how much of a large Missoula flood pours into Grand Coulee and other distributaries.

Six recent 2-D model runs (Denlinger et al., 2021, figs. 6–8) in different ways roughly mimic patterns of flooding in the Channeled Scabland as visible in satellite imagery and determined across decades of field mapping and recently LiDAR imaging. But southward through the flooded system, peak flood levels by any of the model scenarios fall short by 10–50 m from peak-flood limits provable by field evidence (Denlinger

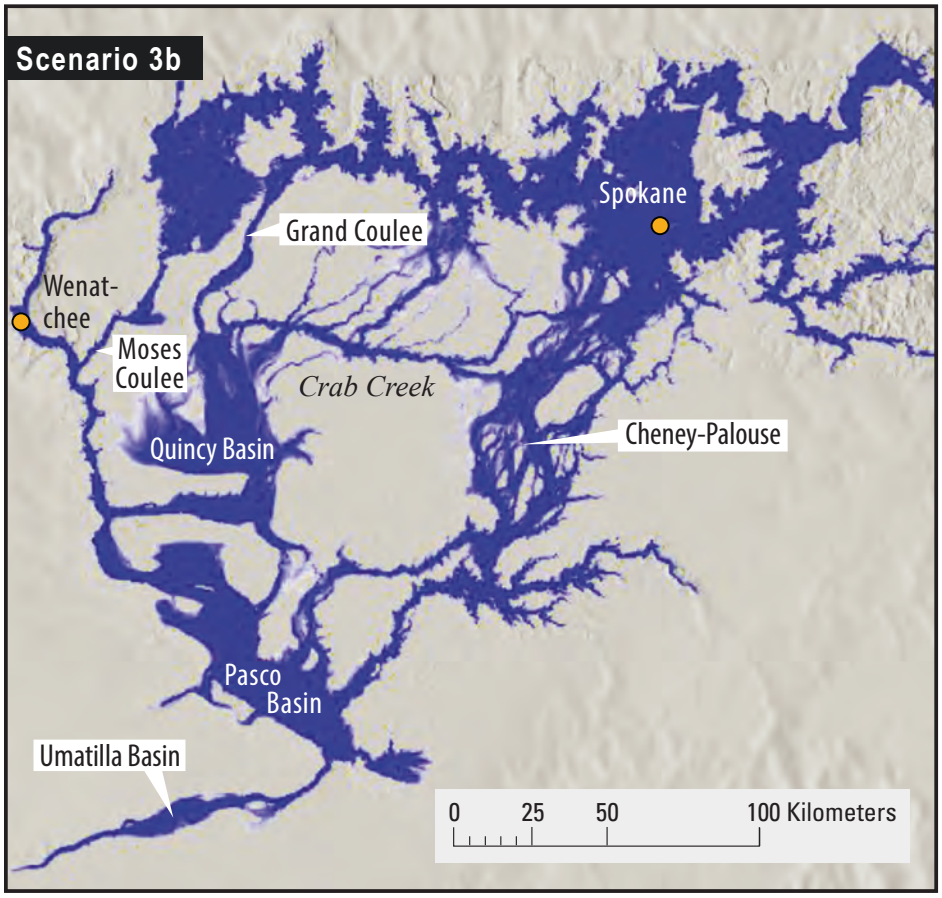


Figure 9. 2-D model of Scenario 3b at 23 h after dambreak. (From Denlinger et al., 2021, fig 7D). Model instantly releases the 2500 km^3 glacial Lake Missoula (at and beyond northeast map margin), which flows over a digitized model of present-day topography. Reconstructed lobes of the Cordilleran ice sheet block valleys on the north. In this scenario, Columbia valley is ice-blocked at mouth of Okanogan valley but Foster valley and thus route to Moses Coulee unblocked. Water that would flow down Columbia valley were it not blocked instead floods into Moses Coulee, Grand Coulee, and low tracts farther east.

et al., 2021, fig. 9). The models so far use only the water of the initial volume of a maximal glacial Lake Missoula. A test by (Denlinger et al., 2021, fig. 10) that incorporates loess from uplands boosts downflood flow volumes only in eastern parts of the system. Other factors being now model-tested would boost downflood volumes farther west and south. For instance, large bulking of flow by sediment incorporated in floods along the main valleys—from Rathdrum valley, from partly sediment-filled glacial Lake Columbia—adding overall volume to flood-flow through downflood parts of the system. A large Missoula flood triggering simultaneous failure of glacial Lake Columbia would add more overall flood volume in water and sediment down Columbia valley.

Aerial Viewing

Many erosional and depositional features of the Channeled Scabland are so large they can be comprehended well only from the air. Detailed modern digital-elevation-model maps do help portray extraordinarily large features. A new book by Bruce Bjornstad (2021) features scores of photographs of scabland features, including of Grand Coulee, shot from an aerial drone that reveal the Channeled Scabland as never before.

GRAND COULEE FIELD TRIP STOPS

Day 1 (Much of This Day is Travel)

Grand Coulee lies in the upper Columbia River drainage in north-central Washington well east of the Cascade Range. From Portland, Oregon, USA, it is about a 340 mile drive to our motel in Coulee Dam near the coulee's northeast end (Fig. 1). En route to the first stop, we pass through Portland basin and cross the Cascade Range through Columbia Gorge. The route then climbs over Columbia Hills anticline and onto Horse Heaven Hills anticline—both formed in Miocene Columbia River basalt. This high area is capped by the Pliocene-Pleistocene Simcoe basaltic volcanic field.

Mileages are pieced together by various methods of variable accuracy, in places only approximate. Trip-odometer mileage is reset one or more times during some days.

Mile	This leg of journey approximate miles by I-84 mileposts.
–1	Portland, Oregon Convention Center, north side.
0	Onto Interstate 84 east. We ride upcurrent on Missoula-floods bars. Light-rail construction

	in the 1980s exposed sandy rhythmites near Portland but farther east basalt cobble to pebble gravel foreset toward the west and southwest. Miles farther east lie flood-moved boulders up to a few meters diameter.
5–8	Just north lies Rocky Butte, an early Pleistocene basaltic volcano. It stood as a high to Missoula floods, which swept out a huge scour depression around it that I-84 and I-205 follow.
8	Cross I-205.
9–15	Missoula-floods bars generally coarsen to boulders upcurrent toward Columbia Gorge.
22	Entering lower end of Columbia Gorge.
24	Crown Point, above on south, a focal point on the historic (1915) Columbia River highway. The largest Missoula floods overran it. For more detail along the Columbia Gorge about Columbia River basalt, Missoula-floods effects, and postflood landslides, see O'Connor et al. (2021, this volume).
32	Multnomah Falls on south.
40	Bonneville Dam. First large dam completed on the Columbia (ca. 1939).
40–44	We skirt Bonneville landslide that slid down from the north ca. 1440 C.E. (Reynolds et al., 2015; O'Connor et al., 2021, this volume) and dammed Columbia River at a higher level for some months and then unleashed a flood.
52–54	Between Wind Mountain (Washington) and Shellrock Mountain (Oregon) is one of several constrictions along Columbia Gorge that together formed a valve limiting westward discharge of largest Missoula floods (Benito and O'Connor, 2003).

(*O'Connor*)

Several constrictions in Columbia Gorge cumulatively choked the largest Missoula floods, raising maximum flood stages far upstream into the capacious Pasco Basin. During largest floods these transitory lakes lasted a few days as the immense flood volumes squeezed through Wallula Gap and Columbia Gorge. Maximum flood stages, as marked by ice-rafted erratics and trim lines eroded into the loess- and fan-covered hillslopes, show that maximum stages were ~365 m asl in Pasco Basin upstream of Wallula Gap but only 20–25 m lower at 340–345 m asl in the Dalles basin 200 km downstream. Through the gorge below The Dalles, maximum stage descended more than 220 m over just 100 km to ~120 m asl in the Portland basin (Benito and O'Connor, 2003; O'Connor et al., 2020). The maximum-stage evidence in conjunction with flow modeling suggests the flow descended through the gorge in steps at constrictions at Rowena Gap, Bingen Gap, Mitchell Point, and Crown Point. As the floods jetted through these constrictions they entrained material from the valley sides and deposited downstream.

The single flow route through the gorge and the constrictions of hydraulically critical or near-critical flow provide a robust setting for discharge estimation. Our step-backwater (1-D) modeling in the 1990s indicated ~10 million m^3/s (Benito and O'Connor, 2003). Later 2-D hydrodynamic modeling is refining and perhaps lowering this value but confirms control by the gorge in limiting the discharge of passing giant floods (Denlinger and O'Connell, 2010; Denlinger et al., 2021). At close to 10 million m^3/s here—and two or three times that at the breached ice dam—the largest Missoula floods were the greatest known freshwater discharges on Earth. These ice-age floods were ~300 times that of the largest historical Columbia River flood, 32,000 m^3/s measured at The Dalles during the freshet of June 1894.

The profound hydraulic control exerted by the gorge was an important factor in the early Spokane-flood controversy. In particular, Ira Allison (1933) documented consistently high maximum water stages east of the gorge upstream into the Pasco Basin. By this observation, he hypothesized an ice blockage, perhaps initiated by a landslide in the gorge, as source of high-water levels upstream in the Channeled Scabland. This explanation, Allison asserted "will make the idea of such a flood more generally acceptable" and "removes the flood from the 'impossible' category." The idea of a persistent gorge blockage creating a persisting "Lake Lewis" in the Pasco Basin was a key element of Flint's (1938) fill-and-incision hypothesis for the origin of the Cheney-Palouse scabland tract, which concluded that instead of by tremendous flood, the scabland channels were cut gradually by "leisurely streams with normal discharge."

Mile	
54	Creeping Wind Mountain landslide from north side constricts Columbia River.
58–59	Mitchell Point (Ore.) and ribs of inclined basalt (Wash.) make another of the bedrock constrictions that together limited discharge and thus the rate largest Missoula floods could drain out from east of the Cascades.
63	Hood River, Ore.
72	Rest area (toilets).
76	Lyle, Wash., on north side of river.
77–81	This whole reach through Rowena Gap (Ortley anticline) is a relatively narrow reach of the gorge, another bottleneck to large Missoula floods.
82–86	City of The Dalles, occupying a synclinal open reach of valley.
87–88	The Dalles Dam, completed 1954. It drowned the famous series of rapids and narrows, the "Dalles of the Columbia." between here and Celilo.
97	Celilo village. Historic Celilo Falls was close to this side. Last field stop on Day 4 is on bluffs across the river.
99	Deschutes River. Miller Island opposite on Washington side (see O'Connor writeup for Stop 4.3).

104	Exit to U.S. Highway 97. Turn north toward bridge. (Trip odometer reset just ahead.)
0.0	Cross Columbia River. In middle, **reset trip-odometer miles to 0.0.**
2.1	Intersection with Washington state route (SR) 14. Turn left, staying on U.S. 97 north.
2.6	Intersection. Turn right, continuing on U.S. 97 north.
3–8	Climb Columbia Hills anticline formed in Miocene Columbia River basalt.
12.8	Goldendale.
14–16	Climb into and through Pliocene to early Pleistocene Simcoe Mountains volcanics: basalt to trachyte but mostly basalt to trachybasalt (Hildreth and Fierstein, 2015).
16–23	A few scattered roadcuts expose round-stone pebble-cobble gravel. Quartzite (mainly from Precambrian Belt metasedimentary rocks in north Idaho) are a signature lithology of Columbia River. The river once came this way (Warren, 1941a, 1941b; Waters, 1955), a course later blocked by rising Miocene anticlines that are surmounted by the Simcoe volcanics.
23.5	St. John's Monastery (The Holy Monastery of St. John the Forerunner) on right.

Stop 1.1. St. John's Monastery Shop: Coffee, Pastry (and Toilet) (45.9425, –120.6782)
(*O'Connor and Waitt*)

A brief discussion of quartzitic gravel that we passed to the SW, Satus Pass coming to the N, and a few geologic highlights like Yakima foldbelt and Missoula-floods bars along the coming long drive northeast to Stop 1.2.

Mile	
23.6	Return to U.S. Highway 97 north. Continue climb through Pliocene to early Pleistocene Simcoe volcanics.
29.3	Satus Pass. An earlier idea that this was a wind gap cut by the Columbia River (Warren, 1941b) was discredited by Waters (1955), who points out that all rocks in these roadcuts are local (Simcoe) basalt clasts, including no quartzite or other uniquely Columbia River lithologies. Waters infers that the pass owes to a local stream capture long postdating when Columbia River flowed roughly this way.
30–34	Descend off Horse Heaven anticline and its capping Pliocene-Pleistocene Simcoe volcanics into underlying Miocene Columbia River basalt.
34–49	Highway is atop Columbia River basalt and along Satus Creek. Some roadcuts are through tributary fans that lead down to Satus Creek.
49.3	Cross Satus Creek.
50.3	Approach Dry Creek (Satus Creek tributary from the west). Along this stretch, fresh exposures during 1994 highway regrading showed eight stacked mica-bearing rhythmites normally graded from very fine sand to silt. These beds reveal eight separate Missoula-flood backfloodings from Yakima valley up Satus Creek to these altitudes, 299–311 m.
50.7	Cross Dry Creek bridge.
51–56	Cross Toppenish Ridge anticline in Columbia River basalt, one of the east-west highs of the Yakima foldbelt. Some roadcuts show sedimentary interbeds between basalt flows.
56.4	Yakima valley floor—a broad syncline between anticlinal ridges. Continue on U.S. 97 straight north.
61.0	In Toppenish, approaching large intersection of SR 22. Stay right to continue north, off U.S. 97 and onto SR 22.
61.0	Continue through intersection, now on SR 22. In intersection, **reset trip odometer**.

Mile	
0.0	**Reset trip mileage at Toppenish at U.S. Highway 97 and SR 22.** North on 22.
2.8	Yakima River.
3.2	Highway I-82 at Buena. Enter I-82 east (exit 50).
5–7	Backflood rhythmites along discontinuous cliffs north side I-82. For the seminal report announcing dozens of last-glacial Missoula floods, Waitt (1980) studied exposures of this sort when they were fresh during construction of I-82 in 1978–1979.
21.3	Exit 69 off I-82 to SR 241, north on 241.
21.6	Cross Yakima Valley highway, continuing straight on SR 241.
24–25	Occasional backflood silt deposits. Begin climb from Yakima valley syncline onto Rattlesnake Hills anticline formed in Columbia River basalt—all parts of the Yakima foldbelt.
32.6	Road crest of Rattlesnake Hills anticline; begin descent toward Pasco basin.
35.3	T-intersection with SR 24. Turn right (east) on 24.
43.1	Cold Creek Road. We have descended off the anticline into high-level Missoula-flood beds in a west arm of Pasco basin. This is another synclinal low in the basalt.
45.9	Junction SR 240 to Richland. Turn left, staying on SR 24 atop a high Missoula-flood gravel bar.
47.5	Descend through basalt on a small anticlinal ridge.
49.1	End descent on lower-level younger Missoula-flood gravel bar.
49.5	Rest area on left. Stop for long-awaited toilets. Then continue north on SR 24.

50.7	Cross Columbia River on Vernita Bridge.
51.1	ntersection SR 243. Turn left (west) onto it. Road still rides atop a low-level Missoula-floods gravel bar. To the north is high-level older bar truncated by later smaller Missoula floods.
58.8	Priest Rapids Dam. This site was lower end of a reach of a series of broad rapids falling across basalt flows. This area of type section of Priest Rapids basalts is now mostly beneath water.
59–60	Flood-moved basalt boulders to ~0.8 m diameter culled from adjacent fields (Mattawa area).
63–64	Basalt boulders to 2+ m culled from adjacent fields.
67.7	Very large floodborne boulders below Sentinal Gap.
68–69	Sentinal Gap through Saddle Mountains anticline. Lower parts expose older (Grand Ronde) basalt.
69.9	Lower Crab Creek. This valley was the main outlet to the Columbia of floodflow shunted down Grand Coulee and through Quincy basin. Crab Creek is now crowded to south side of valley by a large late flood down the Columbia that poured southeast into the valley mouth (Fig. 10).
71.2	Beverly on east.
71.3	Railroad overpass.
74.8	Wanapum Dam.
75.3	Parking area on right (gate with sign "IXI LLC"). Walk south beside road ~500 yards to tallest rhythmite outcrop on east side. Beware of traffic, semi trucks, etc. Please walk single file on one side of road, off pavement as much as possible.

Stop 1.2. Rhythmites and Mount St. Helens "S" Ash at Wanapum Site (46.8874, –119.9552)
(*Waitt and Atwater*)

Sand-and-silt rhythmites of late floods, nestled into scabland (Fig. 10), juxtapose contrasting effects of highly erosional catastrophic floods and lesser, frequent ones that deposit here. Mullineaux et al. (1978) called attention to the rhythmites for containing tephra layers of Mount St. Helens Sg and So (Mullineaux, 1996). Moody (1987, fig. 52) reported ash layers from each of three sand-to-silt rhythmites. She assigned these tephras to layers Sg and So and sketched 11 rhythmites above the highest ash-bearing one. Smith (1993, fig. 3) grouped the post-tephra rhythmites into eight flood "cycles," each cycle concluding with evidence for subaerial exposure.

The set S tephras may enable matching of flood rhythmites here with flood beds that alternate with varves of glacial Lake Columbia. Upstream of Grand Coulee in Sanpoil valley, a set of interflood varves contains a tephra that has been assigned to set S (Stop 2.5). If this tephra is Sg or So, approximate varve counts in that part of the Sanpoil section give interflood intervals here of two decades at most. In addition, the flood rhythmites here may have counterparts on the floor of upper Grand Coulee (Stop 3.4).

Smith (1993) noted here dominantly southward downvalley-pointing ripple-drift laminations. One usually sees some in lower parts of this section. But during middle and late parts of the late Wisconsin glacial-and-floods sequence (Fig. 6D), the Okanogan ice lobe blocked off flood routes past here from upvalley. These late Missoula floods arrived as Quincy basin water via lower Crab Creek to Beverly, from there backflooding north up Columbia valley to this site. An answer to this directional enigma lies just north: a backflood eddy may have caused downvalley currents.

On the walk back north to parked vehicles, ~150 m north of the tephra section, we cross a deep gully. Though now largely surrounded by scabland, the gully is likely a preglacial small tributary to Columbia River, roughly graded to it. Missoula floodwater via Quincy basin—down lower Crab Creek and backflooding up Columbia valley from Beverly—would backflood up this tributary, thence south into the scabland pocket at the rhythmite site, before rising floodwater could reach the site more directly from the south. The backflood perhaps thus makes a typical clockwise eddy along its right side—in this case around a scabland-basalt high in its center.

Mile	
0.0	**Reset mileage.** From parking area for Stop 1.2 turn right (north) on SR 243.
1–3	Drive through high-relief upland flood-carved scabland in Columbia River basalt.
2.6	Sand Hollow rest area on left (toilet).
2.9	SR 26. Turn left (north).
3.7	Take right lane toward I-90 east.
4.1	Join I-90 east. In next few miles, cross Frenchman Hills anticline (an element of the Yakima foldbelt) that is cut through by Columbia River.
9.6	Take offramp to Silica Road, I-90 exit 143.
10.0	Silica Road. Turn left. Pass beneath I-90 on curvy road northeast.
10.8	Intersection. Turn left onto historic U.S. Route 10 and old "Vantage highway."
11.1	Begin descent through basalt flows (Priest Rapids and underlying Roza flows) into Frenchman Springs cataract. Along the way spectacular views of the main (north) alcove of the cataract.
12.2	Climbing area in small cataract alcove on left (and a toilet). Well-formed wavy columns typical of the Roza basalt flow. Continue descent through thick cliff-forming Frenchman Springs basalt flows.
13.5	Pass west end of (eroded off) Frenchman Springs basalt. Hidden by talus is erodible Vantage sandstone that the Frenchman Springs basalt overlies. This broad esplanade, Babcock Bench, is beveled atop Grande Ronde basalt by Missoula floods readily stripping off the Vantage sandstone.
13.8	Views east up the smaller south alcove of the dual Frenchman Springs cataract. Begin descent through large gravel bar.

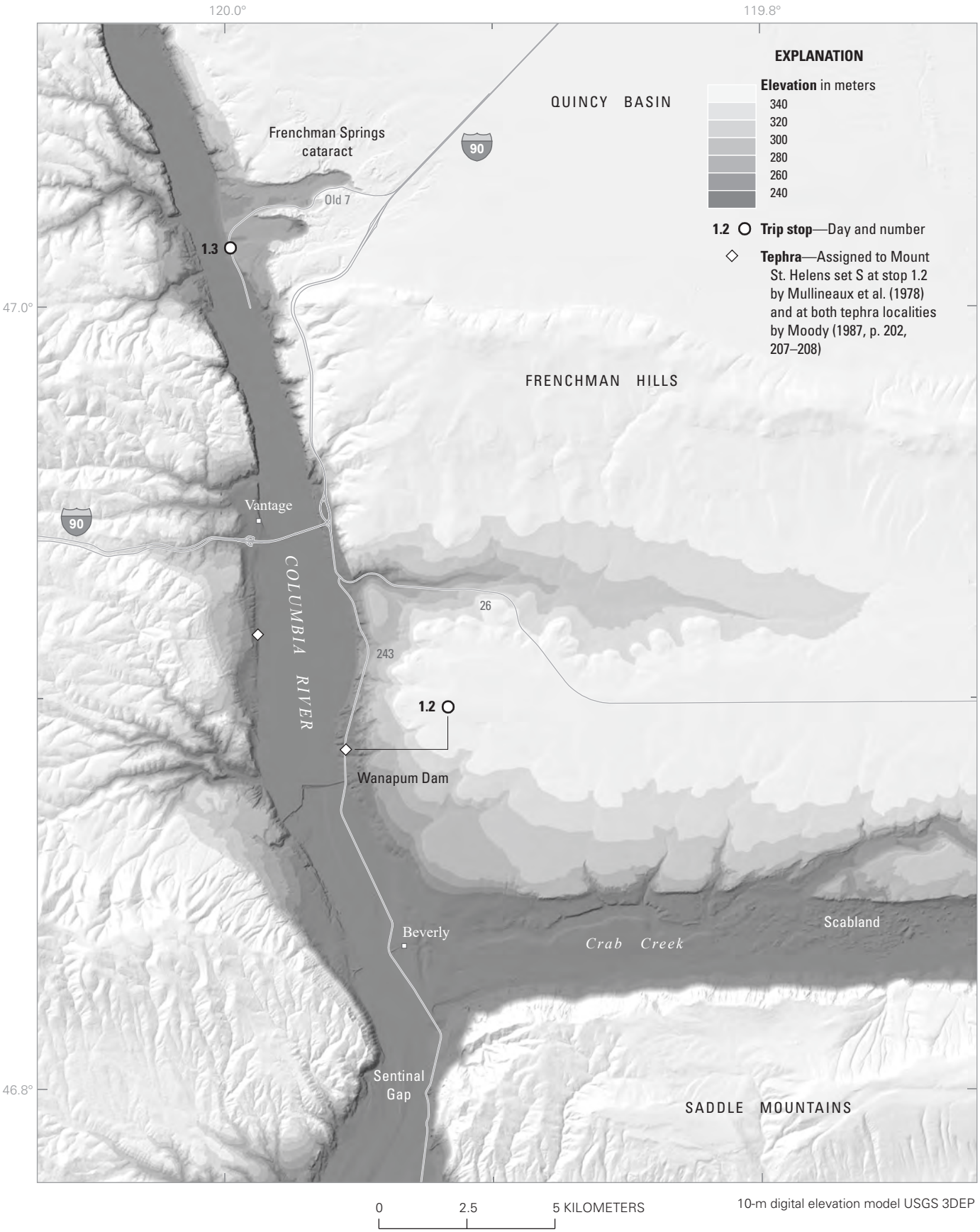


Figure 10. Frenchman Coulee to Sentinal Gap area (*Atwater*).

14.0	Pull off right onto broad shoulder. Park well off road. Walk 50–100 m farther down the road to view roadcut.

Stop 1.3. Frenchman Springs Coulee and Gravel Bar (47.0154, –119.9981)
(*Waitt*)

Only the very largest Missoula floods channeled down Grand Coulee could deliver enough water fast enough to fill Quincy basin to its higher west rim while most floodwater pours out the broad Drumheller Channels on the south (Fig. 6D).

Potholes and Frenchman Springs cataracts drop from ~300 m altitude in three steps over Columbia River basalt flows—Roza, Frenchman Springs, and uppermost Grande Ronde. The Vantage sandstone, 5–13 m thick atop the uppermost Grande Ronde flow, eroded back to form the esplanade of Babcock Bench 0.5–2 km broad—its northern higher parts etched by Missoula floods into locally deep scabland. Arcuate sub-cataract heads, plunge pools, and linear scabland and gravel bars attest that last-glacial floods through Potholes, Frenchman Springs, and Crater cataracts (at blue arrowheads on Fig. 6D) cascaded with great energy west down to ~270 m altitude. Simultaneous backflood via lower Crab Creek, and later down-Columbia flood(s), built river-parallel bars as high as 85 m athwart the bars from these west-running cataracts (Waitt, 2021, figs. 9, 10).

A tall roadcut through the end of a large bar from the south alcove of Frenchman Springs cataract shows chaotically bedded cobble to coarse-boulder gravel 50 m thick. Most clasts are angular to very angular basalt—up to 4 m diameter but mostly smaller. The gravel also holds angular cobbles of calcrete, and a few boulders from the Vantage sandstone. Though bordered by Columbia River, the gravel includes no crystalline rocks from upriver. This bar, like those at Potholes and Crater cataracts, formed under high-energy westward outflow from Quincy basin.

Note a general lack of sand-silt matrix. This contrasts with a gravel body we visit early on Day 4 (Stop 4.2).

Mile	
14.0	Continue down historic U.S. Route 10, viewing lower parts of Frenchman Springs gravel bar.
15.2	End of road at boat ramp. Old highway descends beneath the reservoir held by Wanapum Dam. Before the dam and reservoir, the road ahead led down to a bridge across Columbia River to Vantage.
15.3	Boat-ramp facilities include a one-hole potty. Return up through Frenchman Springs cataract on historic U.S. 10 to I-90.
20.4	Return to I-90, Exit 143. Turn left onto onramp for I-90 east.
23	Highway climbs gradually from cataract head into Quincy basin proper.
26.3	Pass George, Washington.
27.9	Off I-90 at exit 151.
28.4	SR 283. Cross I-90 and turn northeast, SR 283 toward Ephrata.
29–33	Southern Quincy basin flood deposits are sand.
34–36	On clear days view to the northwest shows the ragged glaciated Mount Stuart range part of the North Cascade Range. To its northeast is rounded smoother Naneum Ridge, a huge anticline in Columbia River basalt, also part of the Yakima foldbelt.
37.4	Winchester wasteway irrigation canal (Fig. 11).
37.7	Missoula-flood deposits have coarsened upcurrent to gravel. Round-stone large cobbles of basalt culled from fields.
42.8	Join SR 28. Continue straight (north), now on SR 28, toward Ephrata.
47.2	Ephrata. Intersection SR 282. Continue north on SR 28.
53.4	Intersection SR 17. Turn left (north) toward Soap Lake.
54	Soap Lake, Wash.
54–57	Soap Lake (alkaline). Discussion about this early on Day 4.
57	Enter lower Grand Coulee. For next 15 miles we roughly follow the limb of Coulee monocline.
58–63	Lenore Lake on left. Alkaline, but less so than Soap Lake.
66–68	Blue Lake. Less alkaline than Lenore Lake.
69–70	Park Lake. Essentially fresh water.
70–71	Ascend onto higher basalt flows.
72.7	Turn Right into Dry Falls parking lot.

Stop 1.4. Dry Falls (47.6062, –119.3649)
(*Waitt*)

Grand Coulee, the deepest of the great scabland channels, consists of two tandem canyons described in Bretz's (1932) eloquent monograph. Dry Falls twin cataract, 122 m high and 1.8 km wide—twice Niagara's height and a third wider—defines the head of lower Grand Coulee (Fig. 12). Lower cataracts of the group reach 5 km farther east. High scabland reaching 10.5 km east from the cataracts shows that even deep and wide lower Grand Coulee did not contain largest floods. Floods gouged out widest and deepest scabland channels by cataract retreat. Longitudinal bars on the floor of the gorge below Dry Falls are studded with enormous basalt-entablature boulders quarried from the cataract walls.

Bretz's early concept required Dry Falls cataract to retreat 27 km from Soap Lake during one giant flood (at first called the "Spokane Flood"). The great cataract of upper Grand Coulee receded nearly 40 km. Later evidence for several floods across different glaciations (Bretz et al., 1956; Bretz, 1959, 1969) reduced recession of these cataracts required of any one flood to several kilometers—still huge. Then much later came evidence of scores of megafloods during the last glaciation alone (Waitt,

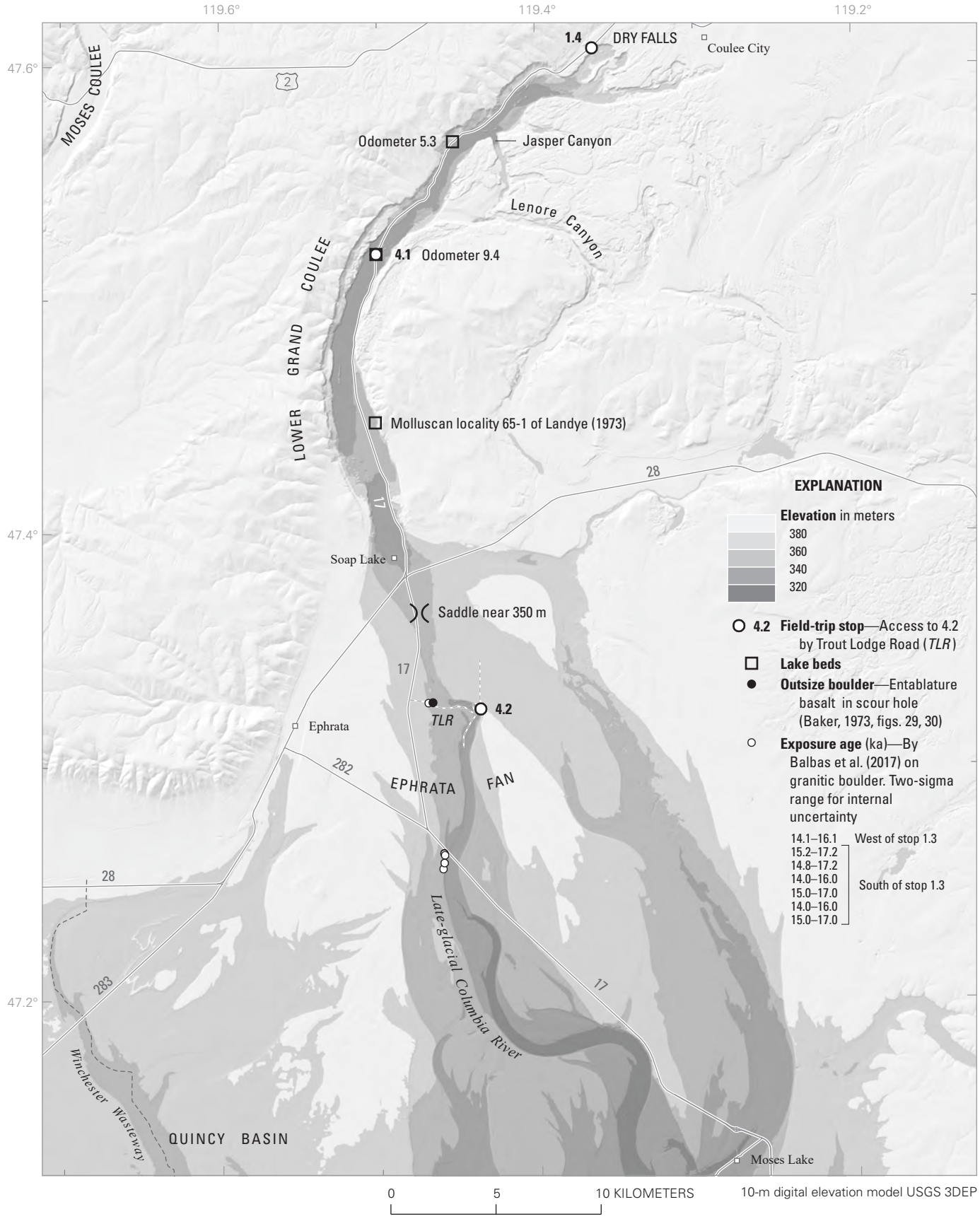


Figure 11. Central and northern Quincy basin (*Atwater*).

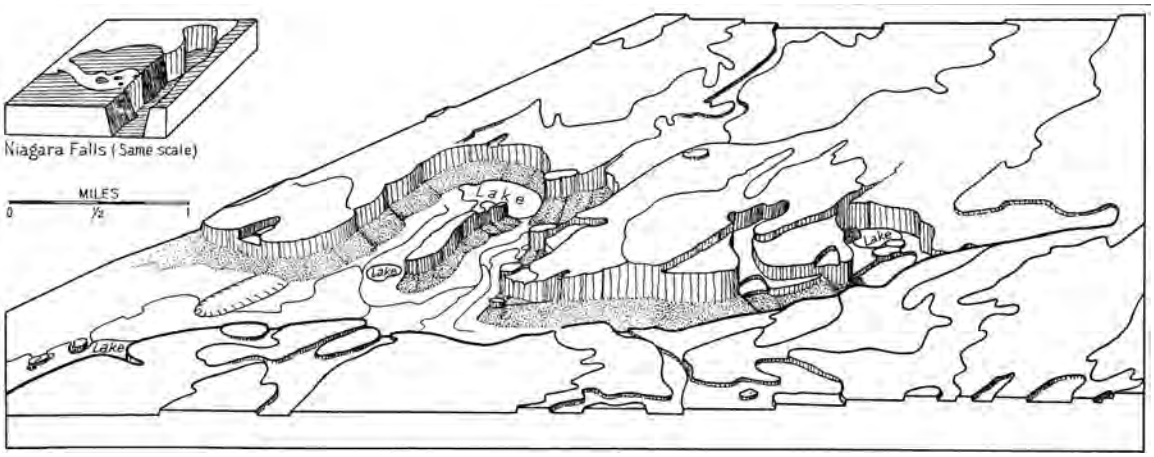


Figure 12. Bretz's (1932, fig. 8) block diagram of Great Cataract group looking north, and compared with Niagara Falls (1 mile is 1.6 km).

1980, 1985). With megaflood erosion during these and during earlier glaciations, Dry Falls cataract needn't recede more than a small fraction of a kilometer during any one flood. With more than a hundred great floods during two or more glaciations, perhaps hundreds of floods among multiple glaciations, eroding great volumes of basalt from the Channeled Scabland becomes an understandable process. Protracted excavation of upper Grand Coulee also becomes likely, as does its completion before the last glaciation.

We shall now ascend upper Grand Coulee, to which we will return on day 3. Granitic basement rocks will emerge in its upper reaches, their basalt cover having been removed by Missoula floods. End moraines of the Okanogan lobe stand out of view to our west on the Waterville Plateau and on the summit of Steamboat Rock (Fig. 13).

Mile

72.8	From Dry Falls parking lot, turn right (north) onto SR 17.
74.8	Intersection with U.S. Highway 2. Turn right onto U.S. 2. Cross Dry Falls Dam, holding in Banks Lake reservoir (Columbia River water is pumped in at north end).
76.4	Outlet canal from Banks Lake for irrigation water to Quincy basin and farther south.
77	Coulee City, Wash.
79.1	Intersection SR 155. Go straight (north), leaving U.S. 2 and onto SR 155.
81	Inclined basalt beds on limb of Coulee monocline. Here we enter upper Grand Coulee. We will explore it from its head southward to near here on Day 3.
96.5	Steamboat Rock stands prominently in mid coulee.
103.2	Electric City, Wash.
104.7	Intersection SR 174. Continue straight on SR 155.
105	Grand Coulee, Wash. Continue on SR 155.
106.4	Grand Coulee Dam. Continue descent of SR 155.
106.9	Outskirts of Coulee Dam, Wash.
107.0	Turn left across median onto Lincoln Ave. to Columbia River Inn (10 Lincoln Ave., Coulee Dam, WA 99116).

Upper Upper Grand Coulee: To Contemplate Next Three Days

(O'Connor)

The present Grand Coulee bedrock threshold near Coulee City held glacial Lake Columbia at ~470 m during late phases of Missoula flooding. Varved lacustrine silt on the floor of upper Grand Coulee upstream of the lake outlet, and in adjacent Columbia valley, suggest a persistent lake draining over the ~470 m threshold (Fig. 14). Rippled sand beds within these lacustrine deposits (Stop 3.4) indicate at least 14 incursions inferred as late, small Missoula floods into this low-level glacial Lake Columbia (Atwater, 1987, p. 188; Waitt, 1994).

Was this 470 m outlet in Grand Coulee a controlling level for glacial Lake Columbia for all the last-glacial period? Or was there significant lowering down to this level during the period of Missoula flooding? When did the retreating cataract break through to Columbia valley and lower the coulee entrance? Stratigraphy and geomorphology indicate a fully deepened coulee early in the last-glacial period. But recent flow modeling suggests that upper Grand Coulee was not incised before last-glacial flooding of Moses Coulee and Columbia valley. The timing and amount of incision of upper Grand Coulee would influence levels of glacial Lake Columbia and flood magnitudes in Grand Coulee and other scablands.

Recent 2-D flow modeling (Denlinger et al., 2021) suggests that a still-blocked upper Grand Coulee is needed to allow

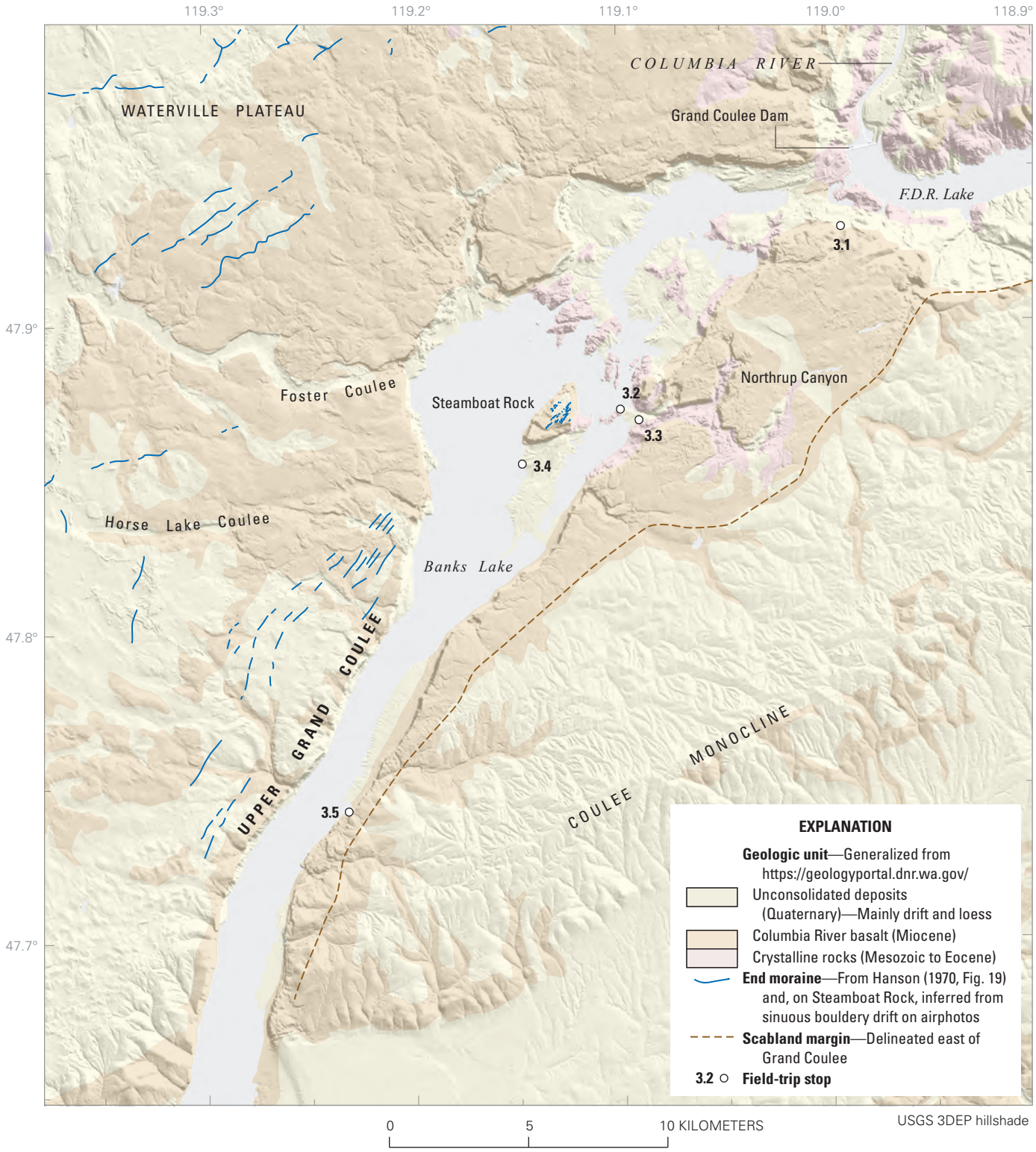
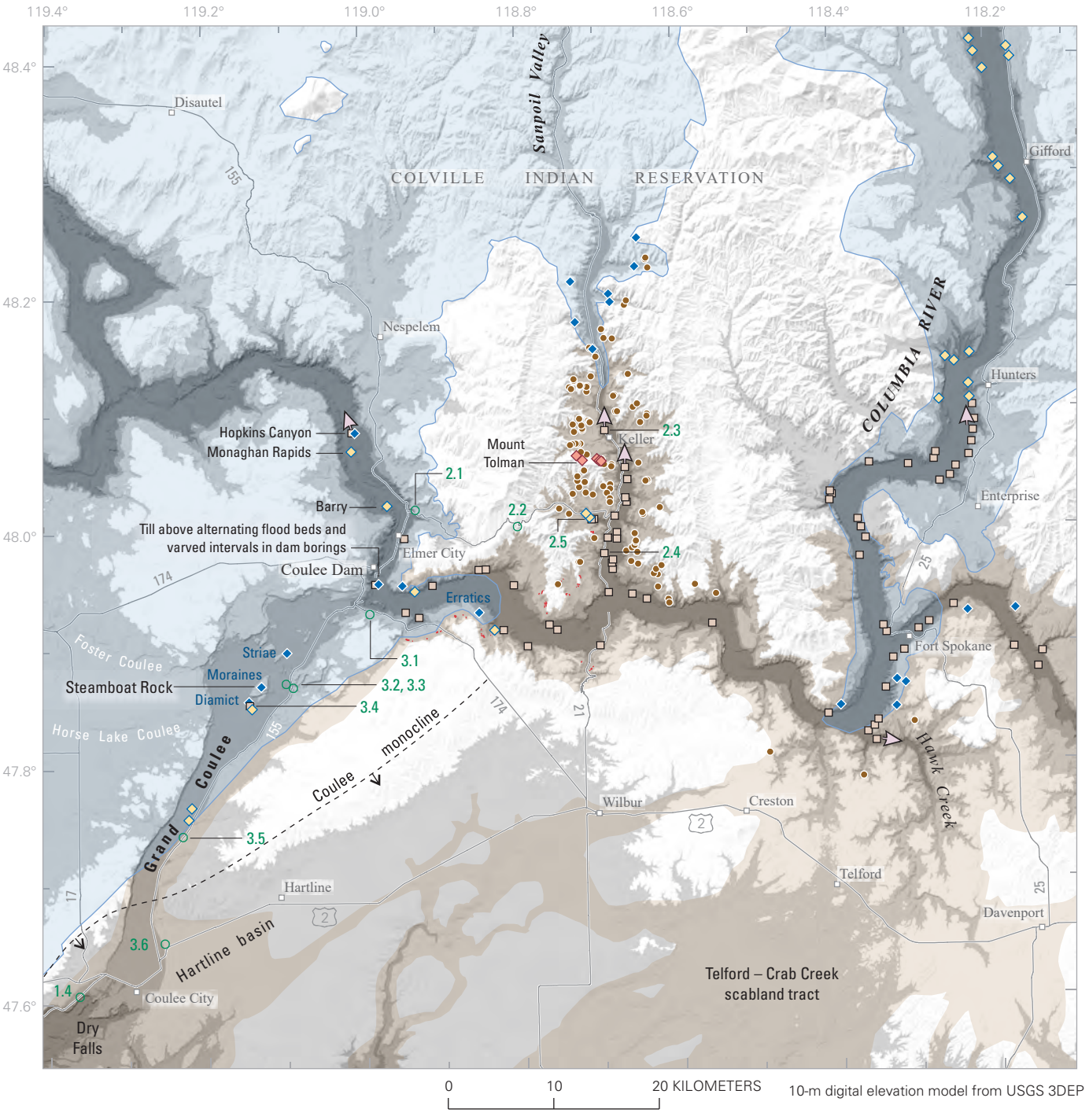


Figure 13. Generalized geologic map of upper Grand Coulee and vicinity (*Atwater*).



EXPLANATION

Cordilleran ice sheet or Missoula flood—Maximum extent as depicted by O'Connor et al. (2020, Fig. 1)
- Okanogan and Columbia River lobes
- Flood area—Shown only beyond ice-sheet terminus

Modern elevation—Dividing contours, in meters, related to glacial-lake levels along Columbia River and tributaries upstream from Okanogan lobe (O'Connor et al., 2020, Fig. 6)
- 715—High-level lake
- 520—Low-level lake
- 470—Grand Coulee threshold above Dry Falls

2.1 ○ **Field trip stop**

Stratified deposits—After Atwater (1986, 1987) and Hanson and Clague (2016)
- Varves without recognized evidence for periodic floods
- Flood beds intercalated with varved intervals
- Ripple-drift laminae indicative of backflooding

Strandline—Deposits observed on Mount Tolman (red diamond; Cochran and Warlow, 1980); lineament on lidar, airphoto, or both (short red line)

Ice-rafted erratic—From Atwater (1986, pl. 2) and O'Connor (unpubl. data)

Glacial feature—Moraine, till, or striated bedrock. Includes till near Fort Spokane (O'Connor, unpubl. data) and coulee striae and till (Day 3)

Figure 14. Glacial features of upper Grand Coulee, Sanpoil valley, and nearby parts of Columbia valley (*Atwater*).

enough flow down Moses Coulee and Columbia valley farther west (Fig. 5) to reach the high levels of flood evidence there. Both routes passed large last-glacial floods early in the overall Missoula flood sequence (Waitt, 2016, 2021). But upper Grand Coulee at its present depth and width siphons off so much floodwater that it leaves too little flow along the western routes to reach high-level flood evidence there. This modeling result implies that the cataract in upper Grand Coulee had not yet retreated to Columbia valley. Hanson (1970) too concluded that large Moses Coulee floods preceded incision of upper Grand Coulee.

Yet geomorphic and stratigraphic evidence suggests that upper Grand Coulee was open to near its present 470 m elevation near the start of the last glacial period (Atwater, 1986). The lacustrine silt on the floor of upper Grand Coulee and prominent terraces at 475–500 m extending up Columbia and Spokane valleys (Fig. 14) show a lasting glacial lake held near the present 470 m threshold in Grand Coulee while Columbia River remained blocked by the Okanogan lobe (Bretz, 1932; Flint, 1936; Atwater, 1987; Kiver and Stradling, 1995). The evidence for short-lived glacial Lake Columbia levels as much as 250 m higher than 470 m is explained by the Okanogan lobe filling and blocking an already incised Grand Coulee.

A hybrid scenario that may resolve opposing interpretations: early last-glacial floods did erode the upper cataract back to Columbia valley—but not before a few large floods passed west through Columbia valley and Moses Coulee. Before the cataract receded to the Columbia, glacial Lake Columbia may have briefly stabilized at the higher ~653 m saddle threshold into Moses Coulee until overrun by ice. Once that outlet was covered and before full cataract retreat, any glacial Lake Columbia outlet would have been farther east, perhaps a saddle where Grand Coulee now lies or a divide saddle farther east. The absence of conspicuous shorelines at high elevations indicates that any such stability was short-lived.

Considering the conflicting flow-modeling results and stratigraphic interpretations, the timing, magnitude, and effects of upper Grand Coulee incision remains an unresolved question for understanding late Wisconsin Missoula flooding.

End Day 1.

Day 2. Columbia and Sanpoil Valleys

(*Atwater and Hanson*)

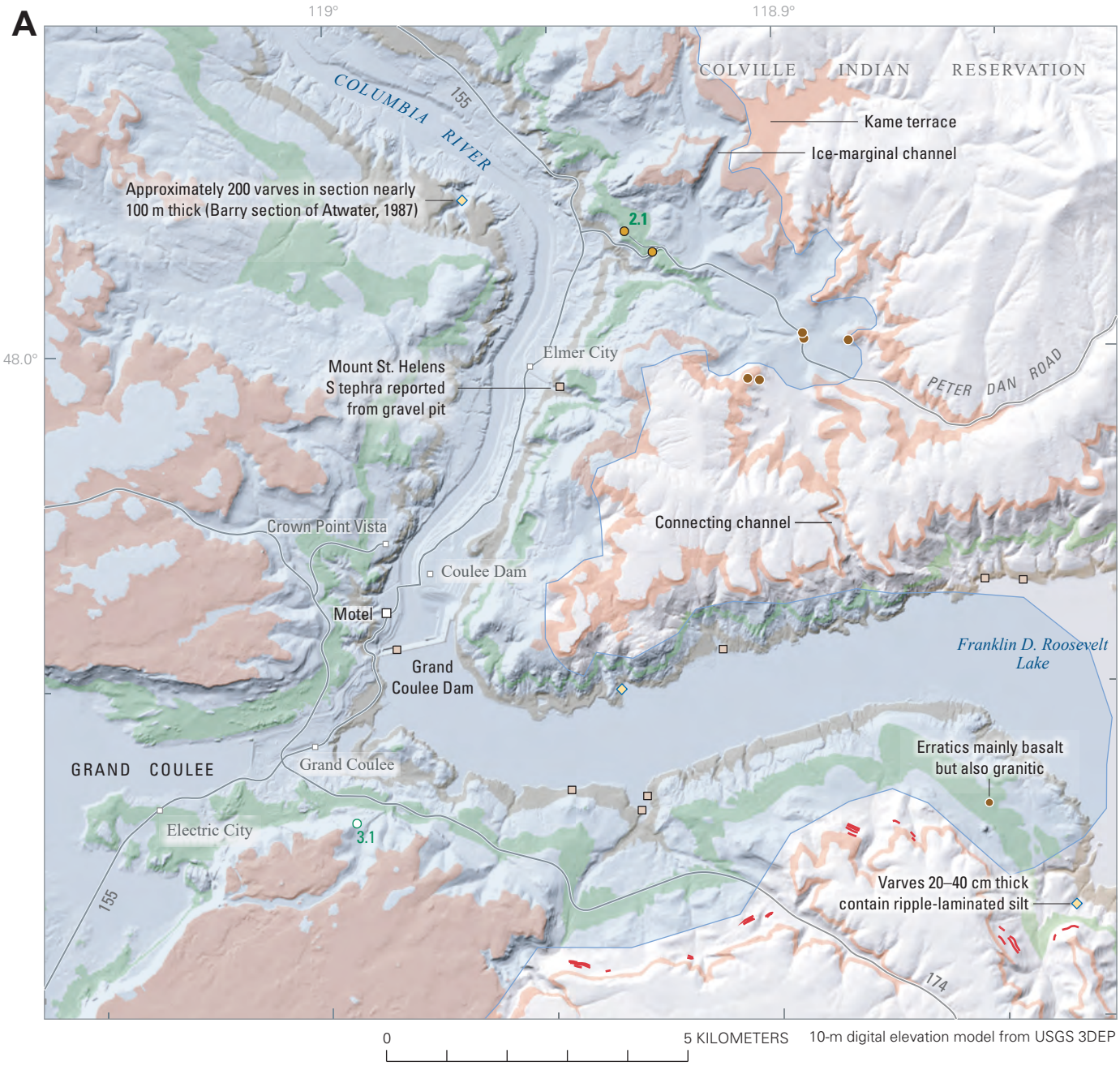
The trip's second day examines glacial-lake clues to Grand Coulee history. Each of five stops displays traces of glacial Lake Columbia, which during the last glaciation received no fewer than ~90 Missoula floods and usually drained through upper Grand Coulee.

At all five stops we are guests, by written permission, of the Confederated Tribes of the Colville Reservation. The federation represents bands from the Chelan, Nez Perce, Colville, Entiat, Lakes, Methow, Moses-Columbia, Nespelem, Okanogan, Palus, San Poil, and Wenatchi tribes. The tribal government granted Atwater a research permit for scouting of the October 2021 trip and for the trip itself. A prior tribal permit covered Hanson's Sanpoil valley field work in 2006. The Bureau of Indian Affairs funded bedrock mapping that brought Atwater to the area in 1980–1985. The Bureau of Reclamation in 1983 drilled borings below Stop 2.5 to sample unexposed parts of a glacial-lake section.

The road log and stop descriptions presuppose access to cited reports on Grand Coulee Dam (Flint and Irwin, 1939) and glacial Lake Columbia (Atwater, 1984, 1986, 1987; Hanson et al., 2015; Hanson and Clague, 2016). Figures in this guide locate today's stops on maps of three scales (Figs. 1, 14, 15), illustrate a tephra layer at Stop 2.5 (Fig. 16), and relate Sanpoil stratigraphy to inferred ice-age history (Fig. 17). The day's road log, which begins at the trip motel, includes a mileage reset between Stops 2.2 and 2.3.

Today's field area illustrates onlap of Columbia River basalt onto pre-Oligocene rocks of the Okanogan highlands (Pardee, 1918). The older rocks include gneiss domes, plutons, dike swarms, and volcanic rocks that together record Eocene extension (Fox and Beck, 1985; Holder et al., 1990; Kruckenberg et al., 2008).

Mile	
0.0	Columbia River Inn. Facing Grand Coulee Dam, turn left (north) after crossing median strip in SR 155.
0.3	Bear right on SR 155, toward Omak, across Columbia River. Photos of pre-dam landscapes and dam construction adjoin walkway on the bridge's south (dam) side.
0.5	Stay on SR 155 by turning left.
1.9	Gravel pit to the right. Road cuts checked in a 1985 reconnaissance included one interval of ~20–25 varves averaging 2 cm in thickness and containing, in some instances, basal fine sand. Dropstones common.
3.2	Elmer City. Out of view to east, the lowest part of a borrow pit exposes gravel foresets with apparent dip toward Columbia River. Above the gravel lie graded sandy beds as thick as 3 m that alternate with eroded, partly dismembered intervals each containing no more than ~25 varves. Six of the graded beds were evident in 1985 and three in 2021, when the pit owner disallowed access. Kiver and Stradling (1995, p. 143) reported collecting here, in 1979, an ash layer they correlated with Mount St. Helens set S.
4.4	Turn right from SR 155 onto Peter Dan Road.
5.5	The road cut at left rises to a gravel-coated bench equivalent in geomorphic position to what Flint (1935, p. 186) called "Nespelem silt terrace." This guide uses "Nespelem terrace" in recognition of the surficial gravel here, at nearby Stop 2.1, and near Stop 2.5. Laminated fine sand low in the cut may represent a shoaling glacial Lake Columbia.



EXPLANATION

Cordilleran ice sheet—At maximum extent in O'Connor et al. (2020, Fig. 1). Ice limit approximate south and east of Grand Coulee Dam

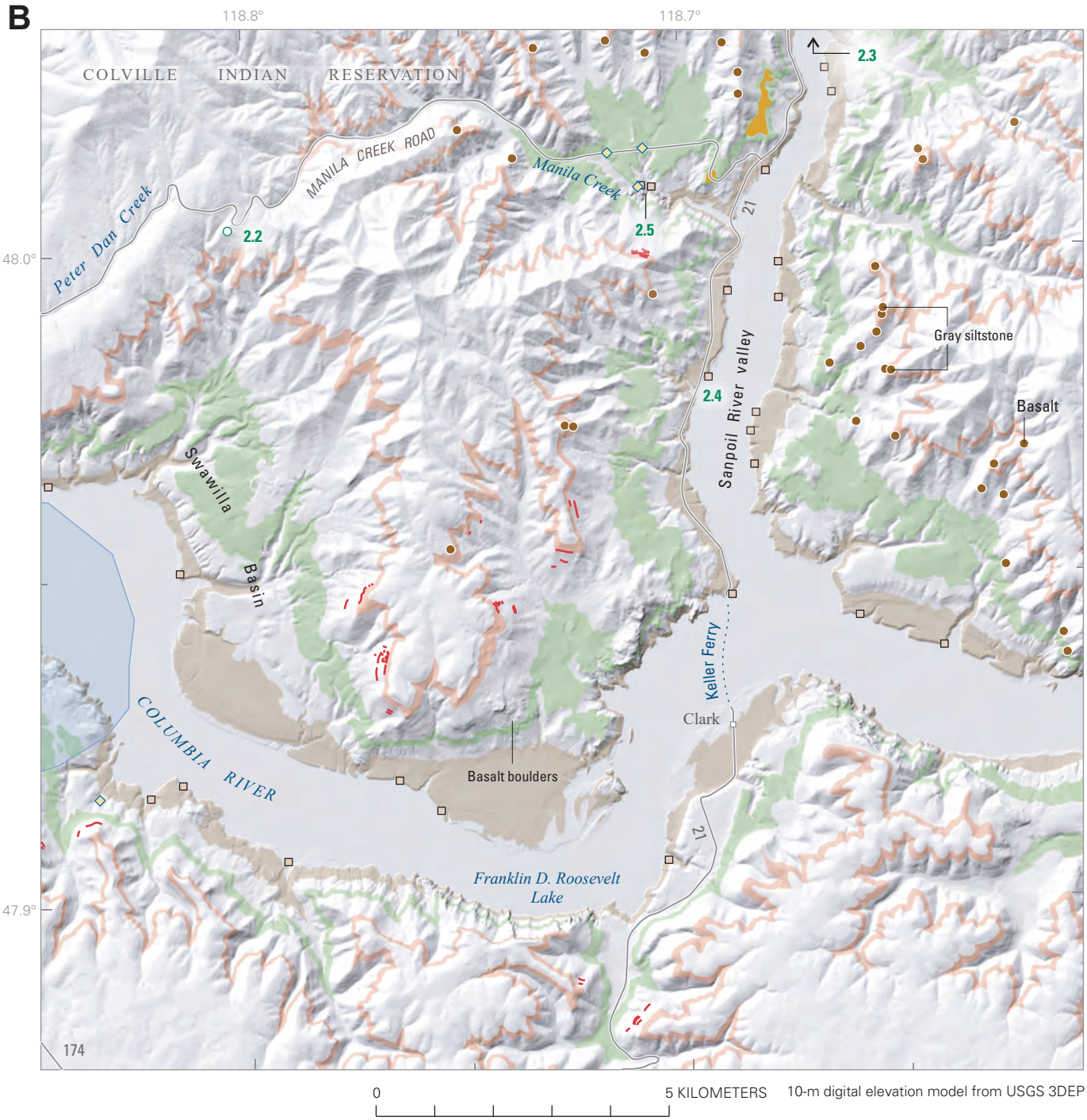
Modern elevation—Each color spans a 40-m elevation range
- 700–740—High-level and flood-swollen Lake Columbia
- 500–540—Low-level Lake Columbia
- 400–440—Terraces formed during incision of lacustrine fill

Strandline—Lineament on lidar, airphoto, or both

Haystack—Basalt erratic mapped near ice margin across Peter Dan Creek

Stratified deposits
- Cobble gravel capping the Nespelem terrace near 740 m—Underlain by laminated very fine sand
- Varves without recognized evidence for periodic floods—Northwest of Barry, varves 20–50 cm thick likely record a few centuries of kettle filling; Barry section of Atwater (1987)
- Sandy floodlaid beds intercalated with varved intervals—Earlier than maximum extension of Okanogan lobe beneath till at Grand Coulee Dam (Flint and Irwin, 1939), later where three cycles interfinger with deltaic gravel in Elmer City. Other sites from Hanson and Clague (2016)

Figure 15. Columbia and Sanpoil valleys for Day 2; scale larger than Figure 14. (A) Head of upper Grand Coulee and environs and setting for Stop 2.1. (*Continued on following page.*)



EXPLANATION

Cordilleran ice sheet—At maximum extent in O'Connor et al. (2020, Fig. 1).

Modern elevation—Each color spans a 40-m elevation range
- 700–740—High-level and flood-swollen Lake Columbia
- 500–540—Low-level Lake Columbia
- 400–440—Terraces formed during incision of lacustrine fill

Strandline—Inferred mainly from lidar and airphotos

Erratic—Ice-rafted (Atwater, 1986, pl. 2A)

Stratified deposits
- Pebbly sand—On terrace north of Manila Creek mouth. Evidence that a Sanpoil River outwash plain isolated Manila Creek valley from Lake Columbia before deglacial incision (Atwater, 1986, p. 11, 14)
- Glacial-lake deposits without recognized evidence for periodic floods
- Sandy floodlaid beds intercalated with varved intervals—After Atwater (1986) and Hanson et al. (2015)

Figure 15 (*Continued*). (B) Map of Sanpoil valley and vicinity and setting for Stops 2.2–2.5 (*Atwater*).

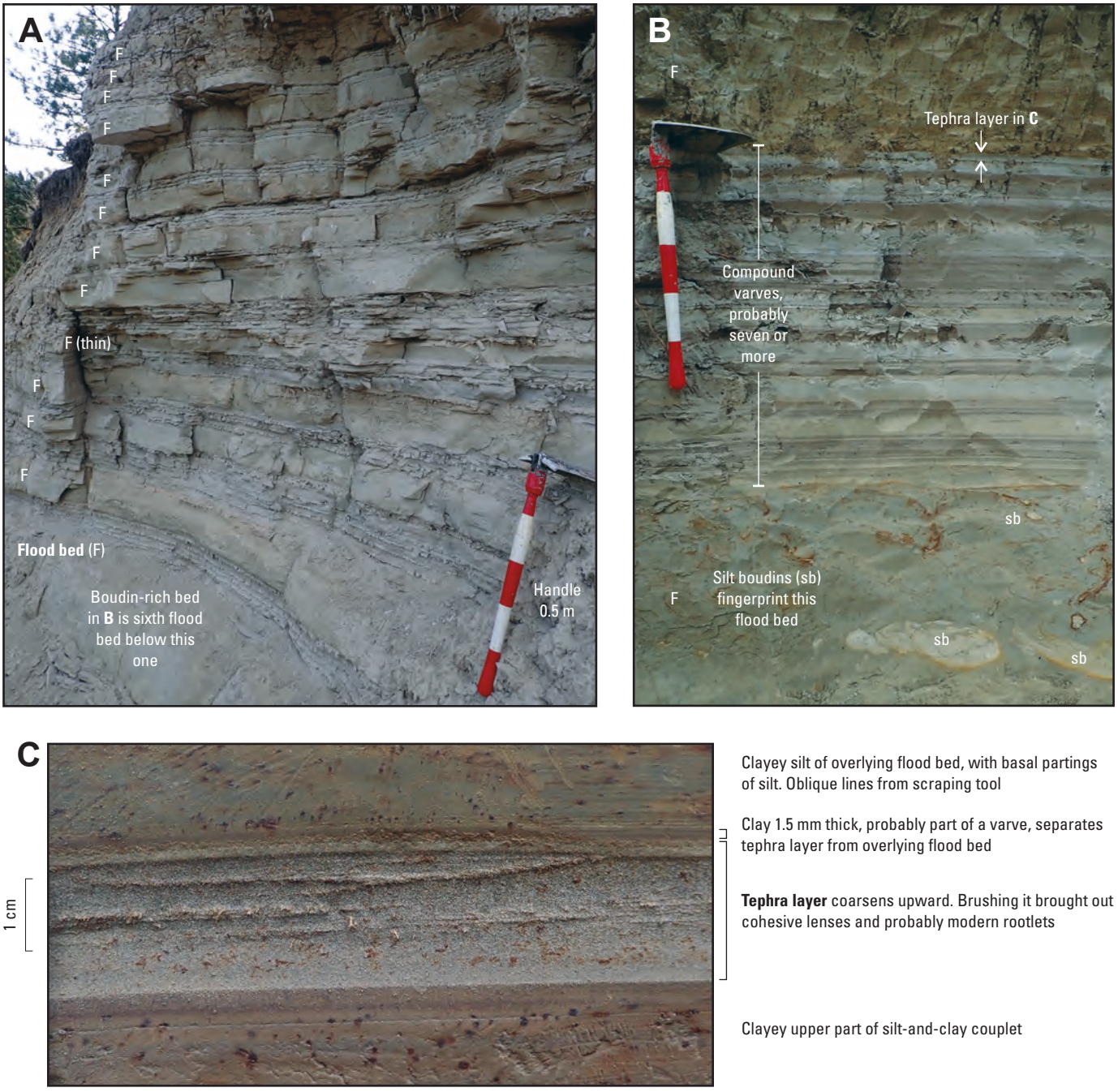


Figure 16. Switchback locality in Sanpoil valley, Stop 2.5. (A) In upper part of Switchback section, 13 silty Missoula-flood beds that alternate with small numbers of varves. Varved intervals here probably formed in a pond recently isolated from Lake Columbia by a Sanpoil River outwash plain. (B) Lower in Switchback section, stratigraphic setting of volcanic-ash horizon tentatively correlated with Mount St. Helens layer Sg. (C) Internal structure of this tephra layer (*Atwater*).

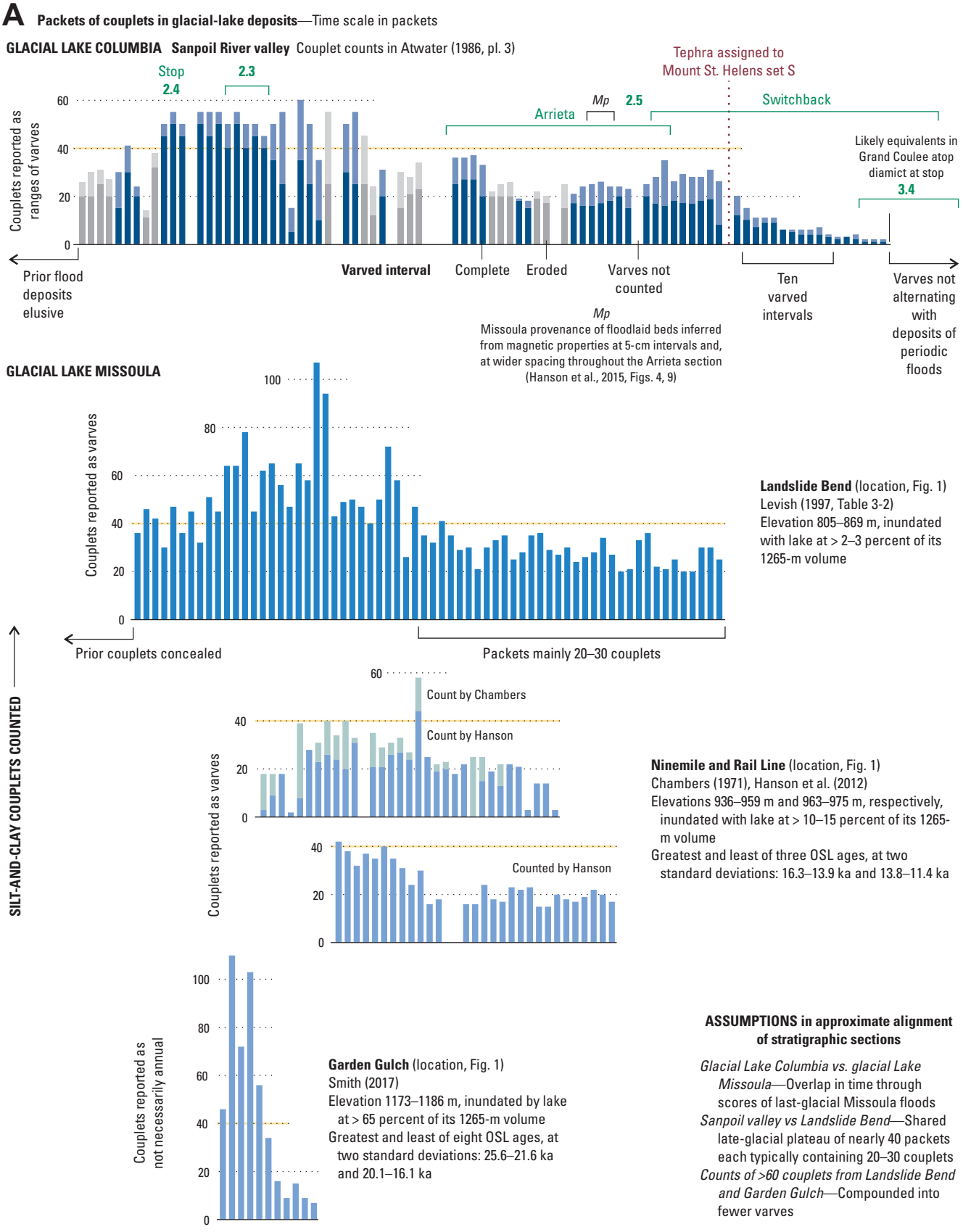
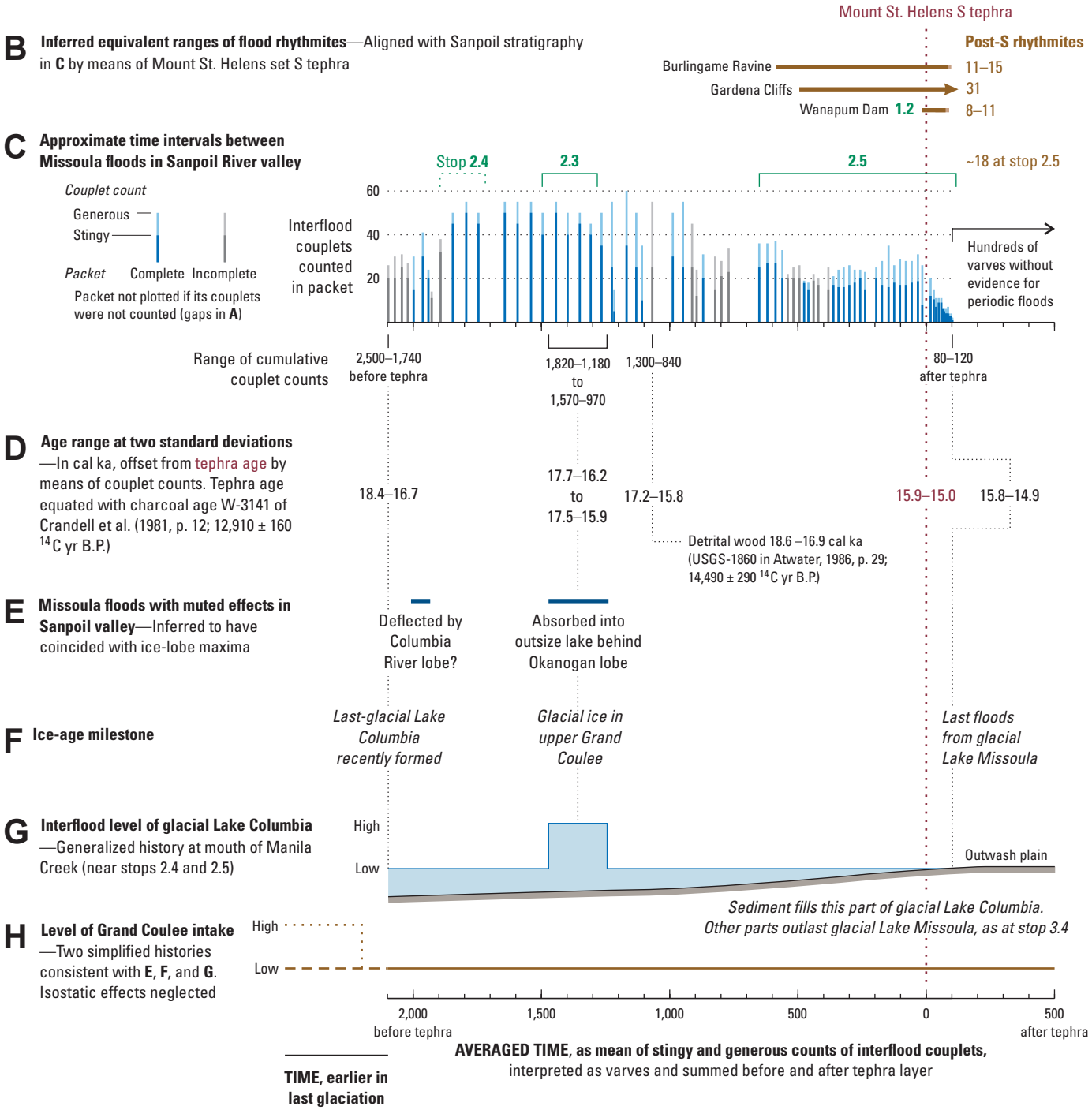


Figure 17. Comparisons among sequences of glacial-lake deposits, Missoula-flood deposits, and ice-sheet positions, locations in Figure 1. (A) Approximate correlations of varved deposits in Sanpoil valley (composite Manila Creek section) with bottom sediments of glacial Lake Missoula. (*Continued on facing page.*)



NOTES

Summed couplet counts—Cumulative ranges in **C** and averages on abscissa include, by interpolation, packets without couplet counts, while neglecting losses to erosion in packets plotted as incomplete (gray), and to truncation along low-angle faults low in section near sloping substrate (perhaps analogous to faults at stop 3.4)

Lake levels in G—Violent flood effects evince low early lake levels, as at stop 2.4. Muted flood effects are evidenced most widely by four anomalously thin flood beds exposed along at least 13 km of the Sanpoil valley. At stop 2.3 the lowest of these beds follows, by one varved packet, sand layers from a maximum advanced Sanpoil glacial tongue

Figure 17 (*Continued*). (B–H) Inferred correlations of composite Manila Creek section with Missoula-flood deposits to the south, extents of the Okanogan lobe, levels of glacial Lake Columbia, and elevation of the intake to upper Grand Coulee (*Atwater*).

5.6	At 48.0154, –118.9252, turn left onto dirt road. Follow it 0.3 mi westward, toward Columbia River, past piled debris from road construction.
5.9	Park for Stop 2.1 in graveled area at 48.0172, –118.9320. Walk northwestward to terrace edge, beyond power lines.

Stop 2.1. Gentle Grade to Coulee Floor (Panorama from 48.0190, –118.9331) (Figs. 5, 14, 15A)

From a Nespelem terrace lip at altitude 530 m, the Columbia valley view here extends from badlands near the townsite of Barry (north) to the Grand Coulee intake (south) (Fig. 14). The Okanogan lobe at its maximum extended beyond the site of Grand Coulee Dam and eastward, behind us, to the area of basalt erratics of road-log mile 8.1. The Nespelem terrace, here a recessional landform inset into the ice margin, projects southward toward the coulee floor as aligned remnants on both valley walls. Also conspicuous are lower benches from deglacial incision.

The Okanogan lobe initially impounded last-glacial Lake Columbia 60 km to our west near Brewster (Fig. 5). Glacial advance up Columbia valley emplaced lodgement till above glacial-lake deposits 10 km to our northwest at Hopkins Canyon (Fig. 14) and to the south at Grand Coulee Dam. Beneath the till in both places, sandy graded beds alternate with intervals of varves. Flint and Irwin (1939, plate 1) called attention to this pattern, which they ascribed to "sudden drainage" of "temporary high-level lakes" that had been dammed by other parts of the ice sheet (p. 671). A modern view identifies glacial Lake Missoula as a source of periodic floods into other lakes (Waitt, 1984) that include glacial Lake Columbia (Atwater, 1984, 1986; Hanson and Clague, 2016).

Upon recession of the Okanogan lobe, the arm of glacial Lake Columbia that reoccupied this part of Columbia valley soon filled with sediment (Flint and Irwin, 1939, p. 669–673). Rounded cobbles and boulders beneath our feet attest to riverine deposition across the former lake floor. The clasts may have been brought in, or at least rinsed by, Missoula floods of late-glacial age that backflooded this part of Columbia valley by running down it to the Okanogan lobe, before reversing direction to drain through the Grand Coulee. Later, hundreds of varves each tens of centimeters thick filled depressions that probably originated as stagnant ice. Uninterrupted by signs of periodic floods, this silt forms badlands visible from here on the far side of the Columbia River, and also out of view to the north (Barry and Monaghan Rapids sections of Atwater, 1987).

It has been thought that a residual glacial Lake Columbia unleashed one or more late-glacial floods that descended Columbia valley as the Okanogan lobe wasted (Waitt, 2016; Balbas et al., 2017; Stop 3.1). But the lake's late-glacial remnants may have held little water, as judged from gravelly evidence for sedimentary filling here and in the Sanpoil valley (road log 19.9 mi, between Stops 2.2 and 2.3), and also from thickly varved fill as in the Barry badlands (Fig. 15A, yellow squares). A search for other post-Missoula water sources might be redoubled.

Mile	
6.3	Return to pavement. At Peter Dan Road. Turn left, heading eastward below a terrace, perhaps a kame, into which the Nespelem terrace is inset.
6.6	Approaching ice margin. Ice-marginal drainage likely accounts for east-facing cliff 1 km north of here (Milliken, 1981, p. 20, 22). An adjoining bench near altitude 720 m probably formed as a kame (Fig. 15A).
7.9	Diamict crops out in road cuts through a lumpy ridge that crosses Peter Dan Creek.
8.1	Basalt haystack at left. Another such erratic crops out just ahead, on the right, mostly enveloped in drift. The flat farther east, though seemingly isolated from Columbia valley by the maximum Okanogan lobe, was subject to backdoor flooding—both by a high-level glacial Lake Columbia and by a flood-swollen glacial Lake Columbia—through a channel 3 km south of here (Fig. 15A).
9.9	Beginning of road cuts into weathered granitic rocks without sign of glaciation. The rest of the day's stops remain outside the ice margin, approaching it most closely at Stop 2.3 (Fig. 14).
13.6	Shoulder parking at right, on outside of hairpin turn, for Stop 2.2. Walk 100 m southwestward to ridge-crest pedestals of granitic rock.

Stop 2.2. Sparse Strandlines (Panorama from 48.0039, –118.8036) (Figs. 14, 15B)

Did a residual rock wall separate upper Grand Coulee from Columbia valley when last-glacial Lake Columbia began (Stops 1.4, 3.2)? A high-level last-glacial Lake Columbia is evidenced both by landforms on view here and by deposits visited next in Sanpoil valley (Stops 2.3, 2.4). But this lake stand is more simply explained by a short-lived barrier of glacial ice (F and G, Fig. 17).

Landforms attest to interflood levels of glacial Lake Columbia that its Grand Coulee outlet controlled. The abundantly recorded level, close to 500 m, is marked by the Nespelem terrace (Stop 2.1), another part of which forms a bench below us here, in Swawilla Basin (green, Fig. 15B). Higher lake levels, as noted by Flint (1935, p. 189), are evidenced by horizontal bands on hillsides. Many are obscure except in low-angle light. Examples seen on aerial photos and lidar are plotted in red on Figures 14 and 15. In addition, a brief highstand near 700 m is confirmed by wave-winnowed angular gravel that Cochran and Warlow (1980) mapped in Sanpoil valley roadcuts (Atwater, 1986, p. 3–7, 17; red diamonds on north side of Mount Tolman, Fig. 14).

At most times of day, the hillsides in view here display sparse signs that high-level glacial Lake Columbia lapped onto their

weathered granitic rock. A few ridge crests flatten near 700 m to our southeast (orange band, Fig. 15B). Benches or notches contour hillsides up to that level below a basalt outlier another five kilometers beyond (red lines, Fig. 15B). These areas are otherwise isolated from basalt stratigraphy that produces ledges and notches, from kame terraces like those near Peter Dan Creek (mileage 6.6), and from Missoula-flood scour that is here impeded by a ridge between Swawilla Basin and Sanpoil valley (Figs. 14, 15B).

Mile	
13.6	Continue eastward into the drainage basin of Manila Creek. The road takes this name on the Sanpoil side of the divide.
18.3	Dirt road at right provides vehicle access to Stop 2.5, bypassed for now.
18.5	Lake beds in road cut approach surface of Nespelem terrace, here silty.
19.9	On both sides of a hairpin turn, road cuts expose pebbly sand that extends upward to the Nespelem terrace (Fig. 15B). This coarse cap probably represents a Sanpoil River outwash plain built into glacial Lake Columbia and across the mouth of Manila Creek (Atwater, 1986, p. 11, 14; pl. 1). Like the terrace gravel at Stop 2.1, this fluvial coating limits late-glacial Lake Columbia as a source of post-Missoula floods.
20.6	T-intersection with SR 21. Turn left onto highway, heading north toward Keller.
22.5	Campground with toilet.
0.0	From the campground, **reset trip odometer** upon returning to SR 21 northbound.
2.9	Keller Grocery.
3.3	Parking for Stop 2.3, in graveled area immediately north of Silver Creek Road.

Stop 2.3. Muted Backflooding (River Bank 48.0845, –118.6910) (Figs. 14, 17)

Evidenced here are Missoula floods that ascended the Sanpoil arm of glacial Lake Columbia shortly before and during the Okanogan lobe occupation of upper Grand Coulee. The outcrop, which Sanpoil River refreshes, corresponds to the Ranger Station locality of Atwater (1986, figs. 6, 24A). In a pattern displayed also at Stops 2.4, 2.5, and 3.4, sandy graded beds alternate with intervals of varves—evidence for periodic floods into glacial Lake Columbia.

That the floods ran northward up Sanpoil valley, consistent with a Missoula source, is shown by ripple-drift laminae in four of the flood beds. Current indicators in the varved intervals, by contrast, point consistently southward—away from an ice-sheet tongue that at its maximum extended nearly this far downvalley. Paleocurrents from backflooding have also been reported from two Sanpoil outcrops south of here (Atwater, 1986, plate 2A) and from other arms of glacial Lake Columbia (Fig. 14, lavender arrows). Three additional lines of Sanpoil evidence further implicate floods from glacial Lake Missoula:

1. A dominant iron mineral probably came from glacial Lake Missoula. Hanson et al. (2015) measured magnetic properties of 25 flood beds below Stop 2.5 (Fig. 17, Arrieta locality). They found that hematite fingerprints sediment from glacial Lake Missoula, most definitively in two flood beds at "Mp" in Figure 17A but probably also in the other 23 flood beds analyzed.
2. A flood bed at Stop 2.5 rests on a volcanic ash layer tentatively identified as Mount St. Helens Sg (Fig. 16) (Hanson, 2013, p. 206). If identified correctly, this marker bed links floods into Sanpoil valley with Missoula floods at Stop 1.2 and at other southern Washington sites where S tephra layers are interbedded with Missoula-flood rhythmites (Figs. 1, 17B).
3. Counts of interflood varves of glacial Lake Columbia may be compatible, within uncertainties in varve counting, with rhythmic bedding of silt and clay that accumulated in glacial Lake Missoula during intervals between its drawdowns (approximate visual alignment in Figure 17A with Landslide Bend, Ninemile, and Rail Line sections). The most complete of the Missoula sections, at Landslide Bend (Fig. 1; Levish, 1997), is low enough for glacial Lake Missoula to have covered it at just a few percent of maximum lake volume. The cutbank here exposes two kinds of evidence for brief rise in glacial Lake Columbia by glacial blockage of upper Grand Coulee. First, four of the flood beds are anomalously thin, both here and at three other Sanpoil sections downstream. These four resemble flood beds above and below in counts of preceding varves (~40–50). But the upper three of the anomalous beds, in addition to being thin, each contain a gray clayey cap, in contrast with a brown cap of flood beds above and below. These anomalies in thickness and color imply muting of flood effects in a deep and extensive lake, given three assumptions: (1) flood size at the outlet of a self-dumping glacial Lake Missoula varied inversely with flood frequency; (2) flood-swollen glacial Lake Columbia accounts for ice-rafted erratics that extend as high as 750 m in Sanpoil valley (Fig. 14); and (3) raising glacial Lake Columbia to this level agitated the lake more if from a starting level near 500 m than from one already near 700 m (Atwater, 1986, p. 35–36).

An Okanogan lobe maximum is further evidenced by an interval of uncommonly thick and sandy varves below those anomalously thin flood beds. This interval, which here extends below river level, is 4.5 m thick 4 km south of here while tapering downvalley from there to a thickness of 1.0 m (Atwater, 1986, p. fig. 9). The flood bed immediately above it is the highest one of ordinary thickness and color below the anomalously thin flood beds. The maximum last-glacial advance of the Sanpoil tongue

is evidenced by morainal boulders 4 km north of here (Atwater, 1986, p. fig. 2, pl. 2A). If the thick, sandy varved interval marks that advance, the paucity of sand in succeeding varved intervals may record calving of the Sanpoil tongue into a glacial Lake Columbia deepened by Okanogan-lobe advance into an already cut Upper Coulee (Figs. 17G, 17H).

Mile

3.3	Turn around in parking area, to head south on SR 21, retracing route to Manila Creek.
8.5	Continue on SR 21 past turnoff to New Manila Creek Road.
10.9	Graveled area at driveway signed 10929. Parking for Stop 2.4.

Stop 2.4. Energetic Early Floods (Lake Shore 47.9800, –118.6950) (Fig. 16)

The muted flood effects at Stop 2.3, taken as indirect evidence for maximum advance of the Okanogan lobe, correlate stratigraphically with upper parts of two waterside sections in this area (French Johns and Henry Kuehne localities of Atwater, 1986, fig. 6, table 2). By extension, advance of the Okanogan lobe into an already-cut Grand Coulee probably postdates topographically lower, gorgeously complex deposits exposed in the pair of bluffs here.

Disruptive Missoula floods produced this complexity while engorging or entraining glacial Lake Columbia, whose approximate water depth here may have tripled. Complex flood beds typify glacial-lake deposits along this reach of Columbia valley. Missoula floods routinely mobilized varves into intrusions and entrained them as rip-up clasts (Hanson and Clague, 2016, Figure 6A–6C). In addition, sand freshly deposited higher up the lake floor commonly ran downslope as turbidity currents while the main flood flows waned—a flow stage marked by upward fining of laminated fine sand through silt into brown silty clay (Atwater, 1986, figs. 8B and 25C).

Some of the deposits entrained likely continued downstream as sediment that Missoula floods discharged from Grand Coulee. On Ephrata Fan, flood sediment derived from glacial Lake Columbia may have contributed to a silty matrix in gravel, and this matrix in turn may have facilitated boulder transport (Stop 4.1). Adding entrained glacial Lake Columbia sediment might help simulated Missoula-flood flows approach downstream high-water indicators (Denlinger et al., 2021).

The pair of bluffs is among many undercut by adjacent Franklin D. Roosevelt Lake, the reservoir behind Grand Coulee Dam, which fills to 393 m. Flood drainage ripples in the northern bluff, dubbed Sage Trig (Atwater, 1986, figs. 8C, 26), provided Shaw et al. (1999) with selective evidence for floods from the north.

Mile

10.9	Turn left, northbound on SR 21, retracing route to New Manila Creek Road.
13.3	Left turn onto New Manila Creek Road. Climb back onto Nespelem terrace
15.6	Turn left onto dirt road that leads to paved remains of Old Manila Creek Road.
16.1	Parking for Stop 2.5.

Stop 2.5. Waning Late Floods (Road Cuts at 48.0094, –118.7092 [Switchback] and 48.0098, –118.7061 [Arrieta]) (Figs. 15B, 16, 17)

The Sanpoil valley has provided stratigraphic evidence for no fewer than 89 Missoula floods of last-glacial age. Half that total, probably including the final floods, was evidenced here during the early 1980s, in roadcuts designated Switchback (near the parked vehicles) and Arrieta (200 m east, downhill). Bureau of Reclamation boreholes at the foot of Arrieta extended the sequence downward through sets of anomalously thin flood beds matched with those at Stop 2.3.

Silt-and-clay couplets in the composite sequence, where intercalated with flood beds, add up to roughly 1,800–2,600 varves (Fig. 17C). This range is intended to capture uncertainty in counting varves that are commonly compound (hence the stingy to generous ranges in Figures 17A and 17C). Varve-counting uncertainty is greater at Switchback than at Arrieta because, although the Arrieta varves accumulated in an arm of glacial Lake Columbia, much of the Switchback sequence was laid down instead in a pond beside a Sanpoil outwash plain that had isolated Manila Creek valley (details, trip mile 19.9).

Still exposed in the Switchback section is evidence for the last Missoula floods. Interflood varves become progressively fewer until just a varve year or two separates beds interpreted as floodlaid (Fig. 16A). Additional varves without evidence for periodic floods, extend upward from there and are exposed more abundantly to the north, where the new Manila Creek Road cuts into the Nespelem terrace (Fig. 15B).

Also notable at Switchback is a volcanic ash layer that may link histories of glacial lakes, their ice dams, and outburst floods. The layer was logged in 1983 for its inverse grading (near nominal altitude 496 m in plate 3 of Atwater, 1986), but it was not seen as tephra until 2006, when recognized as such by René Barendregt, working with Hanson. Tephra analyses by Stephen C. Kuehn, then at University of Alberta, revealed two geochemical populations of shards (Table 1). One Kuehne assigned to Mount St. Helens Sg; the other to the far earlier Mount St. Helens set C. (It is hard to understand much older "C" tephra within the "S" tephra unless by unlikely contaminant entrained at the volcano.) Potential implications of the S tephra in Figure 17:

1. ***Flood frequency and size in southern Washington.*** If the tephra at Switchback is layer Sg, southern reference sections at Burlingame ravine and Gardena cliffs begin late in the last-glacial history of Missoula floods, after interflood periods have declined from four or five decades to two or three (Figs. 17A and 17B). If flood size increased

with interflood period, those sections exclude evidence for the largest Missoula floods.

2. ***Ice-sheet chronology.*** Varve counts below and above the tephra layer enable approximate dating of Okanogan lobe extents and Lake Missoula's demise (Figs. 17C–17F). These proxy estimates hinge in part on the radiocarbon age or ages used to date set S tephra. Among the ages compiled by Clynne et al. (2008, p. 619), the youngest measured on charcoal (12,910 ± 160 ^{14}C yr B.P.; W-3141) corresponds at 95 percent confidence to the interval 15.9–15.0 cal ka (IntCal20 calibration data of Reimer et al., 2020). If this charcoal age is equated with the tephra layer age, varve counts below the tephra imply that the last-glacial Okanogan lobe advanced across the Columbia River by 18.4–16.7 cal ka and occupied upper Grand Coulee for a few varve centuries at most somewhere within 17.7–15.9 cal ka. In addition, the last floods from Lake Missoula occurred sometime in 15.8–14.9 cal ka. By the same token, a piece of detrital wood from water-laid diamict beneath the Arrieta locality, dated by radiocarbon to 18.6–16.9 cal ka, was deposited 17.2–15.8 cal ka—a disconcerting but plausible offset for woody detritus.
3. ***Counts and sources of late floods.*** In sections both here (Switchback) and near Steamboat Rock (northernmost part of Stop 3.4), intercalated flood beds and varved intervals grade upward into thick varves-a gradation that complicates counting of flood beds. From independent counts by Atwater and Hanson, the Switchback section contains evidence for as many as 18 post-tephra floods from Lake Missoula. In southern Washington (Fig. 17B), at higher altitude with respect to local base level, a dozen or so flood rhythmites postdate set S at Burlingame ravine (Waitt, 1980, p. 659), Mabton (Waitt, 1985, p. 1284), and Wanapum Dam (Stop 1.2) (Moody, 1987, p. 206–209; Smith, 1993, Fig. 3), as do 31 along the Walla Walla River at Gardena cliffs (Clague et al., 2003, their Touchet locality of fig. 2, p. 248). Do the additional rhythmites at Gardena cliffs require a flood source other than glacial Lake Missoula?

TABLE 1. GEOCHEMICAL BASIS FOR MATCHING A SANPOIL VALLEY ASH WITH A MOUNT ST. HELENS ASH LAYER

Oxide	1 Mt. St. Helens Sg standard* n = 20		2 Burlingame lower tephra n = 23		3 Zillah lower tephra n = 22		4 Zillah lower tephra n = 34		5a Sanpoil Valley chemical group a n = 19		5b Sanpoil Valley chemical group b n = 7	
	Mean	S.D.†	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
SiO_2	76.50	0.14	76.37	0.25	76.78	0.28	76.44	0.28	76.69	0.31	76.42	0.45
Al_2O_3	13.80	0.09	13.87	0.13	13.82	0.09	13.66	0.13	13.59	0.14	14.32	0.84
Fe_2O_3 or *FeO*	1.29	0.03	1.33	0.09	1.31	0.05	1.31	0.07	1.25	0.10	0.93	0.15
TiO_2	0.16	0.02	0.17	0.02	0.17	0.02	0.15	0.04	0.14	0.04	0.12	0.02
Na_2O	4.11	0.12	3.96	0.17	3.67	0.31	4.07	0.18	4.04	0.20	3.59	0.69
K_2O	2.14	0.04	2.29	0.13	2.24	0.11	2.27	0.08	2.24	0.07	2.63	0.25
MnO	N.D.§		N.D.		N.D.		0.03	0.03	0.05	0.02	0.05	0.03
MgO	0.32	0.02	0.33	0.04	0.32	0.02	0.32	0.03	0.30	0.03	0.25	0.03
CaO	1.59	0.02	1.60	0.06	1.58	0.04	1.66	0.06	1.63	0.08	1.68	0.06
Cl	0.09	0.01	0.09	0.01	0.08	0.01	0.09	0.02	0.08	0.02	0.07	0.02
Total	100.00		100.00		100.00		100.00		100.00		100.00	
Correlated Mount St. Helens ash layer			Sg		Sg		Sg		Sg		Similar to set C; possibly reworked material	
Similarity coefficient#			0.98		0.98				0.93		0.84	

Note: Analyses by Franklin F. Foit Jr. at Washington State University (columns 1–3; Clague et al., 2003, their table DR2) and by Stephen C. Kuehn at the University of Alberta Tephrochronology Laboratory (columns 4 and 5; Hanson, 2013, p. 206). Columns 5a and 5b represent different geochemical populations that Kuehn identified by plotting 26 individual analyses from a single field sample. Samples of Zillah lower tephra (columns 3 and 4) were collected at different times and analyzed separately. The iron oxide is Fe_2O_3 for the comparisons among columns 1–3 and FeO for the comparisons in columns 4, 5a, and 5b.

*Sample location not reported by Clague et al. (2003).

†S.D.—standard deviation.

§N.D.—no data.

#Clague et al. (2003) matched the lower tephras in couplets at Burlingame and Zillah (columns 2 and 3) with the Sg standard in column 1. The lower tephra in the Zillah couplet (column 4) serves as an Sg reference for the 19 analyses grouped geochemically in column 5a. Similarity coefficients in columns 2 and 3 are reported from Clague et al. (2003). Similarity coefficients in columns 5a and 5b were computed with equation (1) of Borchardt et al. (1972).

Mile	
16.1	Retrace path, by way of Old Manila Creek Road and New Manila Creek Road, to SR 21.
18.9	Foot of New Manila Creek Road. Turn right onto SR 21, heading toward Columbia River.
23.8	Embark Keller Ferry. Viewed from ferry, Nespelem terrace spreads widely to the east. Horizontal lines near 700 m can be made out in favorable light.
23.8	Disembark Keller Ferry. Continue south on SR 21, toward Wilbur.
26.7	To the northwest above the floor of Swawilla basin on the far shore, basalt boulders probably delivered by landslide (Fig. 15B).
27.6	Probable strandlines mapped on hillslope to east. Continue south on SR 21 to outskirts of Wilbur. Turn right onto SR 174, to Grand Coulee. In town of Grand Coulee, turn right onto SR 155. Descend hill past dam to motel.

Day 3. Upper Grand Coulee

Day 3 focuses on upper Grand Coulee (Figs. 14, 15, 18–27). Some hypotheses by different coleaders do not necessarily agree with each other. Indeed, different points of view point out a number of enigmas of upper Grand Coulee as we view together some of the evidence.

Reconstruction of Flood Discharge in Upper Grand Coulee (*Lehnigk and Larsen*)

Upper Grand Coulee is the largest canyon in the Channeled Scabland. Yet discharge that drove its incision is largely unconstrained. Bedrock erosion and ice-sheet advance altered the size of the channel that conveyed floods during coulee development, which confounds efforts to reconstruct discharge from high-water evidence. We have run our own new series of 2-D hydraulic models with different channel geometries encountered by floods.

Upper Grand Coulee was incised by headward retreat of a cataract driven by spillover of floodwaters from Columbia valley. To the south, the floodwaters encountered the steep limb of the Coulee monocline. From the monocline, the cataract retreated upstream by plucking jointed basalt, leaving the rock monolith Steamboat Rock as evidence of its passage (Bretz, 1932). The cataract retreated farther headward until cutting through the drainage divide into the Columbia valley (Bretz, 1923, 1932). A catastrophic flood would have been released through the coulee and been exceptionally large if glacial Lake Columbia drained during Lake Missoula flood (Bretz, 1969, p. 527). From upper Grand Coulee, floodwaters expanded into Hartline basin, depositing an immense bar. The basaltic composition of the gravel indicates the bar accumulated during cataract retreat (Stop 3.6). During the last glaciation, the Okanogan lobe's advance into upper Grand Coulee raised the level of glacial Lake Columbia (Waitt and Thorson, 1983; Atwater, 1986). Glacial striae and erratics on Steamboat's summit indicate ice covered it (Bretz, 1932, p. 35). The ice apparently reached an inferred maximum extent on the high scabland tract along the eastern rim of the coulee (Bretz, 1932; Hanson, 1970; Kovanen and Slaymaker, 2004).

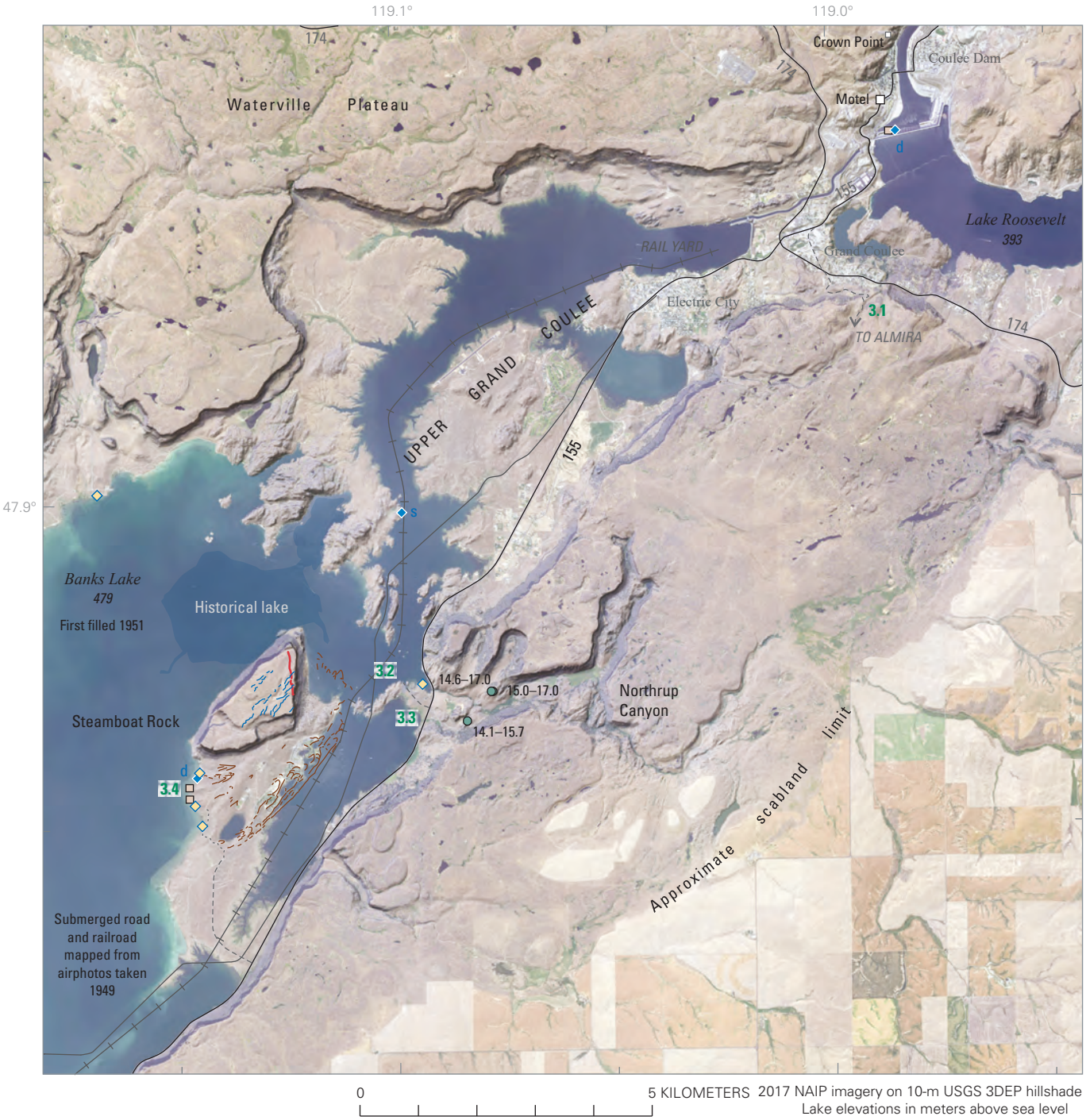
The boundary between flood-scoured basalt and loess on the east rim of upper Grand Coulee defines the maximum elevation of floodwaters. These high-water marks occurred either during a prior glaciation (Atwater, 1986, p. 37) or early during the last glaciation (O'Connor et al., 2020, p. 18–19). It is uncertain whether the high-water evidence occurred when (a) the cataract was retreating before the coulee reached its present depth, (b) the Okanogan Lobe occupied the coulee and covered Steamboat Rock, diverting flow to the east, or (c) a large Missoula flood entered the fully formed coulee. Recent modeling of Lake Missoula outbursts through upper Grand Coulee suggests the last scenario is unlikely, as 2-D simulated floods did not fully inundate the high scabland on the east rim (Denlinger et al., 2021). High-water marks adjacent to the Hartline bar also constrains the maximum flood stage in the lower part of upper Grand Coulee. However, inferred discharge at the time of Hartline bar depends on whether Dry Falls had receded to its current position at the head of lower Grand Coulee or whether the falls were farther south. Bretz (1923, p. 644) determined that the bar was deposited before lower Grand Coulee had been eroded.

An earlier paleo-flood reconstruction in upper Grand Coulee yielded an estimated discharge of $12–14 \times 10^6$ m^3 s^{-1}, if present-day topography was filled to the elevation of high-water marks (Harpel et al., 2000; Waitt et al., 2000; Waitt et al., 2009; O'Connor et al., 2020). However, if lowering of canyon floors during flooding caused high-water marks to become abandoned, discharge is over-estimated (Larsen and Lamb, 2016). Hence we may extract a more nuanced history of discharge by reconstructing topography encountered by floods during different stages in the development of upper Grand Coulee.

We reconstructed topography and ran several two-dimensional hydraulic models to quantify paleo-flood discharge in Grand Coulee during plausible stages of canyon development. The topography at the time of cataract retreat was generated by reconstructing the channel bed elevation between Steamboat Rock and the paleo-divide to constrain the flood discharge responsible for incising upper Grand Coulee to its current depth. We simulated the maximum flood discharge when the Okanogan lobe occupied Grand Coulee by routing flow onto the east rim, flow confined on the west by a wall of Cordilleran ice. For each scenario we ran a series of simulations with steady-state discharge and determined which discharge best matched high-water evidence on the east rim of the coulee or in Hartline basin, based on high-water marks reported by O'Connor et al. (2020) and Baker (1973).

Road Log and Stops

Mile	
0.0	Columbia River Inn. Facing Grand Coulee Dam, bear right onto SR 155 south.



EXPLANATION

- **Dune field**—Parabolic and linear ridges on airphoto taken 1949
- **Glacial-lake deposits** without recognized evidence for periodic floods—Exposed more widely than shown
- **Sandy floodlaid beds** intercalated with varved intervals—Fig. 23
- s d **Evidence for grounded ice**—Diamict (d) at Grand Coulee Dam (Flint and Irwin, 1939) and at stop 3.4 (Fig. 7B); striae (s) beside Banks Lake at low water (Fig. 7A; observed by Bruce Bjornstad); end moraines on Steamboat Rock (modified from Crosby and Carson, 1999; lines show sinuous traces in Google Earth)
- **Lateral spread**—Sakungen on Steamboat Rock (Crosby and Carson, 1999), apparently draped by end moraines
- 15.0–17.0 **Exposure age** (ka)—Measured by Balbas et al. (2017) on granitic bedrock. Range at two standard deviations for internal error
- 3.1 **Trip stop**—Day 3

Figure 18. Head of upper Grand Coulee and vicinity (*Atwater*).



Figure 19. Oblique aerial view northwest across the flood-breached head of upper Grand Coulee, the floor of which is perched almost 200 m above the Columbia River (upper right) (*Bjornstad*).

1.8	Left onto Spokane Way.
2.3	Cross SR 174 (the modern highway) on a dogleg left, still Spokane Way, signed toward Almira. This is the old highway to Wilbur.
3.0	Park on broad shoulders of road's "S" curves.

Stop 3.1. Overlook Head of Upper Grand Coulee and Columbia Valley (47.9308, –118.9921)
(*Bjornstad*)

When ice of the Okanogan lobe advanced to block the Columbia River, megafloods from glacial Lake Missoula were diverted across a high divide (~680 m) near the head of Grand Coulee (Figs. 18, 19) (Waitt 2021). With each flood a giant recessional cataract that began 40 km downstream eroded headward. After repeated floods, all arguably within pre-late Wisconsin time, the cataract broke through the head of the coulee suddenly lowering the flood intake 200 m or more. Later, ca. 15 ka, the Okanogan lobe advanced to fill the upper end of Grand Coulee, temporarily rerouting floodwaters eastward down the Cheney-Palouse and Telford-Crab Creek tracts (Figs. 5, 6D), as well as nearby Northrup Canyon. Subsequently, partial retreat of the Okanogan lobe allowed the last dozen or so smaller floods from glacial Lake Missoula to drain via Grand Coulee (Atwater, 1987; Waitt et al., 2009). For several hundred years after the last Missoula flood, glacial Lake Columbia continued to occupy the Columbia valley and upper Grand Coulee based on varved sequences preserved within. Finally, at ca. 14 ka, a last outburst from the sudden breakup of the Okanogan lobe released the contents of glacial Lake Columbia solely down the Columbia valley, permanently cutting off the flow of surface water via Grand Coulee.

(*Waitt*)

Grand Coulee Dam is the centerpiece of the Columbia Basin irrigation system. Columbia water pumped up to Banks Lake in upper Grand Coulee distributes by gravity to vast areas of Quincy, Othello, and Pasco basins—Columbia water flowing to basins that haven't felt it since the Missoula floods.

Flint and Irwin (1939) in the excavations of the Grand Coulee damsite and farther downvalley, and Jones et al. (1961) downvalley, recognize lower and upper glaciolacustrine beds separated by till. This stratigraphic sandwich shows glacial Lake Columbia dammed during both the advance and retreat of the late Wisconsin Okanogan lobe. Flint and Irwin (1939) explained the varved intervals alternating with coarser or thicker beds by waxing and waning of the Okanogan lobe downvalley, and episodic drainage of glacial lakes upvalley. Atwater (1986, 1987) later proved these sedimentary alternations to be from Missoula floods from the east engorging varve-making glacial Lake Columbia.

Traces of shorelines at the high level (715 m) of glacial Lake Columbia are scant (Day 2) (Atwater, 1986, p. 6–7). This implies that the lake stood high only briefly—when Okanogan ice blocked the head of upper Grand Coulee. The great cataract must have cut through to the Columbia *before* the late Wisconsin glaciation. Thus, Grand Coulee conveyed most and perhaps all late Wisconsin Missoula floods large and small (Fig. 6, B–D).

Step-Backwater (1-D) modeling. Great flood down an ice-free Columbia valley had in the 1990s been calculated by 1-D methods (Harpel et al., 2000)—rerun in 2017 (Fig. 8)—to fill the valley up to the limits of field evidence such as highest ice-rafted erratics (fig. 7). Grand Coulee's intake in its current shape 183 m above Columbia valley received this peak flow more than 153 m deep during this ice-free stage of Columbia valley.

But maximum-flood indicators around the head of Grand Coulee—upper limits of scabland, loess scarps, and erratics—are more than 100 m above the 1-D modeled Columbia valley water-surface altitude around 640 m. The high-flood limits near Grand Coulee's head cannot have been reached during an open-Columbia setting. This high-level flooding instead must have been when the Okanogan lobe had plugged upper Grand Coulee, or the Columbia downvalley, and turned all Missoula floodwaters into high-level spillways.

2-D modeling. A shallow-water 2-D "dambreak" model has been applied to the Missoula floods with the Columbia below Grand Coulee blocked by the Okanogan lobe (Denlinger and O'Connell, 2010)—the prevailing setting during most last-glacial floods.

At 23 h after dambreak, floodwater pours deeper than 200 m into upper Grand Coulee at more than 20 m/s through the relatively narrow 1 km nozzle at coulee's head (Fig. 20). This scenario brings water up to about the limit of field evidence, including over Steamboat Rock's scabbed summit later capped by glacial drift during the Okanogan lobe's maximum stand. Water in Columbia valley just below Grand Coulee is deeper than 250 m. Flow vectors are almost nil in dead-end Columbia valley against the glacier.

This hydraulic model shows that glacial Lake Missoula alone contained enough water to flood nearly up to the limits of field evidence—no additional mysterious water source farther north required.

A new generation of more robust 2-D hydraulic models—by David George, Roger Denlinger, and Charles Cannon (all USGS)—is being run on closer topographic grids than earlier and for several different hypothetical configurations of the Okanogan lobe and glacial Lake Columbia (Fig. 9) (e.g., Denlinger et al., 2021; O'Connor et al., 2020). This ongoing project is far from complete, but generally the 2-D methodology yields smaller peak discharges throughout the flooded system than did the earlier 1-D methodology along some of the major channels.

Mile	
3.0	Continue up road.
3,2	On straight reach with wide shoulders, turn around. Descend Spokane Way.
3.4	Pass Stop 3.1.
4.6	Left onto SR 155 south.
5.0	Intersections with SR 174. Stay on SR 155 south.

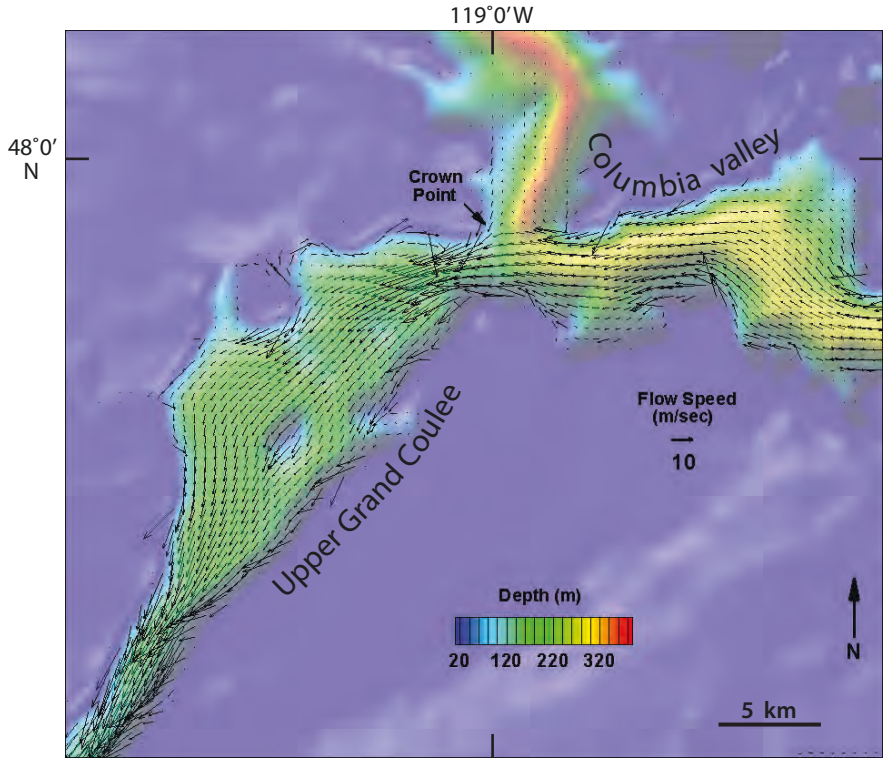


Figure 20. Depth and velocity of Missoula flood through glacial Lake Columbia and down upper Grand Coulee by 2-D hydraulic model of Denlinger and O'Connell (2010) at time 23 h after dambreak (from Waitt et al., 2009, fig. 34). Latitude and longitude ticks and scale approximate, placed on figure manually, but compare to map Figs. 14 and 15. Velocity (depicted by length of black arrows) drops nearly to nil down the dead-end Columbia valley blocked by ice farther downvalley but speeds up through the relative narrows in upper Grand Coulee.

5.4	North dam for irrigation project Banks Lake.
8.4	Dissected flat near 495 m altitude. Likely continuation of Nespelem terrace.
8.7	Knolls of Mesozoic to lower Cenozoic granitic rocks right and left.
9.8	Jones Bay turnoff. The slot to south in granitic rocks provides an elevated temporary spillway for glacial Lake Columbia when the Okanogan ice lobe is banked against west side.
11.1	Varved lake beds in lakeshore bluff ahead. This section probably correlates with the post-flood varves at Stop 3.4.
11.7	Turn right into state park road to Northrup Point. Park near boat ramp and picnic area. Toilets at picnic area. Short uphill walk on trail that heads near boat-ramp toilet.

Stop 3.2. Steamboat Rock Outlook: Granite Knoll (47.8724, –119.0992)
(*Waitt and Atwater*)

Rock hills that rise from the floor of upper Grand Coulee escaped removal during cataract retreat. Many are made up of crystalline basement that projects beneath the grandest of the hills, Steamboat Rock (Fig. 13). Its steep flanks, like the coulee walls east and west, expose Miocene basalt mapped as Grande Ronde (lowest), Roza, and Priest Rapids (Swanson et al., 1979, plate 4).

Steamboat's broad summit records shaping by flood water and glacial ice (Bretz, 1932, p. 35–37). Scabland features include a coulee that traverses the butte east-west and which was left hanging by cataract retreat. Glacial striae trending away from Waterville Plateau are inscribed into basalt in the south, and morainal ridges in the northeast (Fig. 18) lack recognized signs of modification by floods (Crosby and Carson, 1999, p. 6). Bretz (1932, p. 35) thought that "the making of Grand Coulee" required a minimum of two "glacial epochs," one for catastrophic flooding and a second for glacial override. Crosby and Carson called upon two glacial advances among three times of flooding, with the last of the floods surrounding but not overtopping a fully eroded Steamboat Rock.

Nearby landforms and deposits have provided further evidence that Steamboat Rock's erosion preceded its most recent glaciation. On crystalline-rock knolls on the coulee floor stretching 5 km to our northwest, the "crumbling granite" lacking "good striated surfaces" (Bretz, 1932, p. 36) does retain smoothing and rounding that Bretz found consistent with glacial flow from Waterville Plateau. To our southwest at Stop 3.4, just down-coulee from Steamboat Rock, glacial-lake deposits rest unconformably on diamict interpreted as lodgement till (Atwater, 1987, p. 188).

New Evidence for Glaciation within Upper Grand Coulee
(*Bjornstad*)

There has long been evidence for glaciation atop the Waterville Plateau and Steamboat (Bretz 1932, 1969; Crosby and

Carson 1999; Bjornstad and Kiver 2012) but a general lack of conclusive evidence reported on the upper coulee floor. This could suggest that upper Grand Coulee had not been occupied by ice during the last (late Wisconsin) glaciation.

In August 2011 during a kayak reconnaissance of granite inselbergs along Banks Lake 2.6 km northeast of Steamboat Rock, I found and photographed a surface on massive to slightly fractured granite that shows parallel striations and polish characteristic of glaciated bedrock (Fig. 7A). On the near-vertical to slightly overhanging exposure trending 145 (S35E), striations dip southeast 10–25 degrees. The down-coulee dipping striations are conspicuous on the lichen-free surface below the summer waterline. The striae continue up above summer high lake level but are visually masked by a heavy cover of lichens. In September 2020 while revisiting the area, I found two nearby sites with likely glacial polish and grooves.

Why so few examples of ice contact on the floor of upper Grand Coulee? All three sites may have been covered in lacustrine silt from ancient glacial Lake Columbia that protected them since the late Wisconsin from exposure and weathering. Since Banks Lake was first filled in 1951, much silt has been washed away by waves along the lakeshore, exposing long-covered rock surfaces. The general lack of observable striations elsewhere on granitic inselbergs may owe to the millennia of weathering since the Pleistocene.

Upper Grand Coulee 2-D Hydraulic Modeling
(*Lehnigk and Larsen*)

When the cataract was retreating through upper Grand Coulee and the channel floor was near the elevation of the Steamboat Rock summit, the high-water marks along the eastern rim are inundated by a discharge of at least 2.8×10^6 m^3 s^{-1} (Fig. 21A). In contrast, for the present-day topography, the high-water mark on the eastern rim is inundated at a discharge of 17×10^6 m^3 s^{-1} (Fig. 21B); flow is allowed to exit the model domain via Foster Coulee in this scenario. It is important to note that the 17×10^6 m^3 s^{-1} discharge exceeds high-water evidence in Hartline basin and elsewhere in lower Grand Coulee (Fig. 21B). A discharge of 17×10^6 m^3 s^{-1} is equivalent to the peak discharge inferred in the Spokane valley (O'Connor and Baker, 1992), roughly 150 km to the east.

The breach of the divide with Columbia valley would potentially have triggered a large flood from the drainage of glacial Lake Columbia (Bretz, 1969, p. 527). Although the stage of glacial Lake Columbia at the time of the breach is unknown, it is plausible that the breach was associated with a Missoula flood (Bretz, 1969, p. 527). Hence we simulated a flood by instantaneously removing a dam impounding a "flood swollen" glacial Lake Columbia with a stage of 750 m based on the elevation of ice-rafted erratics (Atwater, 1986). Given there is no evidence of flooding at higher elevations, drainage of the 750 m stage glacial Lake Columbia is the largest possible paleo-flood through Grand Coulee; such a flood could have occurred during a prior glaciation or early in the history of Lake Columbia during the last glaciation. The peak discharge of the lake-draining flood attenuates downstream from 12.8×10^6 m^3 s^{-1} at the breach to 6.9×10^6 m^3 s^{-1} where flow exits lower Grand Coulee and flows into the Quincy basin. Because the initial lake stage is higher than the elevation of the east rim, floodwaters initially inundate the rim, but are routed to the coulee floor via Northrup canyon. The maximum dam-release flood stage within upper Grand Coulee is below the elevation of the eastern rim and the summit of Steamboat Rock is not inundated. Near Steamboat Rock, where high-water marks were used to constrain the maximum discharge in the present-day topography, steady-state discharge model, the maximum discharge is 7.8×10^6 m^3 s^{-1} (Fig. 21C).

When the Okanogan lobe blocked upper Grand Coulee, the high-water marks along the eastern rim are nearly inundated by a discharge of 0.25×10^6 m^3 s^{-1} (Fig. 21D). Whether flow was routed on the eastern rim scabland when the Okanogan lobe occupied Grand Coulee is unknown, and a topic for geochronological investigation.

Collectively, these results indicate the 2.8×10^6 m^3 s^{-1} discharge that caused canyon incision via cataract retreat was only ~16% of that required to fill the canyon to the high-water marks. The maximum 7.8×10^6 m^3 s^{-1} outburst flood discharge from the drainage of glacial Lake Columbia is less than half of the discharge that fills upper Grand Coulee to the high-water evidence at the same location. These discharges are lower than previously reported values from 1-D modeling for upper Grand Coulee (Harpel et al., 2000; Waitt et al., 2000, 2009; O'Connor et al., 2020). Given our findings that drainage of the 750-m-stage glacial Lake Columbia does not inundate the east rim scabland, nor do floods from glacial Lake Missoula (Denlinger et al., 2021), the high-water marks must have been emplaced when the cataract was retreating through the coulee (Waitt, 2021) or, alternatively, when or if the Okanogan lobe forced flow onto the eastern rim. The high-water evidence and present-day topography hence do not provide robust constraints on paleo-flood stage and discharge, because the elevation of the current channel floor differs from when the high-water evidence was emplaced. The maximum discharge when the floor of upper Grand Coulee was blocked with ice was slightly higher than 0.25×10^6 m^3 s^{-1}, indicating no large floods were traversing Grand Coulee when the Okanogan lobe was at its maximum extent. Our discharge estimate is higher than a value of 0.13×10^6 m^3 s^{-1} that is based on glacial Lake Columbia flood-bed sedimentology, though that value was considered to be an under-estimate (Atwater, 1987). The deposits are interpreted to represent later, low-magnitude Missoula floods (Atwater, 1987; Stop 3.4). If floodwaters were routed along the east rim by the Okanogan lobe, the low discharges predicted by our modeling is consistent with occupation of Grand Coulee by ice late in the sequence of last glacial Missoula floods (Stop 2.3).

Steamboat Rock. The high-water evidence and hydraulic model results predict that waterfall retreat in upper Grand Coulee was driven by a discharge of at least 2.8×10^6 m^3 s^{-1}, which is lower than most discharges reported for the Channeled Scablands (e.g., O'Connor et al., 2020). To address whether this low discharge was capable of eroding the Columbia River basalt, we

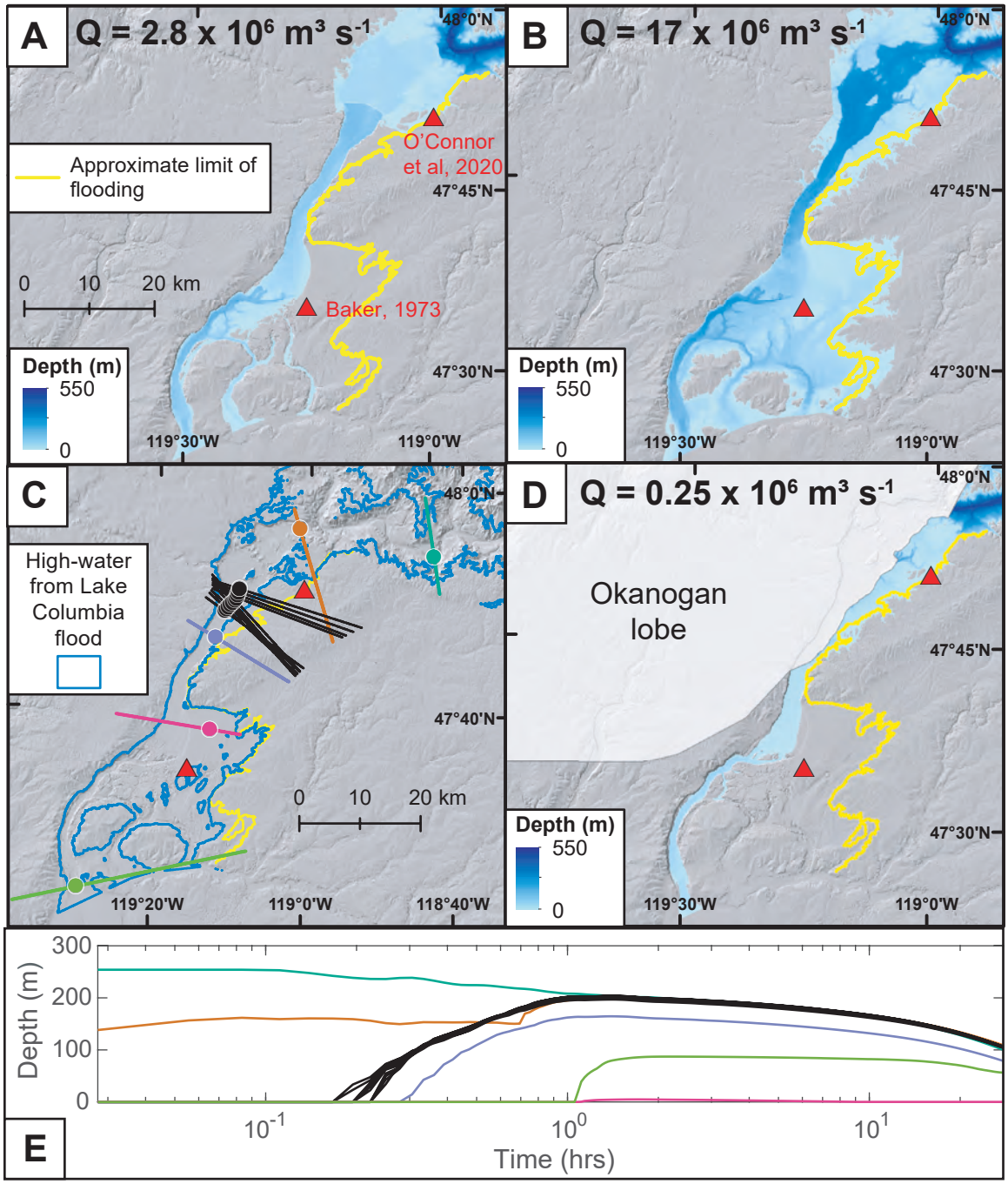


Figure 21. Flood inundation, flow depth, and discharge (Q) for simulated floods in Grand Coulee. (A) A steady-state discharge of 2.8×10^6 m^3 s^{-1} on topography with a reconstructed cataract at Steamboat Rock nearly inundates high-water marks on the east rim. (B) A steady-state discharge of 17×10^6 m^3 s^{-1} on the present-day topography inundates high-water marks on the east rim. (C) The extent of inundation from a flood produced by the instantaneous removal of a dam impounding glacial Lake Columbia with an initial stage of 750 m. (D) A steady-state discharge of 0.25×10^6 m^3 s^{-1} on topography where the Okanogan lobe blocks the coulee floor nearly inundates high-water marks on the east rim. (E) Flood stage as a function of time during the simulated flood from glacial Lake Columbia. The line colors refer to points shown in C (*Lehnigk and Larsen*).

measured the dimensions of 50 large basalt columns near the base of Steamboat Rock. We used the median column dimension and the torque balance model of Lamb and Dietrich (2009) to estimate that the threshold shear stress for failure of a stack of columns on the downstream face of the cataract was 205 Pa (Fig. 22A). Our predicted threshold is exceeded by the median modeled shear stress value of 820 Pa for points along the cataract brink at the discharge of 2.8×10^6 m^3 s^{-1} (Fig. 22B). The theory-based predictions indicate the relatively low discharge of 2.8×10^6 m^3 s^{-1} is sufficient to topple basalt columns, and drive the retreat of the cataract (Fig. 22C). Interlocking of columns or more complicated fracture patterns found in basalt entablature may require exceedance of a yield strength for failure to occur and there is a need for future work to measure the yield strength of jointed basalt in the field (Lamb and Dietrich, 2009). However, the topography of Steamboat Rock and our modeling results suggest the shear stresses generated by a discharge of ~2.8×10^6 m^3 s^{-1} are great enough to exceed the threshold for failure by toppling and to maintain a vertical waterfall, consistent with the interpretation that "a receding waterfall mechanism would operate readily in the horizontal sheets of Columbia basalt" (Bretz, 1932, p. 48).

Mile

11.7	Return upslope east.
12.4	At SR 155 turn right, then a quick left, continuing east on Park's gravel road.
12.4	Park at gravel pit (gated on road's north).

Stop 3.3. Lower Northrup Canyon Late-Stage Gravel Eddy Bars (47.8684, –119.0905)

(*Waitt*)

There's no stratigraphic record in Grand Coulee of the several dozen gigantic late Wisconsin floods, and pre-Wisconsin ones that did most of the erosional work such as geomorphically freshening the upper east scabland. Perhaps over several glaciations they excavated the coulee and carved scabland about the coulee's head. Within upper and lower Grand Coulee, the late Wisconsin floods left sporadic gravel bars and gigantic boulders. The sequence near the mouth of Northrup Canyon records only fairly late and small elements of the energetic last-glacial Missoula floods.

Gravel pits expose an eddy bar that had built from Grand Coulee into lower Northrup canyon, parts of it perhaps by flows down that canyon. The bar comprises five or six foreset-gravel beds containing outsized 1 m boulders. Each gravel bed grades up into a bed of medium to very fine sand recording a slowing then slack current. The contact between the fine bed and overlying gravel is sharp and erosive. Some gentle foreset bedding dips east into this Northrup valley reentrant, apparently floodflow down Grand Coulee. Other gentle foreset bedding dips west. This may be either by floodflow down Northrup Canyon or in a counterclockwise eddy spinning in this reentrant alongside huge flow down Grand Coulee.

Banked against this bar, a lower bar shows four pebble-gravel beds carrying large boulders, each bed capped by laminated medium to very fine sand. The composite exposures thus show that at least 9–10 separate floods eddied coarse gravel into lower Northrup Canyon 25–40 m above the floor of Grand Coulee. Each vigorous flood ended in slack water.

Mile

12.4	Turn around and return to SR 155.
12.5	SR 155. Turn left (south).
15.9	Turn right onto road to Steamboat Rock State Park.
16.0	Cross old (pre-dams) road up the coulee—that Bretz and Flint drove.

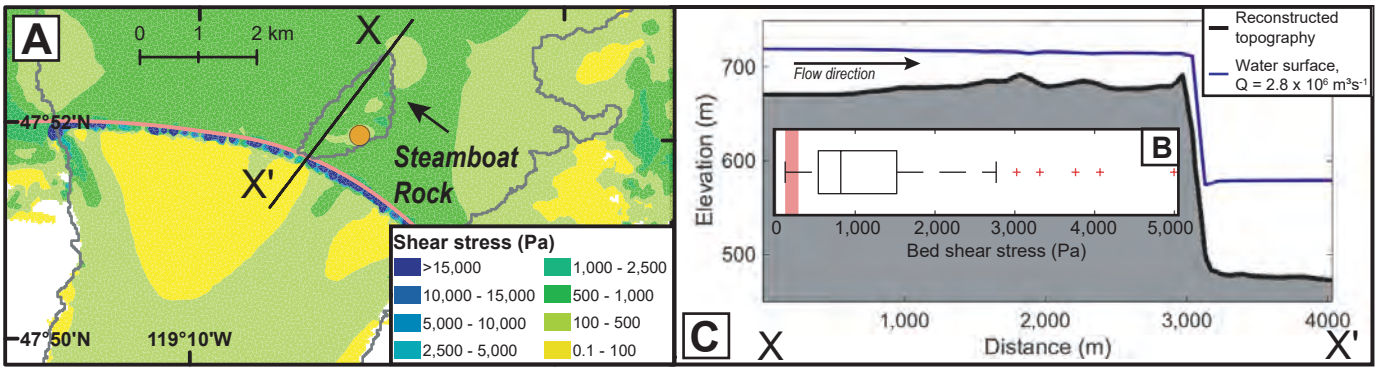


Figure 22. Shear stress thresholds for plucking on the reconstructed waterfall at Steamboat Rock. (A) Bed shear stresses for the high-water-inundating discharge of 2.8×10^6 m^{3}s^{-1} when the cataract would have been retreating at Steamboat Rock. The pink curve shows the location where shear stresses were extracted from the cataract brink; the location where basalt columns were measured to calculate shear stress thresholds for erosion by plucking are indicated with an orange circle. (B) The box plot shows modeled shear stresses along the cataract brink; the median value of 820 Pa exceeds the shear stress threshold for block toppling based on measurements of column dimensions and theory (vertical red line, 205 Pa). (C) Water surface elevation along the cross-section X–X′, indicating headward erosion took place at a cataract or waterfall with a vertical drop in water surface elevation, rather than at a submerged step (*Lehnigk and Larsen*).

17.2 Turn left at dirt (sand-silt) road and park in gravel area.
Hike silt road, or overland northwest, to lake shore ~25 min.
This overland hike shows some 18 m of relief on this apparent pendant bar in the lee of Steamboat rock and scattered large basalt boulders on its surface.

Stop 3.4. Till, Flood Rhythmites, and Varves near Steamboat Rock (47.8582, –119.1456–47.8544, –119.1469)
(*Atwater and Waitt*)

In the lee of Steamboat Rock, stratigraphy exposed in the past seventy years by shore erosion along Banks Lake provides evidence for glacial ice on the coulee floor, frequent floods into a riverine arm of late-glacial Lake Columbia, and continuation of that river lake after those floods had ceased (Fig. 23A).

Glacial ice of the Okanogan lobe, as reconstructed by Bretz (1932, p. 35–38), "pushed partially or completely across the northern part of the upper coulee after the coulee had been eroded to present depths." He cited "a morainal ridge about thirty feet high" atop the summit scabland of Steamboat Rock, and he also noted a coulee-floor ridge that "may be a moraine loop, now silt-covered, made when ice reached the top of Steamboat Rock." That ridge has now been abundantly notched by waves of Banks Lake, first filled in 1951. The deposits exhumed include newly discovered dense black diamict that contains many basaltic cobbles and few granitic ones—apparently lodgement till that holds up steep faces along 50 m of shore at 47.8582, –119.1456 (Figs. 7B, 23B).

Erosive floods were eventually followed by 14 lesser ones evidenced by rhythmites, all but the uppermost two sandy, at 47.8544, –119.1469 (Fig. 23C). Intervening varves provide partial records of a coulee-floor lake that the lesser floods overwhelmed. Loaded, dismembered, and carried were additional varves from interflood times. Counts of surviving varves range from three or four between two flood beds midway up the section (Atwater, 1987, p. 188), to a dozen between the two lowest flood beds (Waitt). The lowest varve remnants contain a puzzling diamict lens ~10 cm by 10 m rich in crystalline-rock stones.

Varves without evidence for periodic floods overlap an unconformity or slide surface that truncates this entire sequence of sandy rhythmites to their southeast. Such varves, more than 100 in all, form cross-cutting packets beside Banks Lake south of the prow of Steamboat Rock (Fig. 23A). In addition, above the diamict in the 50-m exposure cited above, what may be the earliest of these varves succeed thin sandy rhythmites without evident hiatus (Fig. 23B). The sandy deposits, each apparently capped by the winter part of one or two varves, may correspond to the evidence for final Missoula floods at Stop 2.5.

The day's busy schedule excludes a visit to an estimated 180 varves of silt and clay, in a mostly flat-lying sequence uninterrupted by sandy flood deposits, that lap onto a boulder bar farther down the coulee, near Paynes Gulch (north-northeast of Stop 3.5, Fig. 14). Coarse silt in tens of these varves contain ripple-drift laminae, and their foresets indicate flow down-coulee (Atwater, 1987, p. 192–194). The flows likely represent peak yearly discharge in a late-glacial river-lake—the still-diverted Columbia River in the glacial Lake Columbia outlet-that coulee walls confined more near Paynes Gulch than near Steamboat Rock (topography, Fig. 4).

Hanging deltas along the west shore of northern Banks Lake stand some 30 m above the glacial Lake Columbia spillway at Coulee City. Isostatic rebound surely produced this difference at an unknown time with respect to the late-flood rhythmites and post-flood varves of the Upper Coulee.

Interpretations: After glacier ice retreated from the coulee floor and after the last highly erosive Missoula floods down Grand Coulee, there came at least 14 small floods that invaded a lingering river outlet of glacial Lake Columbia. As at Sanpoil Stop 2.5, a year or two separated final Missoula floods, and the post-flood river-lake persisted more than a century thereafter.

Return hike to vehicles ~25 min.

Mile

17.0 Return to SR 155 and turn south.

21.8 East-side longitudinal flood bar 3 km long (N 47.785; W 119.224), its north half strewn by hundreds of large granodiorite as well as basalt boulders.

24.9 In Paynes Gulch area on south part of this bar (N 47.756; W 119.224): Overlying the toe of the floodlaid gravel bar lies 3 m of silt-clay lake deposit including ~180 varves (Atwater 1987).

25.9 Park on right near top of large viewpoint pull-off.

Stop 3.5. Overlook of Central Reach of Coulee (47.7425, –119.2320)
(*Waitt*)

The tall roadcut in basalt gives a close view of some of the common internal structures—colonnade, entablature, occasionally palagonitic base, occasionally rubbly flow top—of the Miocene flood-basalt flows (Fig. 2B) (Mackin, 1961; Reidel et al., 2013a, fig. 14). These structures vary greatly between flows, and within some flows vary along strike of extensive exposures. The wide- and open-jointed colonnade horizons are the most quarryable, column by column, by deep turbulent floods. Some of the palagonitic basal horizons and rubbly tops are also quarryable. The finely jointed but more coherent entablature typically in middle to upper parts of basalt flows hangs together and resists direct erosion. But by quarrying underlying colonnade, a flood undermines entablature, which breaks off to become a deep flood's largest transported rocks—some of them 5–10 m and more across.

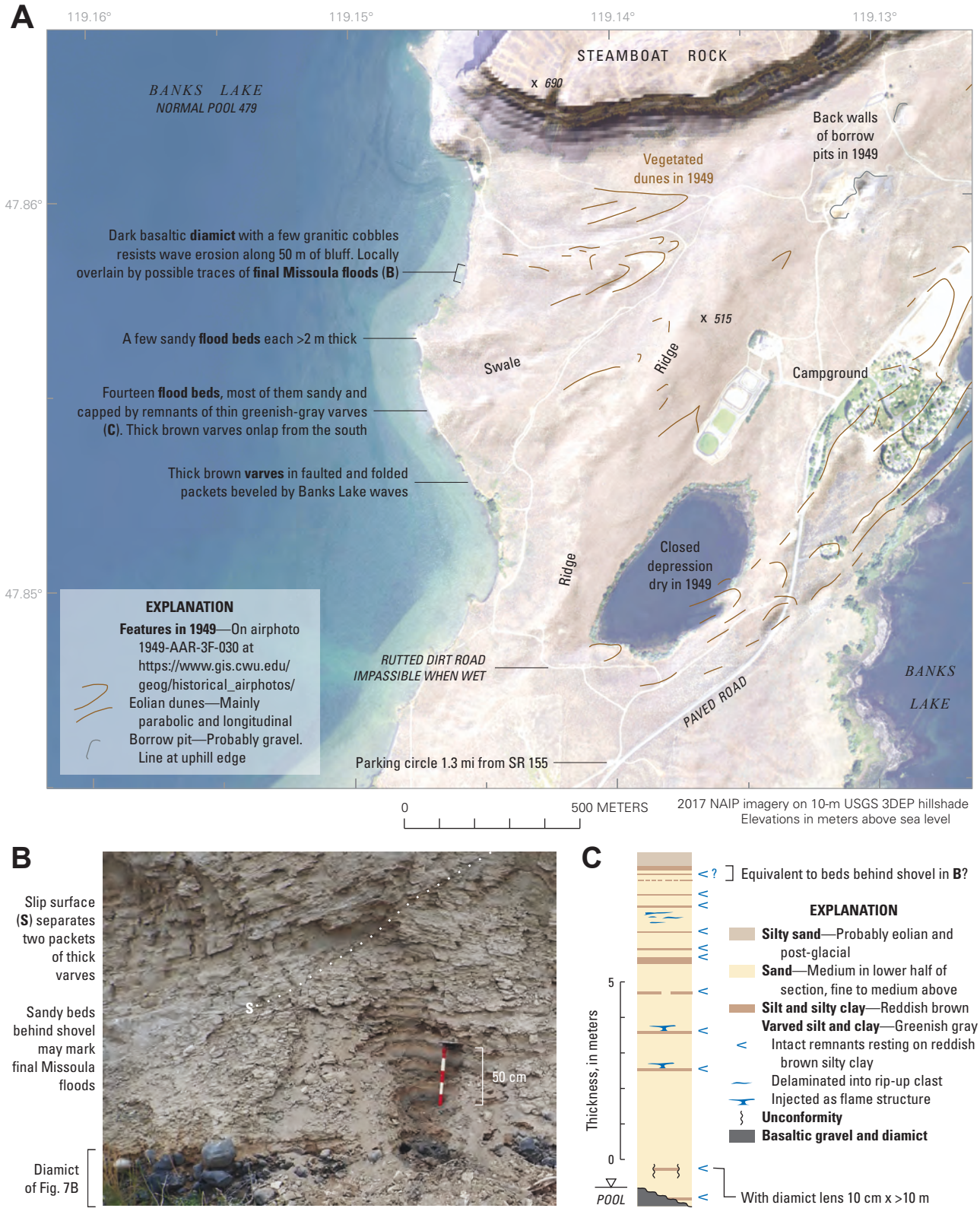


Figure 23. Setting and stratigraphy of glacial-lake deposits south of Steamboat Rock (Stop 3.4) (*Atwater*). (A) Landforms in vicinity of parking area and lake-shore outcrops. (B) Varves atop possible flood beds and last-glacial till. (C) Alternating flood beds and remnant varves, redrawn from Atwater (1987, fig. 5).

Paynes Gulch descends from above peak-flood level on the upside of Coulee monocline down to the eastside upper scabland. From there it faces an uphill grade to surmount the coulee's westside ridge (a low anticline?). This preglacial tributary clearly crossed the monocline southward. The westside ridge is notched by at least six preglacial southeast-draining tributaries that now hang far above Grand Coulee's west wall. All together these tributaries on both sides reveal a preflood master "Coulee creek" draining south about where Grand Coulee later formed (Fig. 4) (Waitt, 2021).

Among the unusual characteristics of upper Grand Coulee itself that Bretz (1932) outlined is the extraordinary fact that the coulee floor from Steamboat Rock to the brink of Dry Falls lacks a gradient. This part of upper Grand Coulee is indeed a closed depression, thus a *reverse* gradient to downcoulee streamflow. Beneath the (now-drowned) coulee floor lies nearly 100 m of gravel that fills a deeper bedrock basin.

(*O'Connor*)

We may discuss an alternative perspective on paleogeography to Waitt's "Coulee creek" and vale (Fig. 4). By this hypothesis, several of the eastside high tributaries like Klobuschur, Devils, and Rusho had connected westward to westside Foster and Horselake coulees, a drainage system later severed by cataract recession during Missoula floods.

Mile	
25.9	Continue south on SR 155.
29.8	In wall of basalt above on right, caves. Floodwaters excavated the more erodible colonnade horizons of basalt flows.
31.2	(47.6680, –119.2727). South 45° dips in the basalt—the steep limb of Coulee monocline. Upper Grand Coulee cataract apparently initiated down this steep slope (Bretz, 1932). During its headward retreat, the cataract increased in height to 150, 200, and 250 m, ultimately twice Dry Falls' height and four times Niagara Falls'. Just west of this site of initial cataract, drilling of the coulee floor revealed a plunge pool filled nearly 100 m with gravel (Bretz et al., 1956). Yet only 1.5 km south the basin floor (now beneath Banks Lake) is scabbed basalt. The north end of a high gravel bar spread into Hartline basin hangs off this high eastern scabland.
33.3	Junction of U.S. Highway 2. Turn left (east) onto it.
33.9	Halfway up the grade (47.6444, –119.2551), roadcut exposes 2–3 m of basaltic gravel (but unsuited to a group stop).
34.1	At top of grade where highway turns east, turn left into hay-stacking area. This place, a safer and less noisy alternative to the highway exposure just passed, is on private land. We are here by permission.

Stop 3.6. Hartline Basin Gravel Expansion Bar (47.6485, –119.2534)

[The actual stop on the field trip, depending on landowner's permission, may be as much as a quarter mile to the north-northwest near a gravel pit.]

(*Bjornstad*)

Where the upper Grand Coulee suddenly exits into the broad Hartline basin near Coulee City is a broad expansion bar 100 m tall referred to by Bretz (1932, p. 23) as "Hartline basin gravel." Growth of the gravel bar likely began in the early Wisconsin as the receding cataract quarried out 75 m or more of basalt bedrock north of the Coulee monocline. Today, the center of the expansion bar is missing, but erosional remnants of the bar exist at a common altitude (550 m) on both sides of the coulee (Fig. 24). An exposure of the bar's interior (Fig. 25) shows mostly angular to subangular gravel and sand entirely composed of basalt (Fig. 26). This composition suggests the bar formed earlier in the development of Grand Coulee prior to unroofing of the granitic basement at the upper end of the coulee (Fig. 4) (Bjornstad and Kiver, 2012). Elsewhere along the west side of the bar, a roadcut exposes several disconformities that separate multiple beds of similar character. Similarly, these deposits were transported into place prior to the breaching event and associated sudden lowering of base level that exposed granitic basement within the upper coulee. An interesting series of five regularly spaced trimlines appear etched into the side of the Hartline expansion bar at the mouth of the coulee (Fig. 26). These may represent trimlines etched into the side of the expansion bar from the last, successively smaller, Missoula floods that escaped down Grand Coulee ca. 15 ka.

(*Lehnigk and Larsen*)

The Hartline basin expansion bar is one of the few depositional landforms likely to record the maximum paleo-flow depth and discharge in Grand Coulee. However, inferring the discharge depends greatly on the downstream flow hydraulics as influenced by Dry Falls, and Bretz (1932) inferred the bar was deposited prior to the retreat of Dry Falls to its current location. When the topography of lower Grand Coulee is reconstructed to its pre-incision state, which eliminates any drawdown effects of Dry Falls, our models predict that the highest nearby evidence of flood stage (552 m) (Baker, 1973, appendix I, site 2 therein) in Hartline basin is inundated by a discharge of 2.5×10^6 m^3 s^{-1} (Fig. 27A). This discharge is comparable to the discharge we infer drove cataract retreat upcoulee near Steamboat Rock (2.8×10^6 m^3 s^{-1}). The similarity in the discharges is consistent with evidence from the basalt composition of gravels (Fig. 25) which indicate the bar was deposited when cataract retreat was occurring upstream. Bretz described the gravel deposit in Hartline basin as "almost perfectly fresh" (Bretz, 1932, p. 25) with respect to the degree of weathering. If the lack of weathering corresponds to deposition of the gravel during the last glaciation, then cataract retreat in upper Grand Coulee was likely still ongoing at that time.

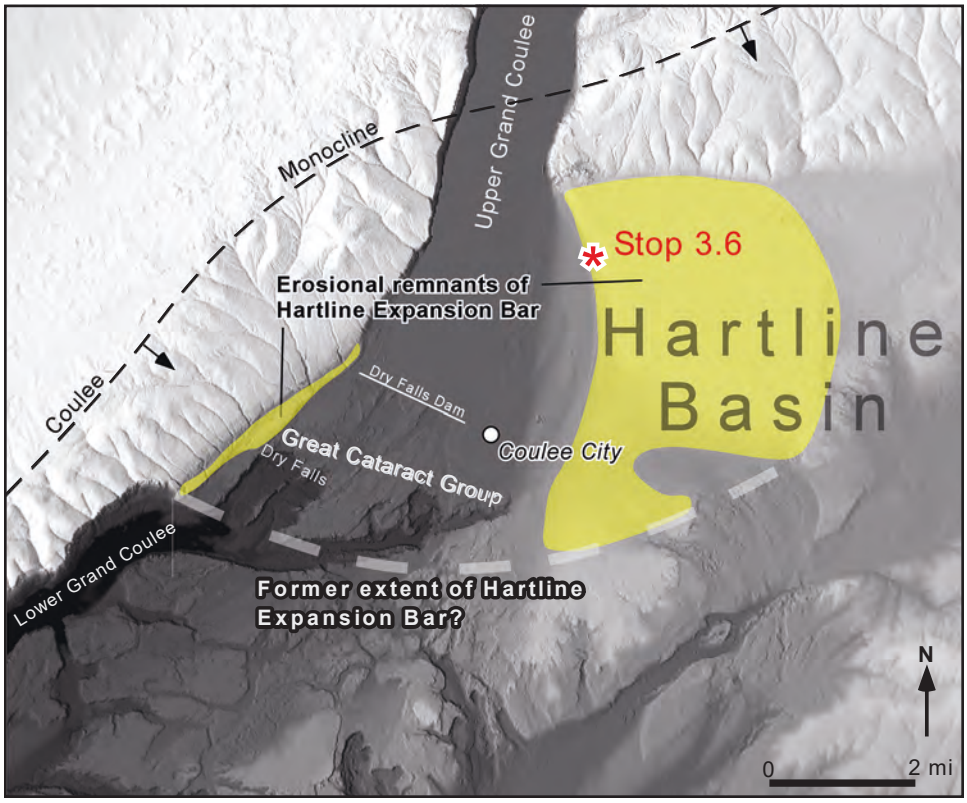


Figure 24. Shaded-relief map showing the former extent of the Hartline expansion flood bar at the mouth of upper Grand Coulee (Bjornstad and Kiver, 2012). The bar developed as the constricted floodwaters temporarily slowed and fanned out into the broad Hartline Basin (*Bjornstad*).

In the present-day topography, a discharge of 7.5×10^6 m^3 s^{-1} is required to inundate the same high-water marks in Hartline basin (Fig. 27B). The peak discharge at the expansion bar during glacial Lake Columbia drainage at the 750 m stage (on the present-day topography) is 7.3×10^6 m^3 s^{-1}. The maximum stage of the lake-draining flood comes close to high-water marks inferred by Bretz (1932) without exceeding them. However, given evidence that the bar was deposited before Dry Falls retreated to its current position (Fig. 24), it is unlikely that the bar was inundated by a 7.5×10^6 m^3 s^{-1} flood, which is required when Dry Falls is at its current location. The peak discharge we predict from the drainage of glacial Lake Columbia is hence likely a maximum value as it is nearly three times greater than what is required to inundate the Hartline bar, though the high-water evidence we used to constrain discharge in Hartline basin provides only a minimum constraint. The threefold variation in discharge that can be inferred for the Hartline basin expansion bar demonstrates how erosion can influence hydraulics and complicate paleo-discharge reconstruction, highlighting the need to understand the timing of both high-water mark emplacement and topographic evolution.

Retrace way on U.S. 2 and SR 155 north to Grand Coulee Dam and the motel.

Day 4. (Much of This Day Is Travel)

From Columbia River Inn drive now-familiar way SR 155 south past Grand Coulee Dam down upper Grand Coulee, across Dry Falls Dam), join SR 17, and drive on south to Dry Falls (our Stop 1.4).

Mile

0.0 On SR 17 south, **reset trip odometer while passing Dry Falls parking lot**.

3.3 **Park Lake.** (*Waitt*) View south of intake of divergent Jasper Canyon, which rejoins Grand Coulee at the Lenore Canyon intake. Bar in Jasper Canyon *rises* 45 m *down*current. The flat rimrock of Jasper Canyon is continuous scabland. Several kilometers southeast is wild scabland landscape with ~140 m of erosional relief (Fig. 28). Bretz (1932) wrote:

> Distributary escape from this part of Grand Coulee was not limited to the gravel-covered area. The whole adjacent summit of High Hill anticline was overrun. Though much of it is higher than the Hartline gravel plain, it is all scabland of very pronounced nature. The basalt yielded by plucking in an almost unbelievable way, and the top and back slope of the fold are scarred in a fashion that must be seen to be appreciated. Scabland channels wind about buttes twenty to fifty feet high in an intricate anastomosis that is wholly unknown in stream patterns. They are not separate streamways; they are subfluvial channels developed beneath one large stream. In this, they have counterparts in some



Figure 25. Flood-eroded margin of the Hartline expansion bar. Borrow pit at lower right exposes coarse basaltic sediments (shown in Fig. 26) transported by megafloods onto the bar. Trimlines at upper left were likely created by the last, successively smaller Missoula floods down Grand Coulee (*Bjornstad*).





Figure 26. Flood deposits exposed in borrow pit located on Figure 25. (A) Foreset-bedded, angular, coarse sand and gravel composed of 100% basalt. (B) Waste pile of unwanted basalt boulders up to 3 m diameter—too large and costly to transport or use as aggregate. (*Bjornstad*).

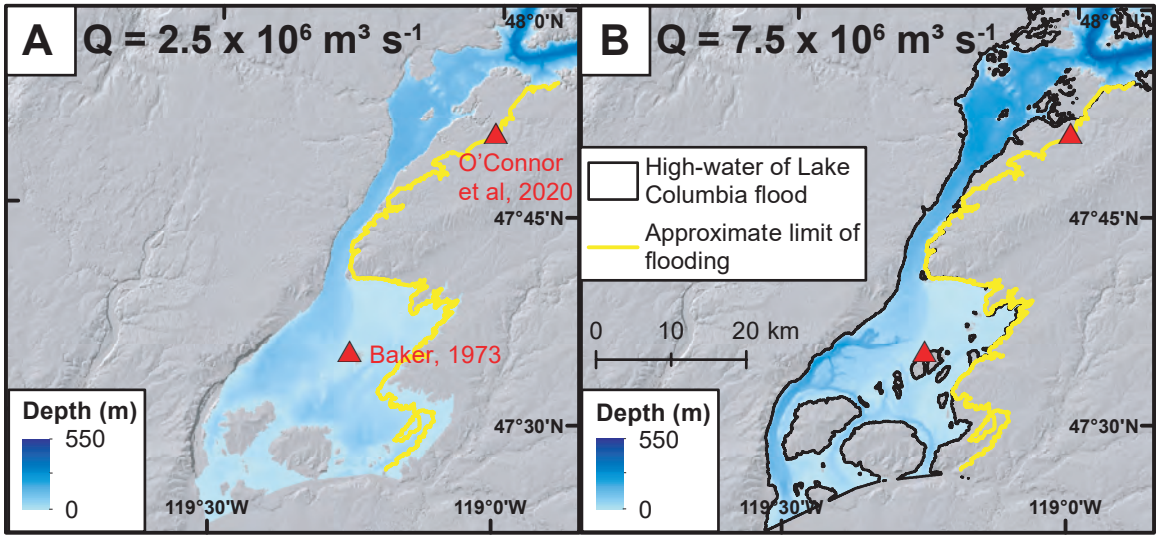


Figure 27. Inundation and discharge (Q) of floods that reach high-water marks in Hartline basin. (A) A steady-state discharge of 2.5 × 10^6 m^3 s^{-1} inundates high-water marks on reconstructed, pre-flood topography in lower Grand Coulee where canyons have been filled in to remove any draw-down effects of Dry Falls. (B) A steady-state discharge of 7.5 × 10^6 m^3 s^{-1} on the present-day Grand Coulee topography inundates high-water marks in Hartline basin. The maximum discharge from the flood that drains glacial Lake Columbia is 7.3 × 10^6 m^3 s^{-1} at the Hartline basin expansion bar (for the present-day topography), which inundates the bar surface and nearly reaches high-water marks inferred by Bretz (1932) (*Lehnigk and Larsen*).

existing Columbia River rapids [The Dalles channels]. No stream bottom has a continuous gradient parallel with the stream surface. Bottom water is constantly flowing upgrade out of deeper holes and over submerged bars. It erodes these holes, and the débris rolls or leaps upgrade out of them to come to rest in hillocks also under the stream surface. A drained river channel is not a valley in miniature. It is a complicated assemblage of hills and hollows. Channeled scabland is river-bottom topography magnified to the proportions of river-valley topography.

On east side of Park Lake, an upper gravel deposit "hangs" as a pendant bar downcurrent off valley-wall basalt; a lower one depends from midcoulee buttes of east-dipping monoclinal basalt. The lower deposit rises downcoulee with 17 m of relief between its crest and a fosse on its east side, and 25 m of relief on this west side. These are the whalebacked shapes by which Bretz repeatedly argued that the gravel cannot be dissected fill terraces but instead are enormous subfluvial bars.

5.2 Light-colored lacustrine silt in roadcut at right. Main exposed horizon 15 m long, 0.2 m thick, 3 m above base of cut. Basal contact horizontal, on basalt gravel of angular pebbles. Silt horizon contains 20–40 successive couplets between 2 mm and 15 mm thick. Graveled parking available beside Blue Lake at mile 5.3.

9.6 Pull into graveled area east side of of highway, north of monoclinal hogback.

Stop 4.1. Laminated Silt West of Lenore Caves (47.5192, –119.5009)
(*Atwater and Waitt*)

A west-side roadcut exposes lacustrine silt above basalt cobble gravel (Fig. 29A) near altitude 350 m. The thickest silt contains roughly 100 couplets (Fig. 29B), mainly 1–2 mm thick. Like thicker ones at mile 5.2, these couplets likely represent a glacially diverted Columbia River that was impounded behind Ephrata fan, south of Soap Lake. This river-lake spilled across a drainage divide today near altitude 350 m (Fig. 11).

If Lake Columbia outlasted glacial Lake Missoula, so did routing of Columbia River through lower Grand Coulee. The lake deposits preserved here evidently postdate the last of the Missoula floods that entrained the river lake and its bottom sediments.

In the upper 10 cm or so of the varved lake beds lies a pair of medium-sand ash layers (Fig. 29C and 29D). The lower one intercalated with lake beds up to 1 cm thick, the upper pumiceous one also within lake beds 4–6 cm thick. They appear to be Glacier Peak layer G (lower bed) and layer B (chemical fingerprinting is in progress). Together the Glacier Peak G and B tephras are dated to ca. 13,650 ka (Kuehn et al., 2009).

About 7 miles (11 km) south near Soap Lake, this same postglacial lake supported mollusks (Landye, 1973) and accumulated

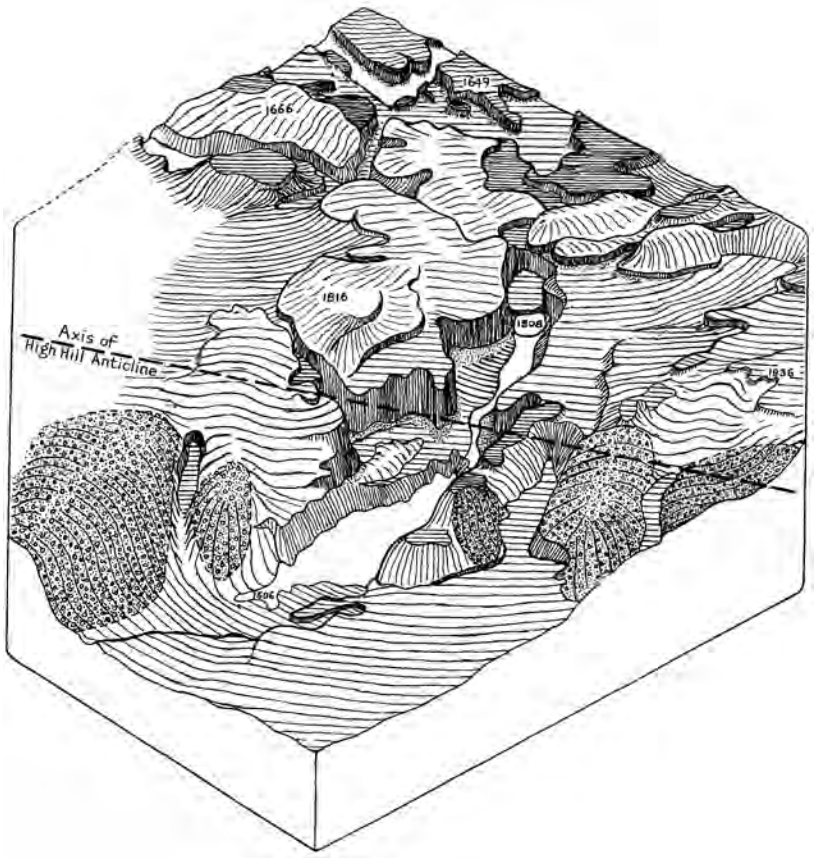


Figure 28. Scabland south or great cataract group (including Dry Falls) and east of lower Grand Coulee as Bretz (1932, fig. 16) pictured it before the eras of modern imagery, digital terrain models, aerial photographs, and large-scale topographic maps. No scale on original drawing, but across the center is roughly 30 km. Bretz's caption: Block diagram of scabland topography on High Hill anticline. Though the anticline above the limit of scabland is a divide, it here has canyons with cataract heads and plunge pools that have receded across the summit almost to Deep Lake. Some of the bars of the southern slope shown with stippling.

Glacier Peak B tephra (Fryxell, 1965; Kuehn et al., 2009, table S1 therein).

Mile

9.8 Gravel road at left leads to Lenore Caves trailhead.

Lenore Caves Area—A Non-Stop

(*Waitt*)

Dip of coulee monocline is ~20° east-southeast. The coulee is cut along the monoclinal axis, the basalt more erodible where it is more fractured (Bretz, 1932). Extreme flows threw water against west wall of curving valley, where widening left preglacial tributaries hanging conspicuously.

Silt overlying basalt as high as 18 m above present lake (alt. 331 m) registers minimum height of a postflood lake. This 349 m altitude is just 1.5 m below apparent upper limit of silt at Soap Lake and 3 m below to lowest point on bar impounding Soap Lake (352 m spillway to Rocky Ford channel). The ponding here seems caused by Soap Lake bar. The many very small floods descending Grand Coulee near the end of Missoula flooding (Stops 2.5 and 3.4) flushed such silt into lower Grand Coulee.

17.5 Halfway along length of Soap Lake, a broad pull-off lies on the right.

Soap Lake Basin—A Non-Stop (47.4058, –119.4905)

(*Waitt*)

Two miles upvalley, the great lower Grand Coulee cataract initiated, down the steep slope of the Coulee monocline. With time it receded along the limb of the monocline 27 km north to Dry Falls. Cataract recession left preglacial tributary valleys hanging 210–270 m above the gorge's floor.

Soap Lake occupies a basin where prodigious current shot out of a rockbound course. Its flood-scoured rock bottom is 65 m below lake surface, and that overlain by 34 m of flood gravel (Bretz et al., 1956). As floodwater abruptly spread into Quincy basin, currents slackened, depositing a crescentic bar around Soap Lake 3–5 km out. Notched in four places, its top lies 25–75 m above lake surface. The lake's rock bottom is 88 m below the lowest notch in the bar's rim, a notch leading to Rocky Ford coulee. With no surface outlet and discharging by evaporation, Soap Lake evolved into alkalinity (pH >> 9), the main salt Na_2CO_3. Alkali-salt encrustations on cliff faces reveal prehistoric alkaline-lake levels at least 9 m above present lake.

Mile

17.5 Continue south on SR 17.

18.8 Town of Soap Lake, Wash. From here south topography *ascends* in the downflood direction.

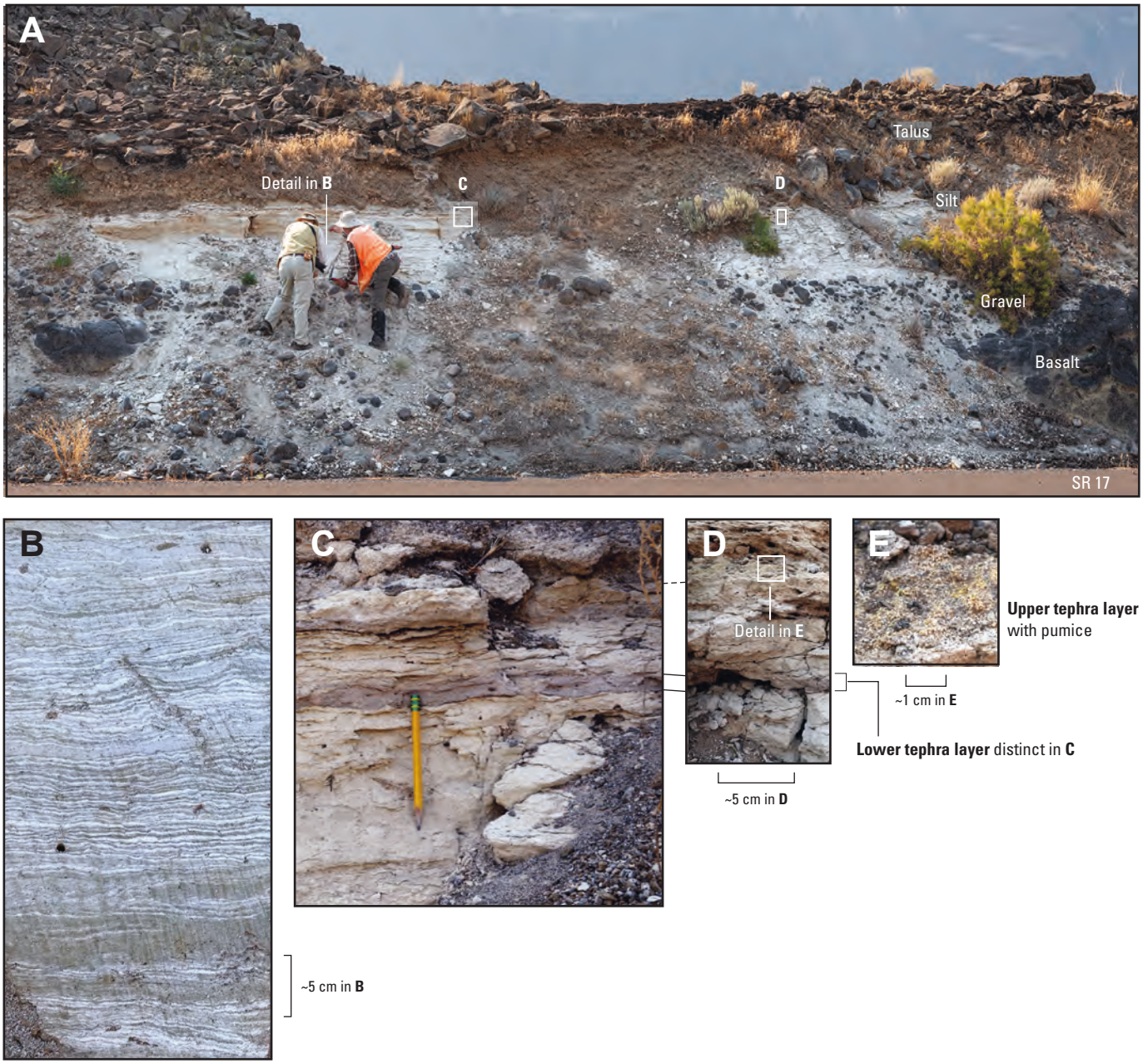


Figure 29. Evidence for a post-glacial lake in lower Grand Coulee (Stop 4.1). (A) Roadcut along SR 17 in which basalt underlies floodlaid gravel, laminated silt, and capping talus. (B) Scraped-down exposure low in the silt, which is uninterrupted by coarser deposits. (C–E) Tephra layers, high in the silt, apparently Glacier Peak G (lower) and B (lapilli; upper). Photos B and E by Nicolaus Zentner and John Stone, respectively (*Atwater and Waitt*).

19.3	Intersection with SR 28. Continue climb out of basin low on SR 17.
20.9	An indistinct notch in the crescentic bar surrounding Soap Lake basin leads into a shallow vale (Rocky Ford coulee) just south of railroad crossing. (Jim Hill's Great Northern Railway pushed west through here in 1893.)
21.5	Boulders to 1.5 m and more scattered on sides and floor of Rocky Ford channel. Seemingly exhumed by late small Missoula floods incising the high bar build by earlier large floods.
22.2	In vicinity of powerline crossing, hundreds of large boulders, up to 2 m and more, almost all basalt.

23.0	Troutlodge Road (gravel) on left. Turn east onto it.
23.7	Outsized boulder of basalt entablature 11 m in intermediate diameter. Large southeast-trending scour depression around it, mapped by Baker (1973, fig. 29).
24.1+	Bouldery bar.
24.5	Columbia River basalt. The giant coarse, thick bar overlies a bedrock basalt basin.
24.9	Low boulder-strewn point bar.
25.1	Turn left (north) as road continues through hatchery.
25.3	Main hatchery buildings. Continue up along tall roadcut.
25.7	At top of grade, park in gravel area on right.

Stop 4.2. Northern Quincy Basin Gravel Bars at Rocky Ford channel (47.3249, –119.4370)
(*Waitt and Atwater*)

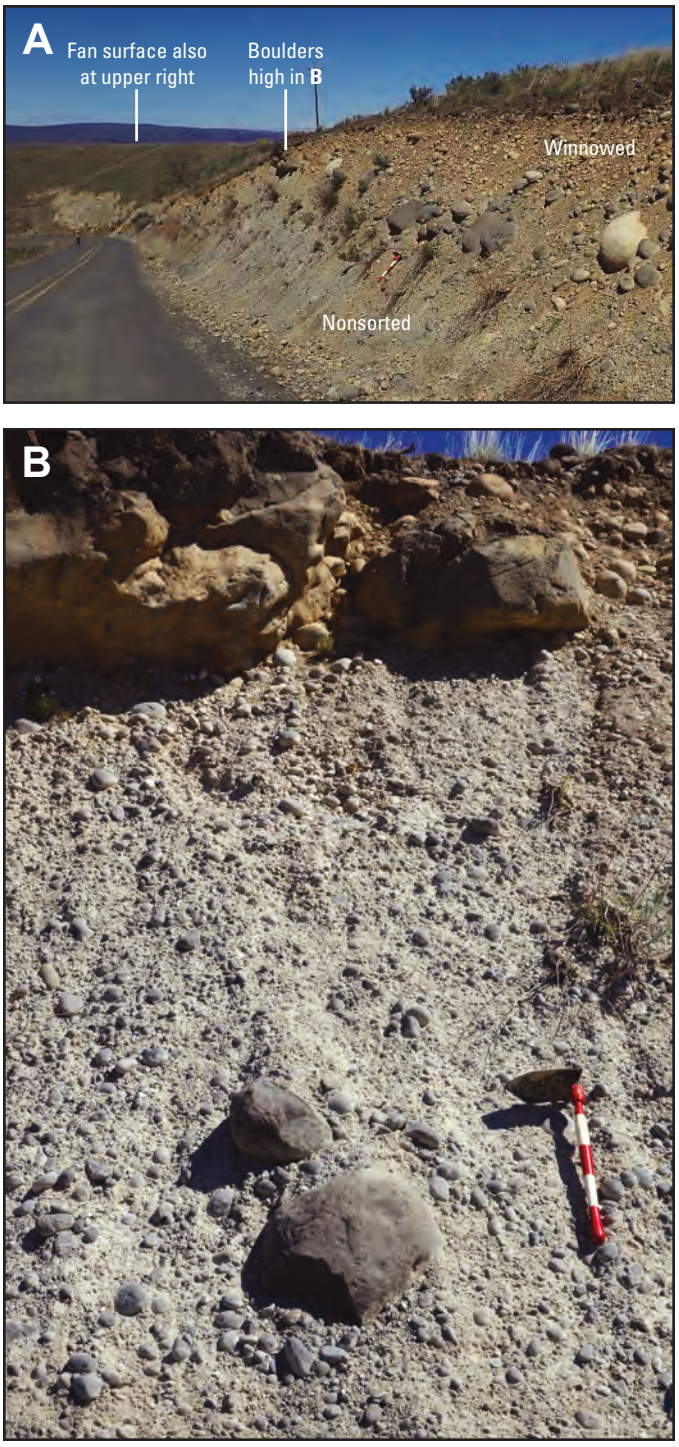
Previously unreported here is evidence that Missoula floods moved gravel not just in watery floods but also in debris flows. Also notable is evidence that later, lesser Missoula floods lowered boulders by tens of meters onto inset terraces that they armor, as a lag.

We stand on a high surface of Ephrata fan, a giant expansion bar built into Quincy Basin by floods exiting the Grand Coulee (Fig. 11). Below us meanders Rocky Ford channel, a late-glacial path of Columbia River aligned with the lowest of four northern notches into the fan. The adjacent road grade exposes well-sorted cobbles and boulders winnowed by flows that rinsed the fan crest. Below, to the west, a flight of boulder-studded terraces rises along Troutlodge Road to SR 17.

Evidence for debris flows crops out high in the roadcut as gravel with a fine matrix. This diamict, several meters thick, lacks internal stratification (Fig. 30). Its lack of sorting contrasts with Missoula-flood gravel reported elsewhere that show obvious bedding and sorting—as do lower parts of this roadcut. The sorted textures of most flood deposits in the region are evidence of a watery flood (less dense than a hyperconcentrated flow or debris flows) and thus capable of winnowing out finer grain sizes, concentrating the gravel fraction.

The diamict here implies that Missoula-flood flow(s), though initially watery, bulked up in many reaches along the way with a great deal of fine sediment. In Grand Coulee available sources of this sediment include the floor of glacial Lake Columbia (Stop 2.4), the terminal moraine of the Okanogan lobe in upper Grand Coulee, and a lake in lower Grand Coulee that the Ephrata fan impounded (Stop 4.1).

Bulking up of initially watery flows was observed in the 1980s at recently erupted volcanoes. Such floods starting with water—from a failing debris-dammed lake, or from snow swiftly melted by flowing hot pyroclasts—entrained enormous quantities of loose sediment in gullies or on steep slopes. The flows changed phase, from liquid to a viscous mix, and the entrained



Striped handle 0.5 m in both photos

Figure 30. Photographs at Stop 4.2. (A) View west of roadcut showing overall stratigraphy. (B) nonstratified poorly sorted floodlaid diamictic gravel with tight, well-consolidated matrix like that of a debris flow. Boulder ~30 cm diameter floats at the top. Gradations on shovel handle 10 cm (*Atwater and Waitt*).

solids increased the flow volume by a factor of two or more (Scott, 1988, 1989; Pierson et al., 1990; Vallance and Iverson, 2015). Such a sediment-dense debris-flow fluid has considerable strength, some debris flows and dense lahars having rafted boulders tens of meters across (e.g., Crandell and Waldron, 1956).

The large boulders here and elsewhere in the Channeled Scabland have traditionally been interpreted as having been transported in traction by water (e.g., Baker, 1973, p. 22–32). But the diamict shows that some Missoula flood flows entrained enough fine sediment, probably from lakes in their path, to reach in places debris-flow density. In that case, many of the boulders flushed out of the Grand Coulee, including the largest ones, may have rafted within or atop dense debris-flow fluid.

Bretz et al. (1956, p. 974) attributed incision of Rocky Ford channel to a few floods that failed to deepen other northern notches of Ephrata fan. And at p. 971:

> A post-flood glacial Grand Coulee river (a detoured Columbia) and/or discharge of a post-glacial lake in Lower Grand Coulee may well have been responsible for the only normal stream-valley pattern in all of Quincy basin's or, indeed, any other scabland gravel deposits.

Baker (1973, p. 39–42) instead called upon a single giant flood whose waning flows ate progressively into the main deposit, leaving behind as a terrace-armoring lag only the boulders they could not remove. The incision and winnowing here most likely concluded during the last of scores of last-glacial floods from a Lake Missoula—a self-dumping lake that during its final centuries drained progressively smaller floods (Waitt, 1985; Atwater, 1986).

Granitic boulders of Ephrata fan, which include much of the boulder lag to our west, were presumed by Bretz et al. (1956, p. 977) to have been derived mainly from the basement rocks of the upper Grand Coulee. Balbas et al. (2017) reported ^{10}Be exposure ages ranging from 14 to 17 ka on seven of the fan's granitic boulders (Fig. 11). All seven are inset some 20–40 m into highest adjoining parts of the fan. Balbas et al. (2017, Data Repository item 2017193, https://doi.org/10.1130/2017193) inferred that the ages represent "a debris-dam failure that held back glacial Lake Columbia." Exposure of the boulders in their current, exhumed positions more likely began when frequent smaller Missoula floods exhumed the boulders but had also become too small to move them.

Mile	
0.0	**Zero odometer.** Return on Troutlodge Road to SR 17.
2.5	SR 17. Turn left (south).
4.0	Many large basalt-entablature 2–5 m boulders. These are the coarsest elements of flood deposits that fine to sand 20–25 miles (32–40 km) to the southwest.
5.0	Very large basalt boulder.
6.2	Intersection SR 282. Turn right toward Ephrata.
10.2	Top of highest gravel bar. Like highest part of bar at Stop 4.2, part of so-called "Ephrata fan" built by largest Missoula floods from lower Grand Coulee through Soap Lake basin. Begin descent off gravel bar west into Ephrata channel.
10.5	We have descended into a floodswept channel cut below highest flood bars.
10.8	Ephrata. Intersection SR 28. Turn left (south) onto SR 28. For next 23 miles our route is reverse direction of our Day-1 route.
15.2	Highway intersection. SR 28 turns west. Continue straight (southwest) onto SR 283.
20.4	Round-stone large cobbles culled from fields.
20.6	Winchester wasteway.
25–2	Surface sediments in southern Quincy basin have fined to sand.
26.4	Enter ramp for I-90 west at exit 151.
37.0	Cross Silica Road. Highway has descended southwest Quincy basin to intake to its overflow outlet Frenchman Springs cataract (field trip Stop 1.3).
38.4	Spoil heap from former diatomite mine.
39–41	Cross Frenchman Hills anticline, uparched Columbia River basalt.
45	Cross Columbia River bridge.
46	Vantage, Wash. Continue west on I-90. In the basalt, a relative synclinal area south of east-west Frenchman Hills anticline. Begin westward climb of a broad north-south anticline that Mackin (1961, fig. 1) calls the Hog Ranch axis. This huge structural high separates the Yakima drainage basin from the Columbia drainage basin.
56–58	Crest of Hog Ranch axis some 600 m above Columbia valley and 300 m above Kittitas Valley to the west. This huge north-south welt is somehow part of the Yakima-foldbelt tectonics (Fig. 2A), but its trend crosses the smaller, sharper east-west anticlines and synclines.
74	Kittitas valley, a broad syncline all but surrounded by anticlinal ridges in Columbia River basalt. At exit 110, leave I-90 and enter I-82 south. **Zero trip odometer at intersection (bridge).**

Mile	
0.0	Highway I-82 south.
3–25	I-82 crosses three large east-southeast–trending anticlines and intervening synclines in Columbia River basalt, part of the Yakima foldbelt (Fig. 2A). In southward succession the anticlines are Manashtash, North Umtanum, and South Umtanum Ridges. Roadcuts across the anticlinal crests show spectacularly complex folding and faulting of basalt, visually highlighted by sedimentary interbeds between some basalt flows.

25–37	Yakima basin synclinal valley.
38–39	Union Gap, a water gap through Yakima Ridge anticline. Enter lower Yakima valley syncline.
50	Exit I-82 at exit 50 at Buena. South on SR 22.
53	Intersection highway U.S. 97. Straight through intersection onto U.S. 97 south. From here to Columbia valley we follow way of trip Day 1 in reverse direction between those miles 61 and 3. We again cross southern elements of the Yakima foldbelt (Fig. 2A).
105+	Descending the long curving grade into Columbia valley, one sees ahead on south valley side the scabbed surfaces of several benches on Columbia River basalt up to 280 m above the river's natural (pre-dam) grade.
111	Approaching intersection of SR 14. Bear right onto SR 14 west. En route to next stop, we have diverse views along road and across to Oregon side of valley of Columbia River basalt, many rubble fans of atop it from higher slopes, and conspicuous to subtle cuts into some of these by the Missoula floods. Missoula-floods descriptive details and interpretation contained in Benito and O'Connor (2003) and for part of the south (Oregon) side of valley field trip details by O'Connor and Waitt (1995). For detailed field presentations along the Columbia Gorge of Columbia River basalt, Missoula-floods effects, and postflood landslides, see O'Connor et al. (2021, this volume).
117.7	Road south to Wishram. Continue ahead on SR 14 west.
118.5	Pull off into broad highway viewpoint (with interpretive signs) on left.

Stop 4.3. Columbia Gorge: Wishram Overlook (45.6629, –120.9566)
(*O'Connor*)

In an upper reach of Columbia Gorge, we look across Lake Celilo, raised behind the Dalles Dam in 1957. It stilled the roar of "The Dalles of the Columbia," a wild stretch of flood- and river-eroded bedrock where the river descended 25 m in 20 km through a series of falls and chutes. Under the south side of pool in front of us was Celilo Falls, where the river fell over the cataract of Horseshoe Falls, a 7 m drop at low flow. This stretch of rapids was a cultural mainstay for the Indigenous nations, attracted by tremendous fisheries of salmon, steelhead, and Pacific lamprey. The fish, resting and leaping to navigate the narrow passages and falls, were netted or speared by Indigenous fishers poised on rickety platforms. The constant wind would then dry and preserve this critical food, part of the "Great Mart of all this Country" recorded by William Clark in 1805 (Moulton, 2002, v. 5, p. 333). The site and remnant fishery remain highly important to the Columbia Plateau Indigenous people, now mostly displaced to the Yakama, Nez Perce, Umatilla, and Warm Springs reservations.

The slots and holes of the Dalles of the Columbia are carved into horizontal to gently west-dipping Columbia River basalt flows. Cataracts like Horseshoe Falls appear to be maintained by undercutting of the weaker flow contacts, promoting toppling and stepwise retreat of the entire flow thickness. Bretz (1924) devoted an entire paper to "The Dalles type of river channel" in which he linked the processes of basalt erosion evident here to the Missoula-flood carved forms of the channeled scabland—including the eroded high benches across the river. He emphasized the role of plucking by swirling vortices in eroding deep holes, not just in the channel but on the flanking basalt surfaces. Soundings show that some of these holes bottom out 75 m below the low-flow river surface, 60 m below sea level.

We are upvalley of the major gorge constrictions. Maximum flood stage was between 340 and 350 m asl, some 165 m above us. Although Missoula floods were mostly confined to the valley of the Columbia, some flood water took short detours, to the south where floods overtopped divides into tributary valleys. One such example is seen to the southwest where Fairbanks Gap decanted floodwater southward into Fifteenmile Creek valley. These divide crossings and the stratigraphy of large delta bars in the backflood tributaries enable counts of the number of floods of certain flood sizes. The col of Fairbanks Gap requires a flow of ~5 million m^3/s to overtop—about half the peak flood discharge for the Columbia Gorge estimated by the older step-backwater modeling of Benito and O'Connor (2003).

The delta bar deposited by flooding through Fairbanks Gap exposes two deposits, indicating at least two floods at least 5 million m^3/s in peak discharge. The gravel delta bar is capped by 20 sand-silt flood beds, inferred left by 20 floods less than 5 million m^3/s that could enter Fifteenmile valley only downstream via lower divides or at the valley mouth at the Columbia. By such reasoning adopted at several places in the gorge, Benito and O'Connor (2003) concluded at least one flood attained ~10 million m^3/s floods, six floods >6.5 million m^3/s, 15 floods >3 million m^3/s, and 10 floods between 1 and 3 million m^3/s. This record is incomplete, only accounting for 25 of ~100 Missoula floods counted elsewhere, but it shows the range of flow magnitudes. More recent and ongoing 2-D flow modeling (Denlinger et al., 2021; O'Connor et al., 2020) likely improve the discharge estimates, but the overall finding of a wide range of flood sizes will stand.

To the east lies bedrock Miller Island that crests 100 m above Lake Celilo, a Missoula flood pendant bar hanging from the island's downstream end. At its upstream end, basalt of Haystack Butte nestles among the Columbia River basalt flows. This basalt cascaded down the colluvial slopes of the Columbia Hills anticline from Haystack Butte, a Pleistocene basaltic volcano on the ridgeline to the east (at communication towers). The lava flow has been recently re-dated with two ^{40}Ar-^{39}Ar ages 0.8315 ± 0.0062 and 0.8365 ± 0.0052 Ma (S.E.K. Bennett and M. Stelten, USGS, personal commun., 2018). At its lowest extent along

the southeast margin of Miller Island, basalt of Haystack Butte flowed into a Columbia River channel, the flat-bottomed flow overlying unweathered quartzite-rich cobble gravel of Columbia River. This contact lies ~18 m above the historical low-water level of the Columbia River but within the range of spring freshets. The 0.84 Ma Basalt of Haystack Butte at near historical river level shows that the Missoula floods did not substantially incise the gorge. This relation and others like it provide evidence that the Columbia Gorge was an established canyon at the time of the Missoula floods, graded to near its modern level. Local widening is evident, though, shown by trimmed colluvial fans and ragged basalt edges.

The basalt of Haystack Butte, preserved on the south edge of Miller Island requires that the deep north channel isolating the tall bedrock island was cut after the lava flow. The likely cause is intense erosion by the Missoula floods, likely by cataract retreat. This feature seems a grand example of a "trenched spur" described by Bretz (1928a) in other parts of the channeled scabland.

Mile	
118.6	Continue west on SR 14. Diverse Missoula-floods effects next 10 miles along this highway, including a stretch of spectacular high-relief scabland. Descriptive details and interpretation for some Missoula-floods elements including high-water indicators contained in Benito and O'Connor (2003). For part of the south (Oregon) side of valley field trip details by O'Connor and Waitt (1995).
128.6	Intersection of U.S. 197. Turn south on 197 toward Columbia River.
131.2	The Dalles bridge.
131.4	Crossing Columbia River. Here just below The Dalles Dam river is nearly at its natural grade.
132	Ramp to I-84 at exit 87. Take it onto I-84 west. From here it is 87 miles in reverse direction from the beginning of Day 1 between miles 0 and 87.
219	Exit I-84 in Portland.
220	Oregon Convention Center, Portland. End of Day 4, end of field trip.

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